Homological Algebra

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1 Chain Complexes

1.1 Complexes of R-modules

1.1.1 Definition

Let R be a ring. A **chain complex** of R modules is a sequence of R-modules $C_\bullet = \{C_n\}_{n\in\mathbb{Z}}$ along with maps $d_n\colon C_n \to C_{n-1}$ such that $d_{n-1} \circ d_n\colon C_n \to C_{n-2}$ is zero for all n. These maps d_n are called the **differentials** of $C_\bullet$.

We often omit the subscripts for the differentials and the chain complex itself; we write d and C instead of d_n and $C_\bullet$ respectively.

Note that $d_n \circ d_{n+1} = 0$ means that $\mathrm{im} d_{n+1} \subseteq \ker d_n$. This justifies the following definition:

1.1.2 Definition

Given a chain complex C, define

(1) the kernel of $d_n\colon C_n \to C_{n-1}$ to be $Z_n = Z_n(C)$, the n**-cycles** of C;

(2) the image of $d_{n+1}\colon C_{n+1} \to C_n$ to be $B_n = B_n(C)$, the n**-boundaries** of C;

(3) the n**th homology module** of C to be their quotient: Z_n/B_n.

1.1.3 Example

Let $C_n = \mathbb{Z}/8\mathbb{Z}$ for $n \geq 0$ and $C_n = 0$ for $n < 0$. Define d_n to map x to $4x$ (modulo 8).

Then we have $d \circ d$ maps x to $8x \equiv 0$, so this forms a chain complex (of $\mathbb{Z}/8\mathbb{Z}$-modules, say). And Z_n is $\mathbb{Z}/2\mathbb{Z}$, B_n is $\mathbb{Z}/4\mathbb{Z}$. Thus we have H_n is (isomorphic to) $\mathbb{Z}/2\mathbb{Z}$.

We would like to build a category of chain complexes of R-modules. To do so we must define morphisms of chain complexes; there is a clear way to do so.

1.1.4 Definition

A **morphism** of chain complexes $u\colon C \to D$ is a collection of R-module morphisms $u_n\colon C_n \to D_n$ which commute with the differential maps: $u_{n-1} \circ d_n = d_{n-1} \circ u_n$.

The category $\mathsf{Ch}(\mathrm{Mod}_R)$ is the category of chain complexes of (right) R-modules.

1.1.5 Proposition

A morphism of chain complexes $u\colon C \to D$ maps boundaries to boundaries and cycles to cycles. Thus defines maps $H_n(C) \to H_n(D)$. Furthermore, for each $n \in \mathbb{Z}$, H_n defines a functor $\mathsf{Ch}(\mathrm{Mod}_R) \to \mathrm{Mod}_R$.

Proof

We have that $udx = dux$, thus for $dx \in B(C)$ its image under u is in $B(D)$. Similarly if $dx = 0$, then $udx = dux = 0$, in particular $ux \in Z(D)$. And so u_n factors through $B_n(C)$ and restricts to cycles, giving us maps $u_n\colon H_n(C) \to H_n(D)$.

This is a functor, as quotienting a morphism is. ■

1.1.6 Example

Let k be a field, and consider k-vector spaces $\{B_n, H_n\}_{n\in\mathbb{Z}}$. Define $C_n = B_n \oplus H_n \oplus B_{n-1}$ and define $d_n\colon C_n \to C_{n-1}$ by projection onto $B_{n-1} \subseteq C_{n-1}$. This is a chain complex: d_n projects onto B_{n-1} which is orthogonal to B_{n-2}, onto which d_{n-1} projects. Further note that $Z_n(C) = B_n \oplus H_n$ and $B_n(C) = B_n$, thus $H_n(C) = H_n$; justifying the choice of names.

If V is a chain complex of k-vector spaces, then note that $V_n \cong B_n(V) \oplus H_n(V) \oplus B_{n-1}(V)$. Thus all chain complexes of k-vector spaces can be constructed in this manner.

1.1.7 Example

Let C be a chain complex of R-modules, and A an R-module. Then we define $d'_n\colon \hom_R(A, C_n) \to \hom_R(A, C_{n-1})$ by mapping f to $d_n f$. This forms a chain complex clearly.

Now if $A = Z_n(C)$ for some n, suppose $H_n(\hom_R(Z_n, C)) = 0$. This means that $B_n(\hom_R(Z_n, C)) = Z_n(\hom_R(Z_n, C))$, i.e. if $df = 0$ then $f = dg$ for all $f\colon Z_n \to C_n$. In particular for inclusion $\iota\colon Z_n \to C_n$; $d\iota = 0$ so $\iota = df$ for some $f\colon Z_n \to C_{n+1}$. But this means that $Z_n \subseteq B_n$ (and thus there is equality): $Z_n = \mathrm{im}\iota = \mathrm{im}df \subseteq \mathrm{im}d = B_n$, and so $H_n(C) = 0$.

1.1.8 Definition

A morphism of chain complexes $C \to D$ is a **quasi-isomorphism** if the maps $H_n(C) \to H_n(D)$ are all isomorphisms.

There is a clear chain complex for all R: the zero chain complex, consisting of only the zero R-module. This complex is a zero object in $\mathsf{Ch}(\mathrm{Mod}_R)$: it is both initial and final.

1.1.9 Proposition

The following are equivalent for every chain complex C:

(1) C is exact;

(2) C is **acyclic**: $H_n(C) = 0$ for all n;

(3) The map $0 \to C$ is a quasi-isomorphism.

Proof

(1) and (2) are clearly equivalent: $H_n(C) = 0$ iff $B_n(C) = Z_n(C)$, which is just what it means for C to be exact. These are also clearly equivalent to (3), by definition. ■

1.1.10 Definition

A **cochain complex** of R-modules is $C^\bullet$, consisting of a collection of R-modules $\{C^n\}_{n\in\mathbb{Z}}$ along with differential maps $d^n\colon C^n \to C^{n+1}$ such that $d \circ d = 0$. As before, $Z^n(C^\bullet) = \ker d^n$ is the module of **n-cocycles** and $B^n(C^\bullet) = \mathrm{im}d^{n-1} \subseteq C^n$ is the module of **n-coboundaries**. $H^n(C^\bullet) = Z^n(C^\bullet)/B^n(C^\bullet)$ is

the n**th cohomology module** of $C^\bullet$.

Morphisms of cochain complexes are defined the same.

If $C_\bullet$ is a chain complex we can define a cochain complex $C^\bullet$ simply by $C^n = C_{-n}$ and $d^n = d_{-n}$.

1.1.11 Definition

A chain complex C is **bounded** if all but finitely many C_n are zero. If $C_n = 0$ for all n other than $a \leq n \leq b$, then C has **amplitude in** $[a, b]$. C is **bounded above** (resp. **bounded below**) if there is a b (resp. a) such that $C_n = 0$ for $n > b$ (resp. $n < a$).

The full subcategories of bounded, bounded above, bounded below chain complexes are denoted Ch_b, Ch_-, Ch_+ respectively. The subcategory of non-negative chain complexes is denoted by $\mathsf{Ch}_{\geq 0}$ (those chain complexes bounded below by 0).

Similar for cochain complexes.

1.2 Operations on chain complexes

The purpose of this section is to show that chain complexes form an **Abelian category**. Recall the following category-theoretic definitions:

1.2.1 Definition

A **preadditive category** is a category whose hom-sets are given the structure of Abelian groups, such that composition distributes over addition: $h(g+g')f = hgf + hg'f$. An **additive functor** is a functor between two preadditive categories whose restrictions to the hom-sets forms an Abelian group homomorphism.

Finite products and coproducts in a preadditive category necessarily coincide; they are called **biproducts**. An **additive category** is a preadditive category with all finite biproducts (equivalently a zero object and binary biproducts).

$\mathsf{Ch} = \mathsf{Ch}(\mathrm{Mod}_R)$ is an additive category. We can add chain complex morphisms in a clear manner: $(u+v)_n = u_n + v_n$. Furthermore as discussed, Ch has a zero object. Ch has all arbitrary direct products and coproducts (and thus all finite biproducts, so it is an additive category).

Indeed, let $\{C_\alpha\}_{\alpha \in I}$ be a collection of chain complexes of R-modules. We define their product $\prod_\alpha C_\alpha$ to have as components the direct product of the components: the nth component is $\prod_\alpha C_{\alpha,n}$. Similarly their coproduct $\bigoplus_\alpha C_\alpha$'s nth component is $\bigoplus_\alpha C_{\alpha,n}$. The differential maps are defined coordinate-wise as well. These satisfy the universal properties of products and coproducts respectively.

1.2.2 Proposition

Products and coproducts commute with the homology functors. That is to say, $H_n(\prod_\alpha C_\alpha) \cong \prod_\alpha H_n(C_\alpha)$ and similar for coproducts.

1.2.3 Definition

A chain complex B is a **subcomplex** of another chain complex C if B_n is a submodule of C_n for every n, and B's differential maps are the restrictions of C's.

If B is a subcomplex of C, then we can define their **quotient complex** C/B whose components are C_n/B_n and whose differential maps are the factorizations of C and B's.

In particular, if $f\colon B \to C$ is a chain map, the kernels $\{\ker f_n\}_n$ form a subcomplex of B denoted $\ker f$. And the cokernels $\{\operatorname{coker} f_n\}_n$ form a quotient complex of C denoted $\operatorname{coker} f$.

1.2.4 Definition

In a category with a zero object (more generally with zero morphisms), the **kernel** of a morphism f is its equalizer with 0. Its **cokernel** is its coequalizer with 0.

In a general category, a **monomorphism** is a morphism which is left-cancelative: $fg = fh$ implies $g = h$. An **epimorphism** is one which is right-cancelative: $gf = hf$ implies $g = h$. In a preadditive category monomorphisms are precisely those morphisms for which $fg = 0$ implies $g = 0$; epimorphisms are those for which $gf = 0$ implies $g = 0$.

Our notion of kernels and cokernels in Ch coincide with the category-theoretic definitions. Furthermore a monomorphism in Ch is precisely a chain morphism whose components are all monomorphisms, similar for epimorphisms.

1.2.5 Definition

An **Abelian category** is an additive category such that

(1) every map has a kernel and cokernel,

(2) every monomorphism is the kernel of its cokernel,

(3) every epimorphism is the cokernel of its kernel.

The prototypical example of an Abelian category is Mod_R (in particular $\mathrm{Ab} = \mathrm{Mod}_{\mathbb{Z}}$).

1.2.6 Definition

Let $\mathcal{C}$ be a category, and $A \in \mathcal{C}$ an object. We say that two monomorphisms $u\colon X \to A$ and $v\colon Y \to A$ are equivalent if there exists an isomorphism $\phi\colon X \to Y$ such that $u = v\phi$. The equivalence classes of these monomorphisms are called **subobjects** of A.

Similarly two epimorphisms $u\colon A \to X$ and $v\colon A \to Y$ are equivalent if there exists an isomorphism $\phi\colon X \to Y$ such that $v = \phi u$. The equivalence classes of these epimorphisms are called **quotient objects** of A.

1.2.7 Definition

In an Abelian category, the **image** of a morphism $f\colon B \to C$ is $\mathrm{im} f = \ker(\mathrm{coker} f)$, considered as a subobject of C. Its **coimage** is $\mathrm{coim} f = \mathrm{coker}(\ker f)$, viewed as a subobject of B.

In Mod_R we define the image of $f\colon B \to C$ to be $\mathrm{im} f = \{fb \mid b \in B\}$ (the coimage is $B/\ker f$). This coincides with the category-theoretic definition, taking the monomorphism to be the inclusion $\mathrm{im} f \to C$. Every map f of R-modules factors as

$$B \xrightarrow{e} \mathrm{im} f \xrightarrow{m} C$$

with e epi and m monic.

1.2.8 Definition

In an Abelian category, a sequence of morphisms $A \xrightarrow{f} B \xrightarrow{g} C$ is **exact** at B if $\mathrm{im} f = \ker g$ (as subobjects of B).

Note that if $\mathcal{A}$ is an Abelian category, we can repeat the discussion of the previous section to define the category of chain complexes on $\mathcal{A}$, $\mathsf{Ch}(\mathcal{A})$. This is an additive category, and homology are additive functors $\mathsf{Ch}(\mathcal{A}) \to \mathcal{A}$.

1.2.9 Theorem

For an Abelian category $\mathcal{A}$, $\mathsf{Ch}(\mathcal{A})$ is itself Abelian as well.

Proof

$\mathsf{Ch}(A)$ has binary biproducts and a zero object; their constructions are the same as before. Now if $f\colon B \to C$ is a chain map, then f is monic iff $f_n\colon B_n \to C_n$ is monic for all n. Indeed, the composite $\ker f \to C$ (inclusion composed with f) is zero, and so if f is monic then $\ker f = 0$ (since $f\iota = 0$ implies $\iota = 0$), and conversely (recall $\ker f = 0$ iff f is monic). Thus if f is monic, then f is isomorphic to the kernel of $C \to C/B$. Similar for epimorphisms. ■

Since $\mathsf{Ch}(\mathcal{A})$ is itself Abelian, we can form the category of chain complexes over $\mathsf{Ch}(\mathcal{A})$. Elements of $\mathsf{Ch}(\mathsf{Ch}(\mathcal{A}))$ are families $\{C_{p,q}\}_{p,q\in\mathbb{Z}}$ with horizontal and vertical differentials $d^h\colon C_{p,q} \to C_{p-1,q}$ and $d^v\colon C_{p,q} \to C_{p,q-1}$ such that

$$d^h \circ d^h = d^v \circ d^v = 0, \qquad d^h \circ d^v = d^v \circ d^h$$

Indeed, a chain complex of chain complexes consist of a family of chain complexes $\left\{\{C_{p,q}, d^h\}_p\right\}_q$ and differentials $d^v\colon C_{\bullet,q} \to C_{\bullet,q-1}$. The first two equalities above say that these are differentials, the last says that d^v is a chain map.

A map of bicomplexes $f\colon C_{\bullet\bullet} \to D_{\bullet\bullet}$ is then a family of maps $f_{p,q}\colon C_{p,q} \to D_{p,q}$ such that $f_{p,\bullet}, f_{\bullet,q}$ are chain maps for all p, q.

For reasons which will become clearer later, we call such a construction a **positive bicomplex**. We define a (signed) bicomplex as follows:

1.2.10 Definition

A **bicomplex** in $\mathcal{A}$ is a family $\{C_{p,q}\}_{p,q\in\mathbb{Z}}$ of objects from $\mathcal{A}$ along with maps $d^h\colon C_{p,q} \to C_{p-1,q}$ and $d^v\colon C_{p,q} \to C_{p,q-1}$ (horizontal and vertical differentials) such that $d^h \circ d^h = d^v \circ d^v = d^v \circ d^h + d^h \circ d^v = 0$.

A bicomplex is **bounded** if C has only finitely many nonzero components along every diagonal $p + q = n$ (e.g. if $C_{p,q} \neq 0$ only when $p, q \geq 0$).

If C_{pq} is a bicomplex, $C_{\bullet q}$ and $C_{p\bullet}$ are chain complexes for all p, q.

Note that due to anticommutativity ($d^v d^h = -d^h d^v$), d^v are not maps $C_{\bullet q} \to C_{\bullet q-1}$, but they can be used to construct them: define

$$f_{pq} = (-1)^p d^v_{pq}\colon C_{pq} \to C_{p,q-1}$$

Then $f_{\bullet q}$ defines a chain map $C_{\bullet q} \to C_{\bullet q-1}$: $d^h f_{pq} = d^h(-1)^p d^v_{pq} = (-1)^{p-1} d^v_{p-1,q} d^h = f_{p-1,q} d^h$. This "sign trick" allows us to identify the category of bicomplexes with $\mathsf{Ch}(\mathsf{Ch}(\mathcal{A}))$ (i.e. this defines an isomorphism of categories).

Let C be a bicomplex, and define the chain complexes $\mathrm{Tot}^\Pi(C)$ and $\mathrm{Tot}^\oplus(C)$ by

$$\mathrm{Tot}^\Pi(C)_n = \prod_{p+q=n} C_{pq}, \qquad \mathrm{Tot}^\oplus(C)_n = \bigoplus_{p+q=n} C_{pq}$$

Then $d = d^h + d^v$ defines maps $\mathrm{Tot}^\Pi(C)_n \to \mathrm{Tot}^\Pi(C)_{n-1}$ and $\mathrm{Tot}^\oplus(C)_n \to \mathrm{Tot}^\oplus(C)_{n-1}$. Explicitly, we map $x \in C_{pq}$ to $(d^h x, d^v x) \in C_{p-1,q} \times C_{p,q-1}$, or more suggestively:

$$d \sum_{p+q=n} x_{pq} = \sum_{p+q=n} (d^h x_{pq} + d^v x_{pq})$$

These are differential maps: $(d^h + d^v)(d^h + d^v) = 0$ precisely because $d^h d^v + d^v d^h = 0$.

And so $\mathrm{Tot}^\Pi(C)$ and $\mathrm{Tot}^\oplus(C)$ define chain complexes, called **total complexes**. If C is bounded, then $\mathrm{Tot}^\Pi(C) = \mathrm{Tot}^\oplus(C)$ (as there are only finitely many nonzero C_{pq} for which $p + q = n$).

The reason for using (signed) bicomplexes is because they simplify the construction of the total complexes. If we were to use a positive bicomplex, then the total complex would have differential $d^h + (-1)^p d^v$.

Note that both total complex constructions form functors $\mathsf{Ch}(\mathsf{Ch}(\mathcal{A})) \to \mathsf{Ch}(\mathcal{A})$. Indeed, if $f_{p,q}\colon C_{p,q} \to D_{p,q}$ is a map of bicomplexes, then it defines a chain map between total complexes in a natural way. Map $(a_{p,q})_{p+q=n} \in \mathrm{Tot}(C)_n$ to $(f_{p,q} a_{p,q})_{p+q=n} \in \mathrm{Tot}(D)_n$. This is indeed a chain map:

$$f(d^h a_{pq} + (-1)^p d^v a_{pq}) = (d^h f a_{pq} + (-1)^p d^v f a_{pq}) = (d^h + (-1)^p d^v)(f a_{pq})$$

Note:

The total complexes do not exist in all Abelian categories! In order form them to exist, the infinite products and coproducts of C_{pq} must exist. If C is bounded, then they do, as the total complexes are equal to the finite biproducts of C_{pq}. But in general, we need $\mathcal{A}$ to be complete (equivalently have arbitrary products) to define $\mathrm{Tot}^{\Pi}(C)$, and cocomplete (arbitrary coproducts) to define $\mathrm{Tot}^{\oplus}(C)$.

In particular the total complexes exist in Mod_R and $\mathsf{Ch}(\mathrm{Mod}_R)$.

1.2.11 Definition

Let C be a chain complex, and n an integer. We define $\tau_{\geq n}C$ to be the subcomplex of C defined by

$$(\tau_{\geq n}C)_i = \begin{cases} 0 & \text{if } i < n \\ Z_n & \text{if } i = n \\ C_n & \text{if } i > n \end{cases}$$

This is called the **truncation** of C below n. The quotient complex $\tau_{<n}C = C/\tau_{\geq n}C$ is the truncation of C above n.

Note that the ith boundary module and ith cycle module for $\tau_{\geq n}C$ are the same as those for C, except when $i < n$ when they are zero. Thus

$$H_i(\tau_{\geq n}C) = \begin{cases} 0 & i < n \\ H_i(C) & i \geq n \end{cases}$$

And so we get

$$H_i(\tau_{<n}C) = \begin{cases} H_i(C) & i < n \\ 0 & i \geq n \end{cases}$$

The more naive method defining truncation does not give as clean of a homology group (defining the nth term to be 0 or C_n, whatever). These are the **brutal truncations** of a chain complex C: $(\sigma_{<n}C)_i = C_i$ when $i < n$ and 0 otherwise. And then $\sigma_{\geq n}C = C/(\sigma_{<n}C)$. Note that

$$H_n(\sigma_{\geq n}C) = C_n/B_n$$

1.2.12 Definition

If C is a chain or cochain complex and p an integer we define a new complex (chain or cochain) $C[p]$ by

$$C[p]_n = C_{n+p} \qquad (\text{resp. } C[p]^n = C^{n-p})$$

with differential $(-1)^p d$. These are called the **pth translate** of C. The sign convention (multiplying by $(-1)^p$ will simplify notation later on).

Clearly we have

$$H_n(C[p]) = H_{n+p}(C) \qquad (\text{resp. } H^n(C[p]) = H^{n-p}(C))$$

We extand translation into a functor by shifting indices on a chain map: if $f\colon C \to D$ is a chain map then we define $f[p]$ by

$$f[p]_n = f_{n+p} \qquad (\text{resp. } f[p]^n = f^{n-p})$$

1.2.13 Example

Let C be a complex, then there is an exact sequence of complexes

$$0 \to Z(C) \to C \xrightarrow{d} B(C)[-1] \to 0$$

Indeed, inclusion is mono and thus has trivial kernel; by definition the kernel of d is the image of $Z(C)$. And the image of d is by definition all boundaries. We must reindex for d to be a chain map.

Similarly we have an exact sequence

$$0 \to H(C) \to C/B(C) \xrightarrow{d} Z(C)[-1] \to H(C)[-1] \to 0$$

1.3 Long exact sequences

Chain complexes are useful as they provide long exact sequences. Our goal in this section is to prove the following:

1.3.1 Theorem

Let $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ be a short exact sequence of chain complexes. Then there are natural maps $\partial_n \colon H_n(C) \to H_{n-1}(A)$ called **connecting homomorphisms** such that

$$\cdots \xrightarrow{g} H_{n+1}(C) \xrightarrow{\partial} H_n(A) \xrightarrow{f} H_n(B) \xrightarrow{g} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{f} \cdots$$

is an exact sequence.

Similarly if $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ is a short exact sequence of cochain complexes, then there are natural maps $\partial_n \colon H^n(C) \to H^{n+1}(A)$ and a long exact sequence:

$$\cdots \xrightarrow{g} H^{n-1}(C) \xrightarrow{\partial} H^n(A) \xrightarrow{f} H^n(B) \xrightarrow{g} H^n(C) \xrightarrow{\partial} H^{n+1}(A) \xrightarrow{f} \cdots$$

1.3.2 Example

Note that if $0 \to A \to B \to C \to 0$ is a short exact sequence such that two of the three complexes A, B, C are exact (acyclic) then so is the third. If A, C are exact then since we have an exact sequence $0 = H_n(A) \to H_n(B) \to H_n(C) = 0$, we must have $H_n(B) = 0$ so B is exact. The other cases are due to the above theorem: we will always have an exact sequence of the form $0 \to H_n(X) \to 0$ for $X = A, B, C$.

We will require the following lemma:

1.3.3 Lemma (Snake lemma)

Consider a commutative diagram of R-modules of the form:

$$\begin{tikzcd}
 & A' \arrow[r, "\alpha"'] \arrow[d, "f"'] & B' \arrow[r, "p"'] \arrow[d, "g"'] & C' \arrow[r] \arrow[d, "h"'] & 0 \\
0 \arrow[r] & A \arrow[r, "i"] & B \arrow[r, "\beta"] & C &
\end{tikzcd}$$

If the rows are exact, then there is an exact sequence

$$0 \to \ker f \to \ker g \to \ker h \xrightarrow{\partial} \operatorname{coker} f \to \operatorname{coker} g \to \operatorname{coker} h \to 0$$

where ∂ is defined by $\partial(c) = i^{-1}gp^{-1}(c)$ for $c \in \ker h$. Moreover, if $\alpha\colon A' \to B'$ is monic then so is $\ker f \to \ker g$; if $\beta\colon B \to C$ is onto then so is $\operatorname{coker} g \to \operatorname{coker} h$.

Proof

The top row is exact, so p is surjective. Thus for $c \in \ker h$ there is an $x \in B'$ such that $p(x) = c$. Note then that $\beta gx = hpx = hc = 0$ and so $gx \in \ker\beta = \operatorname{im} i$ so $i^{-1}gx$ exists. But this doesn't ensure that this is well-defined: if $p(x) = p(y) = c$ we still need to show that $i^{-1}gx = i^{-1}gy$.

Note that i^{-1} here maps into the cokernel of f, that is to say we map to $i^{-1} \bullet + \operatorname{im} f$. So a more exact way of writing ∂ would be by

$$\partial(c) = i^{-1}gp^{-1}(c) + \operatorname{im} f$$

Suppose $px = 0$ then $x \in \ker p = \operatorname{im}\alpha$ so $x = \alpha(a)$ for $a \in A'$. Then we have $gx = ifa$ and so $i^{-1}gx = fa \in \operatorname{im} f$, and so $i^{-1}gx = 0$ in $\operatorname{coker} f$. Thus if $p(x) = p(y)$ then $p(x-y) = 0$ and so $i^{-1}gx - i^{-1}gy \in \operatorname{im} f$ meaning they define the same coset in $\operatorname{coker} f$; ∂ is well-defined. It is also clearly a homomorphism, as if $p(x) = c$ and $p(y) = c'$ then $\partial(c + c') = i^{-1}gx + i^{-1}gy = \partial(c) + \partial(c')$.

Now $0 \to \ker f \xrightarrow{\alpha} \ker g \xrightarrow{p} \ker h$ is an exact sequence, and so is $\operatorname{coker} f \xrightarrow{i} \operatorname{coker} g \xrightarrow{\beta} \operatorname{coker} h \to 0$. Adding ∂ allows us to connect these two exact sequences. ■

We can extend the snake lemma to arbitrary Abelian categories. The Freyd-Mitchell embedding theorem allows us to embed small Abelian categories into categories of modules. So take the Abelian subcategory generated by the commutative, embed it into a category of modules, and the result is given (details are omitted).

Now we construct the natural maps $\partial\colon H_n(C) \to H_{n-1}(A)$ in our goal theorem, associated with $0 \to A \to B \to C \to 0$. Consider the commutative diagram (we write dA_{n+1} for $B_n(A)$ since we already have a chain complex B_n)

$$\begin{tikzcd}
 & A_n/dA_{n+1} \arrow[r] \arrow[d, "d_A"'] & B_n/dB_{n+1} \arrow[r] \arrow[d, "d_B"'] & C_n/dC_{n+1} \arrow[r] \arrow[d, "d_C"'] & 0 \\
0 \arrow[r] & Z_{n-1}(A) \arrow[r] & Z_{n-1}(B) \arrow[r] & Z_{n-1}(C) &
\end{tikzcd}$$

The rows are exact. The kernel of d_A is $H_n(A)$, its cokernel is $H_{n-1}(A)$. Thus the snake lemma yields an exact sequence

$$H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to H_{n-1}(B) \to H_{n-1}(C)$$

pasting these sequences together gives us the desired long exact sequence. ■

Using the snake lemma's construction, we can understand ∂. Consider the short exact sequence

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

then $\partial(z) = f^{-1}dg^{-1}(z)$. That is, for $[z] \in H_n(C)$ consider its lift $b \in B_n$ (meaning $g(b) = z$), apply d and then f^{-1}: $\partial(z) = f^{-1}(db)$.

Now ∂ is natural in the following sense: consider the category $\mathcal{S}$ whose objects are short exact sequences of chain complexes in $\mathcal{A}$, and whose morphisms commute with these. Similarly there is a category $\mathcal{L}$ of long exact sequences in $\mathcal{A}$.

1.3.4 Proposition

The long exact sequence is a functor from $\mathcal{S}$ to $\mathcal{L}$.

We omit the proof here; it is a direct result of the construction of the connecting morphisms (from the snake lemma) and definitions.

1.3.5 Example

Suppose $f: C \to D$ is a morphism of chain complexes such that $\ker f, \operatorname{coker} f$ are acylic. Then f is a quasi-isomorphism. Indeed, we have an exact sequence

$$\ker f \to C \xrightarrow{f} D \to \operatorname{coker} f$$

This gives us an exact sequence of homology modules

$$0 = H_n(\ker f) \to H_n(C) \xrightarrow{f} H_n(D) \to H_n(\operatorname{coker} f) = 0$$

meaning that the induced morphism between homology modules must be an isomorphism: f is quasi-isomorphic.

1.3.6 Example

Let $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ be a short exact sequence of bicomplexes. This gives us a short exact sequence of total complexes. Indeed, f induces a map $\operatorname{Tot}(A) \to \operatorname{Tot}(B)$ by mapping (a_{pq}) to $(f_{pq}a_{pq})$; the induced maps still form into a short exact sequence. Thus if $\operatorname{Tot}(A)$ or $\operatorname{Tot}(C)$ are acylic, then the other two are quasi-isomorphic (since the exact sequence of homology modules is of the form $0 \to X \to Y \to 0$).

1.4 Chain homotopies

Recall that chain complexes of vector spaces over a field decompose as $C_n = Z_n \oplus B'_n$ where $B'_n \cong C_n/Z_n \cong d(C_n) = B_{n-1}$, and $Z_n = B_n \oplus H'_n$ where $H'_n \cong Z_n/B_n = H_n(C)$. So we have compositions

$$C_n \xrightarrow{\text{proj}} Z_n \xrightarrow{\text{proj}} B_n \cong B'_{n+1} \subseteq C_{n+1}$$

giving us splitting maps $s_n: C_n \to C_{n+1}$ such that $d = dsd$:

$$C_{n+1} \xrightarrow{d} C_n \to Z_n \to B_n \cong B'_{n+1} \subseteq C_{n+1} \xrightarrow{d} C_n$$

the map $C_{n+1} \to B_n$ simply maps a to da. The isomorphism $B_n \cong B'_{n+1}$ maps da to a's projection onto B'_{n+1} (orthogonal to Z'_{n+1}; following the isomorphisms), which d then maps to da (since da is equal to d on the projection). Thus $dsd = d$, as desired.

The composition ds is the projection of C_n onto B_n, and sd the projection onto B'_n. Thus $ds + sd$ is an endomorphism of C_n whose kernel is H'_n, isomorphic to the homology $H_n(C)$. Its image is $B_n \oplus B'_n$ and as such its cokernel is isomorphic to $H_n(C)$ as well. So the kernel and cokernel of $ds + sd$ are isomorphic to the homology complex $H_\bullet(C)$ (the differentials here are zero, so the homology modules are the same as C's). And so the chain maps $H_\bullet(C) \to C$ and $C \to H_\bullet(C)$ are quasi-isomorphisms. Furthermore, since the (co)kernel of $ds + sd$ is the homology complex, C is exact (acylic) iff $ds + sd$ is an isomorphism.

This is not true over an arbitrary ring R, so we give this a name.

1.4.1 Definition

A complex C is called **split** if there are maps $s_n\colon C_n \to C_{n+1}$ such that $d = dsd$. These maps s_n are called **splitting maps**. If C is further acylic, then C is said to be **split exact**.

1.4.2 Proposition

A chain complex C is split iff $C_n \cong Z_n \oplus B'_n$ with $Z_n = B_n \oplus H'_n$.

Proof

If C is split then define B'_n to be the image of $s_{n-1}d_n$. Since $dsd = d$, we have that $\ker sd = \ker d = Z_n$ and $sdsd = sd$, so sd is a projection operator whose image is B'_n and whose kernel is Z_n, thus $C_n = Z_n \oplus B'_n$ as desired. Similarly, ds is a projection operator whose image is B_n; restricting it to Z_n gives us $Z_n = B_n \oplus H'_n$.

If $C_n \cong Z_n \oplus B'_n$ and $Z_n = B_n \oplus H'_n$ then we have $B'_n \cong B_{n-1}$. And so we can map $C_n \to C_{n+1}$ along $B_n \cong B'_{n+1}$; $(x, y, z) \in B_n \oplus H'_n \oplus B'_n$ maps to $(0, 0, x) \in B_{n+1} \oplus H'_{n+1} \oplus B'_{n+1}$. Then dsd maps $(x, y, z) \in C_{n+1}$ to $(z, 0, 0) \in C_n$ to $(0, 0, z) \in C_{n+1}$ to $(z, 0, 0) \in C_n$, which is the same as $d(x, y, z)$; $dsd = d$. ■

1.4.3 Proposition

A chain complex C is split exact iff there exist maps such that $ds + sd$ is the identity.

Proof

If $ds + sd = \mathrm{id}$ then we get that $(ds + sd)d = d$, but the left-hand side is $dsd + sd^2 = dsd$, so C is split. And C is split then we showed that $C_n \cong B_n \oplus H'_n \oplus B'_n$ with ds, sd the projections onto B_n, B'_n respectively. C is then split exact iff $H_n(C) \cong H'_n = 0$, which is iff the projection onto $B_n \oplus B'_n$ is the identity. This is $ds + sd$, and so C is split exact iff $ds + sd$ is the identity, as desired. ■

1.4.4 Example

Consider the complex

$$\cdots \xrightarrow{\cdot 2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\cdot 2} \cdots$$

This is an acylic complex: the image and kernel of multiplication by 2 are $\{0, 2\}$. But it is not split exact: $ds + sd$ can never be the identity.

Now suppose we have two chain complexes C, D and maps $s_n\colon C_n \to D_{n+1}$. Define $f_n\colon C_n \to D_n$ by $f_n = d_{n+1}s_n + s_{n-1}d_n$: $f = ds + sd$. Then notice that

$$df = d(ds + sd) = dsd = (ds + sd)d = fd$$

so $f = ds + sd$ is a chain map $C \to D$.

1.4.5 Definition

A chain map $f: C \to D$ is **null homotopic** if there are maps $s_n: C_n \to D_{n+1}$ such that $f = ds + sd$. The maps s_n are called a **chain contraction** of f.

Immediately we get:

1.4.6 Proposition

C is split exact iff the identity is null homotopic.

Since the sum of chain maps is a chain map, the sum of any chain map with a null homotopic map is again a chain map.

1.4.7 Definition

We say that two maps $f, g: C \to D$ are **chain homotopic** if their difference is null homotopic. If $f - g = sd + ds$ then the maps s_n are called a **chain homotopy** from f to g. Two maps $f: C \rightleftarrows D: g$ form a **chain homotopy equivalence** if their compositions are chain homotopic to the respective identities. C and D are **chain homotopy equivalent** if there exists a chain homotopy equivalence between them.

In particular f is null homotopic iff it is chain homotopic to the zero map.

1.4.8 Lemma

If $f: C \to D$ is null homotopic then the induced maps $f: H_n(C) \to H_n(D)$ are zero. And so if f, g are chain homotopic their induced maps are equal.

Proof

The latter claim follows from the first: the induced map $H_n(f - g)$ is equal to $H_n(f) - H_n(g)$. So if $f - g$ is null homotopic, then $H_n(f) - H_n(g) = 0$.

So suppose $f = ds + sd$, then for $a \in Z_n(C)$ we have $fa = dsa + sda = dsa \in B_n(D)$. Thus f maps cycles to boundaries, and so its induced map is zero. ■

1.4.9 Example

If C is split, then the chain complex $H_\bullet(C)$ and C are chain homotopy equivalent. Indeed, consider $f: C \to C$ given by $f = \mathrm{id} - ds - sd$ (the difference between a chain map and a null homotopy). Then $df = 0$, so f maps into Z_n, that is it defines a map $f: C \to H_\bullet(C)$ (alternatively recall that $C_n \cong B_n \oplus H_n \oplus B'_n$, this is the projection onto H_n). Then define $g: H_\bullet(C) \to C$ by inclusion (recalling that $C_n \cong B_n \oplus H_n \oplus B'_n$).

Then fg is just the identity on $H_\bullet(C)$. And gf is projection onto H_n; it is $\mathrm{id} - ds - sd$ which is homotopy equivalent to id. Thus f, g form a chain homotopy equivalence. ■

Notice that chain homotopy equivalence is an equivalence relation: it is clearly reflexive (taking $s = 0$) and symmetric (taking $-s$). And if $f - g = sd + ds$ and $g - h = s'd + ds'$ then $f - h = (s + s')d + d(s + s')$, so it is transitive. This equivalence relation preserves addition: if $f - f' = sd + ds$ and $g - g' = s'd + ds'$ then $(f + g) - (f' + g') = (s + s')d + d(s + s')$.

Let us construct the category K whose objects are chain complexes and whose morphisms $C \to D$ are equivalence classes of chain maps. We need to show that composition is well-defined: if $f - g = sd + ds$ and u is another chain map then $uf - ug = usd + uds = usd + dus$, so uf and ug are homotopic (similar for fu, gu). Thus if $f \sim g$ and $u \sim v$ we have $fu \sim gu \sim gv$, so composition is well-defined.

This turns K into an additive category; it has biproducts. It is not an Abelian category as it doesn't necessarily have kernels.

Since we showed that chain homotopic maps have the same image under homology functors, homology gives functors $\mathsf{K} \to \mathcal{A}$. In particular chain homotopic equivalent complexes have isomorphic homology (since homotopy equivalence is isomorphism in K):

1.4.10 Theorem

If C and D are chain homotopic equivalent, then $H_\bullet(C) \cong H_\bullet(D)$.

1.5 Mapping cones and cylinders

1.5.1 Definition

Let $f\colon B \to C$ be a map of chain complexes. The **mapping cone** of f is the chain complex $\operatorname{cone}(f)$ whose nth component is $B_{n-1} \oplus C_n$. The differential is given by $d(b,c) = (-d(b), d(c) - f(b))$.

For cochain complexes, the nth component is $B^{n+1} \oplus C^n$, and the differential has the same formula.

We can represent the differential by the matrix $\begin{pmatrix} -d_B & 0 \\ -f & +d_C \end{pmatrix}$.

1.5.2 Example

Let $\operatorname{cone}(C)$ denote the mapping cone of the identity map id_C; its differential is given by $d(b,c) = (-db, dc - b)$. Then $s(b,c) = (-c, 0)$ is a splitting map $\operatorname{cone}(C)_n = C_{n-1} \oplus C_n \to C_n \oplus C_{n+1} = \operatorname{cone}(C)_{n+1}$:

$$(ds + sd)(b,c) = d(-c,0) + s(-db, dc - b) = (dc, c) + (b - dc, 0) = (b, c)$$

so $\operatorname{cone}(C)$ is split exact.

1.5.3 Proposition

Let $f\colon C \to D$ be a map of complexes. Then f is null homotopic iff it extends to a map $(-s, f)\colon \operatorname{cone}(C) \to D$.

Proof

Note that $(-s, f)$ is a chain map iff $d(-s,f)(b,c) = (-s,f)d(b,c)$ for all b, c. These are equal to

$$d(-s,f)(b,c) = -ds(b) + df(c), \qquad (-s,f)d(b,c) = (-s,f)(-db, dc - b) = sd(b) + fd(c) - f(b)$$

since f is a chain map $fd = df$, and so these are equal iff

$$-ds(b) = sd(b) - f(b)$$

i.e. iff $f = sd + ds$, as desired. ■

Consider a chain map $f\colon H_\bullet(B) \to H_\bullet(C)$, then there is a short exact sequence

$$0 \to C \xrightarrow{\iota} \operatorname{cone}(f) \xrightarrow{\delta} B[-1] \to 0$$

where the left map maps $c \in C_n$ to $(0,c) \in \operatorname{cone}(f)_n$, and the right maps (b,c) to $-b$. These are chain maps: $d(0,c) = (0, dc)$ and $d(b,c) = (-db, dc - fb) \mapsto -db$ (recall the differential of $B[-1]$ is $-d_B$ by the sign

convention). The left map is an injection, the right a surjection, and the image of the left is the kernel of the second (the component C_n).

And so this gives us a long exact sequence (recalling that $H_{n+1}(B[-1]) = H_n(B)$)

$$\cdots \to H_{n+1}(\mathrm{cone} f) \xrightarrow{\delta} H_n(B) \xrightarrow{\partial} H_n(C) \to H_n(\mathrm{cone} f) \xrightarrow{\delta} H_{n-1}(B) \to \cdots$$

Let us understand the connecting maps ∂. Recall that $\partial = \iota^{-1} d_{\mathrm{cone}(f)} \delta^{-1}$, so for a cycle $b \in B_n$ we can lift it to $(-b, 0) \in \mathrm{cone}(f)$. Then applying the differential gives $(db, fb) = (0, fb)$ which maps to fb. Thus we have $\partial([b]) = f([b])$ for all $[b] \in H_\bullet(B)$, so our connecting map is simply f.

We have proven the following.

1.5.4 Proposition

For any map $f\colon H_\bullet(B) \to H_\bullet(C)$ we have a long exact sequence

$$\cdots \to H_{n+1}(\mathrm{cone} f) \to H_n(B) \xrightarrow{f} H_n(C) \to H_n(\mathrm{cone} f) \to H_{n-1}(B) \xrightarrow{f} \cdots$$

1.5.5 Corollary

$f\colon B \to C$ is a quasi-isomorphism iff $\mathrm{cone} f$ is acyclic.

Proof

If $\mathrm{cone} f$ is acyclic the above long exact sequence gives us exact sequences $0 \to H_n(B) \xrightarrow{f} H_n(C) \to 0$, so the induced map is an isomorphism. And if f is a quasi-isomorphism, we must have that π and δ are zero on the homology modules, since $\ker\delta = \mathrm{im}\pi = 0$ we must have that $H_n(\mathrm{cone} f) = 0$, so $\mathrm{cone} f$ is acyclic. ■

We have a similar construction:

1.5.6 Definition

Let $f\colon B \to C$ be a chain map, then define the **mapping cylinder** of f to be the complex $\mathrm{cyl}(f)$ whose nth component is $B_n \oplus B_{n-1} \oplus C_n$. The differential is

$$d(b, b', c) = (d(b) + b', -d(b'), d(c) - f(b'))$$

We can represent the differential as the matrix

$$d = \begin{pmatrix} d_B & \mathrm{id}_B & 0 \\ 0 & -d_B & 0 \\ 0 & -f & d_C \end{pmatrix}$$

We see that this is a complex:

$$d^2 = \begin{pmatrix} d_B^2 & d_B - d_B & 0 \\ 0 & d_B^2 & 0 \\ 0 & f d_B - d_C f & d_C^2 \end{pmatrix} = 0$$

1.5.7 Proposition

Let $\mathrm{cyl}(C)$ be the mapping cylinder of id_C. Then $f, g\colon C \to D$ are chain homotopic iff they extend to a map $(f, s, g)\colon \mathrm{cyl}(C) \to D$.

Proof

(f, s, g) is a chain map iff $d(f, s, g)(b, b', c) = (f, s, g)d(b, b', c)$ for all b, b', c. These are equal to

$$d(f, s, g)(b, b', c) = df(b)+ds(b')+dg(c), \qquad (f, s, g)d(b, b', c) = (f, s, g)(d(b)+b', -d(b'), d(c)-b') = fd(b)+f(b')-sd(b')+gd($$

These are equal iff $f(b') - sd(b') - g(b') = ds(b')$ i.e. iff $f - g = ds + sd$, as desired. ■

1.5.8 Lemma

The subcomplex whose elements are of the form $(0, 0, c)$ is isomorphic to C and the corresponding inclusion $\alpha\colon C \to \mathrm{cyl}(f)$ is a quasi-isomorphism.

Proof

Clearly $\alpha\colon c \mapsto (0, 0, c)$ is an injection whose image is this subcomplex. We need to show that α is a chain map:

$$d\alpha(c) = d(0, 0, c) = (0, 0, dc) = \alpha(dc)$$

as desired.

Now, up to isomorphism, $\mathrm{cyl}(f)/\alpha(C)$ has components of the form $B_n \oplus B_{n-1}$ and the differential is given by $d(b, b') = (d(b) + b', -d(b'))$. This is clearly isomorphic to $\mathrm{cone}(-\mathrm{id}_B)$, which we showed is split exact (we showed $\mathrm{cone}(\mathrm{id}_B)$ is). So we have a short exact sequence

$$0 \to C \xrightarrow{\alpha} \mathrm{cyl}(f) \to \mathrm{cone}(-\mathrm{id}_B) \to 0$$

and the homology sequence gives us $0 = H_{n+1}(\mathrm{cone}(-\mathrm{id}_B)) \to H_n(C) \xrightarrow{\alpha} H_n(\mathrm{cyl}(f)) \to H_n(\mathrm{cone}(-\mathrm{id}_B)) = 0$, that is to say α is a quasi-isomorphism. ■

We can actually strengthen this:

1.5.9 Lemma

α is actually a chain homotopy equivalence (thus an isomorphism in K, so in particular a quasi-isomorphism).

Proof

Define $\beta\colon \mathrm{cyl}(f) \to C$ by $\beta(b, b', c) = f(b) + c$. This is a chain map:

$$d\beta(b, b', c) = df(b)+dc, \qquad \beta d(b, b', c) = \beta(d(b)+b', -d(b'), d(c)-f(b')) = f(db+b')+dc-f(b') = fd(b)+dc$$

Now $\beta\alpha = \mathrm{id}_C$ and defining $s_n\colon \mathrm{cyl}(f)_n \to \mathrm{cyl}(f)_{n+1}$ by $s(b, b', c) = (0, b, 0)$ gives a chain homotopy from $\alpha\beta$ to id:

$$ds(b, b', c)+sd(b, b', c) = d(0, b, 0)+s(d(b)+b', -d(b'), d(c)-f(b')) = (b, -d(b), -f(b))+(0, d(b)+b', 0) = (b, b', -f(b))$$

and

$$\alpha\beta(b, b', c) = (0, 0, f(b) + c)$$

thus $\mathrm{id} - \alpha\beta = ds + sd$ as desired. ■

To summarize, $\alpha\colon C \rightleftarrows \mathrm{cyl}(C)\colon \beta$ form a chain homotopy equivalence where

$$\alpha(c) = (0,0,c), \qquad \beta(b,b',c) = f(b) + c$$

Now consider the map $\iota\colon B \to \mathrm{cyl}(f)$ by $\iota(b) = (b,0,0)$. This is a chain map:

$$d\iota(b) = d(b,0,0) = (db,0,0) = \iota(db)$$

and so the subcomplex of $\mathrm{cyl}(f)$ whose elements are of the form $(b,0,0)$ is isomorphic to B. And the quotient complex $\mathrm{cyl}(f)/\iota(B)$ is isomorphic to $\mathrm{cone}(f)$: its terms are of the form $B_{n-1} \oplus C_n$ and the differential is given by $d(b',c) = (-d(b'), d(c) - f(b'))$.

Consider then the composite $B \xrightarrow{\iota} \mathrm{cyl}(f) \xrightarrow{\beta} C$: this is just f. Thus homology $f\colon H(B) \to H(C)$ factors through $H(B) \to H(\mathrm{cyl}(f))$. Then we can construct a commutative diagram:

$$\begin{tikzcd}
 & & C & & & \\
0 \arrow[r] & B \arrow[r] \arrow[ur, "f"] & \mathrm{cyl}(f) \arrow[r] \arrow[u, "\beta"'] & \mathrm{cone}(f) \arrow[r] \arrow[d, leftrightarrow, "="] & 0 & \\
0 \arrow[r] & C \arrow[r] \arrow[u, "\alpha"] & \mathrm{cone}(f) \arrow[r, "\delta"] & B[-1] \arrow[r] & 0 &
\end{tikzcd}$$

The rows are exact (the map $\mathrm{cyl}(f) \to \mathrm{cone}(f)$ is given by the isomorphism $\mathrm{cyl}(f)/\iota(B) \cong \mathrm{cone}(f)$). Thus we have long exact sequences (recall that the connecting morphisms of the bottom homology are the induced maps of f):

$$\begin{tikzcd}
\cdots \arrow[r, "-\partial"] & H_n(B) \arrow[r] \arrow[d, leftrightarrow, "="] \arrow[dr, "f"] & H_n(\mathrm{cyl}(f)) \arrow[r] & H_n(\mathrm{cone}(f)) \arrow[r, "-\partial"] \arrow[d, leftrightarrow, "="] & H_{n-1}(B) \arrow[r] \arrow[d, leftrightarrow, "="] & \cdots \\
\cdots \arrow[r] & H_{n+1}(B[-1]) \arrow[r, "f"'] & H_n(C) \arrow[r] \arrow[u, leftrightarrow, "\cong"] & H_n(\mathrm{cone}(f)) \arrow[r, "\delta"'] & H_n(B[-1]) \arrow[r, "f"'] & \cdots
\end{tikzcd}$$

We take the inverse of the top connecting map precisely because we want this diagram to commute.

To check that this diagram commutes we need only check that the right square with $-\partial$ and δ does. ∂ works as follows: let $[(b,c)]$ be a cycle in $\mathrm{cone}(f)$, so $db = 0$ and $dc = fb$. Then we lift it to $\mathrm{cyl}(f)$ to $(0,b,c)$ and then apply the differential $d(0,b,c) = (b, -db, dc - fb) = (b,0,0)$. Thus ∂ maps $[(b,c)] \in H(B)$ to $[b] \in H_{n-1}(B)$. δ maps $[(b,c)]$ to $[-b]$, i.e. $\delta = -\partial$ as desired.

And so these diagrams are isomorphic.

Now suppose we have a short exact sequence $0 \to B \xrightarrow{f} C \xrightarrow{g} D \to 0$. We define a chain map $\phi\colon \mathrm{cone}(f) \to D$ by $\phi(b,c) = g(c)$:

$$\phi d(b,c) = \phi(-db, dc - fb) = gdc - gfb = gdc = d\phi(b,c)$$

Then we have long exact sequences

$$\begin{tikzcd}
0 \arrow[r] & B \arrow[r, "f"] \arrow[d, leftrightarrow, "="] & C \arrow[r, "g"] & D \arrow[r] & 0 \\
0 \arrow[r] & B \arrow[r] & \mathrm{cyl}(f) \arrow[r] \arrow[u, "\beta"'] & \mathrm{cone}(f) \arrow[r] \arrow[u, "\phi"'] \arrow[d, leftrightarrow, "="] & 0 \\
 & & 0 \arrow[r] \arrow[u, "\alpha"] & \mathrm{cone}(f) \arrow[r, "\delta"] & B[-1] \arrow[r] & 0
\end{tikzcd}$$

Since our map from short to long exact sequences is a functor, and furthermore from the 5-lemma (a special case saying that if the rows of a two-row five-column commutative diagram are exact, and the outer four vertical maps are isomorphisms, then so is the innermost) ϕ is a quasi-isomorphism, since β is.

Thus the rows all generate isomorphic long exact sequences (though we must invert the connecting morphism of the top row, as per the previous argument).

1.5.10 Proposition

Let $f\colon B \to C$ be a chain map, and let $v\colon C \to \operatorname{cone}(f)$ be the inclusion $c \mapsto (0, c)$. There is a chain homotopy equivalence $\operatorname{cone}(v) \to B[-1]$.

Proof

The nth component of $\operatorname{cone}(v)$ is $C_{n-1} \oplus B_{n-1} \oplus C_n$ and its differential is

$$d(c', b, c) = (-d(c'), d(b, c) - v(c')) = (-d(c'), -d(b), d(c) - f(b) - c')$$

The maps $\alpha\colon \operatorname{cone}(v) \rightleftarrows B[-1]\colon \beta$ given by $\alpha(c', b, c) = b$ and $\beta(b) = (f(b), b, 0)$ form a chain homotopy equivalence. $\alpha\beta = \mathrm{id}$ and $\beta\alpha(c', b, c) = (-f(b), b, 0)$ is chain homotopic to id. Define $s_n\colon C_{n-1} \oplus B_{n-1} \oplus C_n \to C_n \oplus B_n \oplus C_{n+1}$ by $s_n(c', b, c) = (-c, 0, 0)$. Then

$$ds(c', b, c) + sd(c', b, c) = d(-c, 0, 0) + s(-dc', -db, dc - fb - c') = (dc, 0, c) + (c' + fb - dc, 0, 0) = (c' + fb, 0, c)$$

which is equal to $(c', b, c) - \beta\alpha(c', b, c) = (c' + fb, 0, c)$, meaning id and $\beta\alpha$ are chain homotopic. ■

1.5.11 Example

Suppose C and C' are split complexes with splitting maps s, s'. Then $\sigma(c, c') = (-sc, s'c' - s'fsc)$ is a splitting map of $\operatorname{cone}(f)$ for $f\colon C \to C'$ iff the induced map $H(f)$ is zero. Indeed:

$$d\sigma d(c, c') = d\sigma(-dc, dc' - fc) = d(sdc, s'dc' - s'fc + s'fsdc) = (-dc, ds'dc' - ds'fc + ds'fsdc - fsdc)$$

Since s' is split, $ds'd = d$ so this is equal to $(-dc, dc' - (ds'f - ds'fsd + fsd)c)$, which is equal to d iff $ds'f - ds'fsd + fsd = f$. Now since C is split, we can embed $H(C)$ into it with projection $\mathrm{id} - ds - sd$, and so $H(f)$ is zero iff $f = fds + fsd$. So if $H(f) = 0$ then we have

$$ds'f - ds'fsd + fsd = ds'(f - fsd) + fsd = ds'fds + fsd = dfs + fsd = f$$

(since $fd = df$) as desired. Now if σ is split then for $z \in Z_n(C)$ we have that $fz = ds'fz \in B_n(C')$, so f is zero on the homology modules.

1.6 More on Abelian categories

We begin by mentioning an important theorem in the study of Abelian categories, without proof:

1.6.1 Theorem (Freyd-Mitchell)

If $\mathcal{A}$ is a small Abelian category, then there exists an exact fully faithful functor from $\mathcal{A}$ to Mod_R for some ring R.

What is an *exact* functor? We will define this soon.

1.6.2 Lemma

Let $\mathcal{C} \subseteq \mathcal{A}$ be a full subcategory of the Abelian category $\mathcal{A}$.

(1) $\mathcal{C}$ is additive iff $\mathcal{C}$ has $\mathcal{A}$'s zero objects and is closed under $\mathcal{A}$'s biproducts.

(2) $\mathcal{C}$ is itself Abelian and an exact subcategory of $\mathcal{A}$ iff $\mathcal{C}$ is additive and closed under $\ker, \operatorname{coker}$.

1.6.3 Example

(1) Mod_R is Abelian and the subcategory of finitely-generated R-modules forms an additive category. It is Abelian iff R is Noetherian.

(2) Ab is Abelian, and the subcategory of torsion-free Abelian groups is an additive subcategory. The subcategory of p-groups is Abelian, so is the subcategory of finite p-groups.

1.6.4 Proposition

Let $\mathcal{A}$ be an Abelian category and $\mathcal{C}$ an arbitrary category. The **functor category** $[\mathcal{C}, \mathcal{A}]$ is Abelian.

Proof

The zero object in $[\mathcal{C}, \mathcal{A}]$ is the constant functor to $0 \in \mathcal{A}$. For two functors $F, G: \mathcal{C} \to \mathcal{A}$ define $F \oplus G: \mathcal{C} \to \mathcal{A}$ to map $F \oplus G(c) = Fc \oplus Gc$ and $f: c \to d$ to the matrix $\mathrm{diag}(Ff, Gf)$. This is a biproduct. The kernel of $\alpha: F \Rightarrow G$ is $(\ker\alpha)(c) = \ker(\alpha(c))$, similar for cokernels. ■

1.6.5 Definition

For a topological space X, consider the partial order $\mathcal{U}$ of open subsets of X, considered as a category. A **presheaf** (with values in $\mathcal{A}$) is a contravariant functor $F: \mathcal{U}^{\mathsf{op}} \to \mathcal{A}$ such that $F\emptyset = 0$. The full subcategory of $[\mathcal{U}^{\mathsf{op}}, \mathcal{A}]$ of presheaves is denoted $\mathsf{Psh}_{\mathcal{A}}(X)$ (some authors do not require $F\emptyset = 0$, in which case the category of presheaves is just the functor category). This is an Abelian category.

An example of a presheaf with values in $\mathrm{Mod}_{\mathbb{R}}$ is the functor C^0 mapping an open set U to the set $C^0(U)$ of continuous functions $U \to \mathbb{R}$. For an inclusion $U \subseteq V$, the morphism $C^0(V) \to C^0(U)$ is given by restricting functions to U.

For a presheaf F, for $U \subseteq V$ we call the induced morphism $FV \to FU$ a **restriction**, and for $f \in FV$ its image under this morphism is also denoted $f|_U$.

1.6.6 Definition

A **sheaf** on a topological space X is a presheaf F which satisfies the following **sheaf axiom**: let $\{U_i\}$ be an open cover of U, an open subset of X. If $\{f_i \in F(U_i)\}$ are such that f_i and f_j agree on $U_i \cap U_j$ (i.e. their restrictions onto $U_i \cap U_j$ are equal), then there is a unique $f \in F(U)$ whose restriction to U_i is f_i.

Note by uniqueness if $f|_{U_i} = g|_{U_i}$ for all i then $f = g$. The sheaf axiom is equivalent to saying that if $f \in F(U)$ vanishes on every U_i, then $f = 0$. Thus we can write the sheaf axiom in element-free form. It is equivalent to saying that for any open U and open cover $\{U_i\}_i$ of U, the following sequence is exact:

$$0 \to F(U) \to \prod_i F(U_i) \to \prod_{i,j} F(U_i \cap U_j)$$

the first map maps $f \in F(U)$ to $(f|_{U_i})_i$, the second maps $(f_i)_i$ to $(f_i|_{U_i \cap U_j} - f_j|_{U_i \cap U_j})_{i,j}$.

1.6.7 Example

Let A be an Abelian group. For any open subset U of X, consider $A(U)$ the set of continuous maps from

U to the discrete topological space on A. $A(V) \to A(U)$ given by restriction forms this into a sheaf.

The category of sheaves, $\mathsf{sh}(X)$, is an Abelian category contained in $\mathsf{Psh}(X)$; but it is not an Abelian subcategory: cokernels in the two categories differ.

For a topological space X, let $\mathcal{O}$ be the sheaf which maps U to $\mathcal{O}(U)$ the Abelian group of continuous maps $U \to \mathbb{C}$. Similarly define $\mathcal{O}^\times(U)$ to be the Abelian group of continuous maps $U \to \mathbb{C}^\times$. Then we have a short exact sequence of sheaves:

$$0 \to \mathbb{Z} \xrightarrow{2\pi i} \mathcal{O} \xrightarrow{\exp} \mathcal{O}^\times \to 0$$

This is not exact in $\mathsf{Psh}(X)$, though, when $X = \mathbb{C}^\times$.

1.6.8 Definition

An additive functor $F\colon \mathcal{A} \to \mathcal{B}$ between Abelian categories is **left exact** (resp. **right exact**) if for every short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$, $0 \to FA \to FB \to FC$ (resp. $FA \to FB \to FC \to 0$) is exact in $\mathcal{B}$. F is then **exact** if it is both left- and right-exact (meaning it preserves short exact sequences).

1.6.9 Example

The inclusion $\mathsf{sh}(X) \to \mathsf{Psh}(X)$ is left-exact (but not right-exact). There is an exact functor $\mathsf{Psh}(X) \to \mathsf{sh}(X)$ called **sheafification**.

1.6.10 Proposition

An additive functor $F\colon \mathcal{A} \to \mathcal{B}$ is left exact (resp. right exact) iff the exactness of $0 \to A \to B \to C$ (resp. $A \to B \to C \to 0$) implies the exactness of $0 \to FA \to FB \to FC$ (resp. $FA \to FB \to FC \to 0$).

Proof

We prove this in categories of modules, and use Freyd-Mitchell to justify the extension to Abelian categories. $0 \to A \to B \to C$ is exact iff $0 \to A \to B \to C' \to 0$ is exact where C' is the image of the final map in C. If F is left exact then $0 \to FA \to FB \to FC'$ is exact, now we claim $FC' \to FC$ is a monomorphism and thus $0 \to FA \to FB \to FC$ is exact as well. Indeed, $0 \to C' \to C \to C/C' \to 0$ is exact and thus since F is left exact $0 \to FC' \to FC$ is exact, meaning $FC' \to FC$ is a monomorphism.

Now suppose F preserves exactness of sequences of the form $0 \to A \to B \to C$, then in particular when we can append $\to 0$. ■

1.6.11 Proposition

Let $\mathcal{A}$ be an Abelian category, then $\hom_{\mathcal{A}}(M, \bullet)$ is a left exact functor $\mathcal{A} \to \mathrm{Ab}$ for every $M \in \mathcal{A}$.

Proof

Suppose we have $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ in $\mathcal{A}$, we want to show that the following is exact

$$0 \to \hom(M, A) \xrightarrow{f_*} \hom(M, B) \xrightarrow{g_*} \hom(M, C)$$

where $f_*\alpha = f \circ \alpha$. Note that if $f_*\alpha = 0$ then $\alpha = 0$ since f is monic, thus f_* is monic. Furthermore since $g \circ f = 0$, $g_*f_*(\alpha) = g \circ f \circ \alpha = 0$ so $g_*f_* = 0$. Now if $\beta \in \hom(M, B)$ is in g_*'s kernel then $g \circ \beta = 0$ so $\beta(M) \subseteq \ker g = \operatorname{im} f$, thus β factors through f, meaning it is in f_*'s image as desired. ■

1.6.12 Corollary

$\hom_{\mathcal{A}}(\bullet, M)$ is left exact.

Proof

$\hom_{\mathcal{A}}(\bullet, M) = \hom_{\mathcal{A}^{\mathrm{op}}}(M, \bullet)$.

1.6.13 Definition

We define the **Yoneda embedding** $\mathsf{y}\colon \mathcal{A} \to [\mathcal{A}^{\mathrm{op}}, \mathrm{Ab}]$ for any additive category by mapping $A \mapsto h_A = \hom_{\mathcal{A}}(\bullet, A)$ and $f\colon A \to B$ to $f_*\alpha = f \circ \alpha$.

Considering the Yoneda embedding in classical category theory shows that this is a fully faithful embedding. When $\mathcal{A}$ is Abelian y is left exact since $\mathsf{y}(A) = h_A$ is.

1.6.14 Lemma (Yoneda)

The Yoneda embedding reflects exactness: $A \xrightarrow{\alpha} B \xrightarrow{\beta} C$ in $\mathcal{A}$ is exact if for every M the following sequence is exact

$$\hom(M, A) \xrightarrow{\alpha_*} \hom(M, B) \xrightarrow{\beta_*} \hom(M, C)$$

Proof

Let $M = A$ then we have $\beta\alpha = \beta_*\alpha_*\mathrm{id} = 0$. Taking $M = \ker\beta$ we see that the inclusion $\iota\colon \ker\beta \to B$ satisfies $\beta_*\iota = \beta\iota = 0$. Thus there is $\sigma \in \hom(M, A)$ such that $\iota = \alpha_*\sigma = \alpha\sigma$, so $\ker\beta = \operatorname{im}\iota \subseteq \operatorname{im}\alpha$ as desired. ■

2 Derived Functors

2.1 δ-functors

Recall that morphisms of exact sequences are collections of morphisms which commute with everything. That is to say, if we have two exact sequences $\{A_n, f_n\}_n$ and $\{B_n, g_n\}_n$ (i.e. $f_n\colon A_n \to A_{n+1}$ and $\operatorname{im} f_n = \ker f_{n+1}$), a morphism $A \to B$ is a collection $\{s_n\colon A_n \to B_n\}_n$ such that $sf = gs$ (i.e. view the exact sequences as chain complexes).

2.1.1 Definition

A **homological** (resp. **cohomological**) **δ-functor** from $\mathcal{A}$ to $\mathcal{B}$ (Abelian categories) is a collection of additive functors $T_n\colon \mathcal{A} \to \mathcal{B}$ (resp. $T^n\colon \mathcal{A} \to \mathcal{B}$) for $n \geq 0$, along with morphisms $\delta_n\colon T_n(C) \to T_{n-1}(A)$ (resp. $\delta^n\colon T^n(C) \to T^{n+1}(A)$) for every short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$.

A δ-functor must satisfy the following (set $T_n = T^n = 0$ for $n < 0$):

(1) for a short exact sequence $0 \to A \to B \to C \to 0$, we get a long exact sequence

$$\cdots \to T_{n+1}(C) \xrightarrow{\delta} T_n(A) \to T_n(B) \to T_n(C) \xrightarrow{\delta} T_{n-1}(A) \to \cdots$$

(ascending indices for cohomological δ-functors.) In particular T_0 is right exact (since $T_{-1} = 0$) and T^0 is left exact (T_n, T^n in general are not: we have an exact sequence $T_n(A) \to T_n(B) \to T_n(C)$ but we cannot append 0).

(2) For a morphism of short exact sequences, δ commutes with the induced morphisms. That is, if we have a morphism between short exact sequences $0 \to A' \to B' \to C' \to 0$ and $0 \to A \to B \to C \to 0$, we get commutative squares

$$\begin{tikzcd} T_n(C') \arrow[r, "\delta"] \arrow[d] & T_{n-1}(A') \arrow[d] \\ T_n(C) \arrow[r, "\delta"] & T_{n-1}(A) \end{tikzcd}$$

for every $n \geq 0$.

2.1.2 Proposition

Homology gives a homological δ-functor $H_\bullet \colon \mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathcal{A}$; cohomology gives a cohomological δ-functor $H^\bullet \colon \mathsf{Ch}^{\geq 0}(\mathcal{A}) \to \mathcal{A}$.

Proof

The δ morphisms are the connecting maps ∂. The axioms are satisfied because the connecting maps are functorial. ■

2.1.3 Proposition

Let $\mathcal{S}(\mathcal{A})$ be the category of short exact sequences in $\mathcal{A}$. If T is a δ-functor then δ_n is a natural transformation from T_n to T_{n-1} (viewed as functors mapping $0 \to A \to B \to C \to 0$ to T_nC and $T_{n-1}A$ respectively).

Proof

This is immediate from the second axiom of δ-functors. ■

2.1.4 Example

Let $p \in \mathbb{Z}$ then define a homological δ-functor $\mathrm{Ab} \to \mathrm{Ab}$ by $T_0(A) = A/pA$ and $T_1(A) = {}_pA = \{a \in A \mid pa = 0\}$ ($T_n = 0$ for all other n). To get the δ morphisms, the snake lemma on the diagram

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, "p"] & B \arrow[r] \arrow[d, "p"] & C \arrow[r] \arrow[d, "p"] & 0 \\
0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0
\end{tikzcd}$$

Since $\ker p = {}_pA$ and $\operatorname{coker} p = A/pA$ we get an exact sequence

$$0 \to {}_pA \to {}_pB \to {}_pC \xrightarrow{\delta} A/pA \to B/pB \to C/pC \to 0$$

Which is precisely the first axiom. Functorality of the connecting morphism gives us the second axiom. We can also view this as a cohomological δ-functor by swapping T_0 and T_1.

2.1.5 Definition

A morphism $S \to T$ of δ-functors is a system of natural transformations $S_n \to T_n$ (or $S^n \to T^n$) which commute with δ.

A homological δ-functor T is **universal** if given any other homological δ-functor S and a natural transformation $f_0 \colon S_0 \to T_0$, we can uniquely extend f_0 to a morphism $\{f_n\}_n : S \to T$. Dually a cohomological δ-functor T is **universal** if given any other cohomological δ-functor S and natural $f_0 \colon T^0 \to S^0$ it can be uniquely extended to $T \to S$.

2.1.6 Example

We will show later that homology and cohomology are universal.

2.1.7 Proposition

If $F \colon \mathcal{A} \to \mathcal{B}$ is exact, then $T_0 = F$ and $T_n = 0$ for $n \neq 0$ is a universal δ-functor (both homological and cohomological).

Proof

A quick check of the axioms shows that F is indeed a δ-functor (with $\delta = 0$). Let S be another δ-functor (suppose homological) and if $f_0 \colon S_0 \to F$ is natural then define $f_n \colon S_n \to T_n = 0$ to be zero. This is the only way to extend f_0, and it is a morphism. ■

2.2 Projective resolutions

2.2.1 Definition

An object P in an Abelian category $\mathcal{A}$ is **projective** if it satisfies the following universal lifting property: given an epimorphism $g\colon B \to C$ and a map $\gamma\colon P \to C$ there exists a (not necessarily unique) map $\beta\colon P \to B$ such that $\gamma = g \circ \beta$. In a diagram:

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists\beta"'] \arrow[d, "\gamma"] & \\ B \arrow[r] & C \arrow[r] & 0 \end{tikzcd}$$

We are mostly concerned with **projective modules** (projective objects in Mod_R). Clearly free modules are projective: take a basis $(v_i)_i$ and map v_i to a preimage of $\gamma(v_i)$.

2.2.2 Proposition

Direct summands of projective modules are projective.

Proof

If $P = M \oplus M'$ is projective then any map $\gamma\colon M \to C$ can be extended to a map $\gamma\colon P \to C$ and the rest follows. ■

2.2.3 Proposition

An R-module is projective iff it is a direct summand of a free R-module (that is, M is projective iff there exists M' such that $M \oplus M'$ is free).

Proof

Direct summands of a free module are by the previous proposition projective.

Conversely $F(A)$ be the free R-module on the set A, for a module M we can consider $F(M)$ on the set underlying M and we have a surjection $\pi\colon F(M) \to M$. If M is projective then the universal lifting property gives us a map $i\colon M \to F(M)$ such that $\pi i = \mathrm{id}_M$:

$$\begin{tikzcd} & M \arrow[dl, dashed, "\exists i"'] \arrow[d, "\mathrm{id}_M"] & \\ F(M) \arrow[r, "\pi"] & M \arrow[r] & 0 \end{tikzcd}$$

So $i\pi$ is a projection onto $i(M) \cong M$, meaning M is a direct summand of $F(M)$. ■

Over many nice rings (e.g. $\mathbb{Z}$, division rings), projective modules are all free. This is not always the case:

2.2.4 Example

If $R = R_1 \times R_2$ is the product of two rings then $R_1 \times 0, 0 \times R_2$ are projective as their sum is R (a free R-module). $P = R_1 \times 0$ is not free since $(0,1)P = 0$ (meaning for any choice $B \subseteq P$ there is an $r \in R$ such that $rB = 0$ so B cannot be linearly independent). So for example $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ is projective but not free.

2.2.5 Example

Let k be a field and consider the matrix ring $R = M_n(k)$. This has a module k^n (multiplication is given by vector-matrix multiplication), and as an R-module we have $R \cong k^n \oplus \cdots \oplus k^n$, so k^n is projective. A free R-module needs to have dimension dn^2 over k for some $d > 0$, but $\dim_k k^n = n$ so it cannot be free.

The category of finite Abelian groups has no projective objects (other than 0): non-trivial subgroups of free Abelian groups are torsion-free and thus infinite.

2.2.6 Definition

$\mathcal{A}$ has **enough projectives** if for every $A \in \mathcal{A}$ there is an epimorphism $P \to A$ with P projective.

Mod_R has enough projectives, take the epimorphism $F(M) \to M$.

2.2.7 Lemma

M is projective iff $\hom_A(M, \bullet)$ is exact.

Proof

Suppose $\hom(M, \bullet)$ is exact, and we have a surjection $g\colon B \to C$ and a map $\gamma\colon M \to C$. Since g is surjective and $\hom(M, \bullet)$ is exact, g_* is surjective as well. Thus we can lift γ to β such that $\gamma = g_*\beta = g \circ \beta$; M has the universal lifting property, it is projective.

Recall that in general $\hom(M, \bullet)$ is left exact. So to show that it is exact, we must only show that for $0 \to A \to B \xrightarrow{g} C \to 0$, g_* is surjective as well (i.e. we can append $\to 0$). This is precisely what the universal lifting property says. ■

2.2.8 Definition

A chain complex where every component is projective is called a **chain complex of projectives**. It is not necessarily projective in Ch.

2.2.9 Proposition

A chain complex P is projective in Ch iff it is a split exact complex of projectives.

Proof

If P is projective then so is $P[-1]$: a surjection $B \to C$ and a map $P[-1] \to C$ can be shifted up to $B[1] \to C[1]$ and $P \to C[1]$, then the resulting lift can be shifted back down $P[-1] \to B$ ($[p]$ is functorial). We have a short exact sequence

$$0 \to P \to \mathrm{cone}(P) \to P[-1] \to 0$$

Now a surjection onto a projective object gives rise to a projection: we can lift $\mathrm{id}_{P[-1]}$ to $\beta\colon P[-1] \to \mathrm{cone}(P)$ such that $\delta\beta = \mathrm{id}_{P[-1]}$ where δ is the surjection. Then $\beta\delta$ is a projection onto $P[-1]$ whose kernel is P, thus $\mathrm{cone}(P) \cong P \oplus P[-1]$. Since $\mathrm{cone}(P)$ is exact, so too must P be.

Since $\delta(p, q) = -p$ we have that β is of the form $\beta(p) = (-p, s(p))$ for $s\colon P[-1] \to P$ (not necessarily a chain map). Now we claim that s is a splitting map. Since β is a chain map we have

$$d\beta(p) = d(-p, s(p)) = (dp, ds(p) + p), \qquad \beta(-dp) = (dp, -s(dp))$$

So we have $ds + \mathrm{id} = -sd$ so $\mathrm{id} = -sd - ds$ meaning P is split exact.

Now we must show that P's components are all projectives. So we need to show that each B'_n is projective: let $g\colon N \to M$ surjective and $\gamma\colon B'_n \to M$. Considering N and M as chain complexes centered at n with all other terms 0, g gives rise to a surjection of a chain complex, and γ a map from P to M. Then we can lift it to $\beta\colon P \to N$, and we get the desired result.

Recall that if P is split exact then $P_n \cong B_n \oplus B'_n$ for $B'_n \cong B_{n-1}$. The differential is zero on B_n and an isomorphism $B'_n \cong B_{n-1}$, so we can view $P_n \cong B_n \oplus B_{n-1}$ with $d(b, b') = (b', 0)$. Now a map $\gamma\colon P \to C$ is of the form $(\gamma_n, \gamma'_n)\colon P_n \to C_n$ and this is a chain map iff $d\gamma'_n = \gamma_{n-1}$. Indeed: $(\gamma_{n-1}, \gamma'_{n-1})d(b, b') = \gamma_{n-1}(b')$ and $d(\gamma_n(b) + \gamma'_n(b')) = d\gamma_n b + d\gamma'_n b'$. This is true for all b, b' and so take $b = 0$ and we get $\gamma_{n-1} = d\gamma'_n$. And conversely this implies $d\gamma_n = 0$ and so we have equality.

So if P is split exact of projectives, let $(\gamma_n, \gamma'_n)\colon P \to C$ and $g\colon A \to C$ surjective. Each B_n is projective (as it is a direct summand of a projective), so we have maps $\beta_n\colon B_{n-1} \to A_n$ which lift γ'_n along g_n. Notice then that the map $d\beta_{n+1}\colon B_n \to A_n$ lifts γ_n along g_n:

$$g_n d\beta_{n+1} = dg_{n+1}\beta_{n+1} = d\gamma'_{n+1} = \gamma_n$$

Then the map $(d\beta_{n+1}, \beta_n)\colon P \to A$ is a chain map and it lifts $\gamma + \gamma'$ along g as desired. ■

2.2.10 Corollary

If $\mathcal{A}$ has enough injectives, so does $\mathsf{Ch}(\mathcal{A})$.

Proof

Let C be a chain complex, then for each C_n let $g_n\colon B_n \to C_n$ be a surjection from a projective module. Then define $P_n = B_n \oplus B_{n-1}$ with differential mapping $d(b, b') = (b', 0)$. This is a split exact complex of projectives, thus a projective complex. Now $(dg_{n+1}, g_n)\colon P \to C$ is a chain map as discussed in the previous proof, and is surjective as desired. ■

2.2.11 Definition

A short exact sequence $0 \to A \to B \to C \to 0$ is called **split** if there is an isomorphism $f\colon B \to A \oplus C$ that commutes with the short exact sequence (with inclusion $\iota\colon A \to A \oplus C$ and projection $\pi\colon A \oplus C \to C$). Explicitly, the following diagram commutes:

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[dr, "\iota"'] & B \arrow[r] \arrow[d, "f"] & C \arrow[r] & 0 \\
 & & A \oplus C \arrow[ur, "\pi"'] & &
\end{tikzcd}$$

2.2.12 Lemma (Splitting lemma)

For a split exact sequence $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$, the following are equivalent:

(1) the sequence splits;

(2) the sequence is **left split**: there exists a morphism $t\colon B \to A$ such that $tf = \mathrm{id}_A$;

(3) the sequence is **right split**: there exists a morphism $s\colon C \to B$ such that $gs = \mathrm{id}_C$.

Proof

The sequence being split implies being left and right split: define $t: A \oplus C \cong B \to A$ to project onto A and $s: C \to B \cong A \oplus C$ to embed C into B. If the sequence is left split then $B = \ker t \oplus \operatorname{im} f$ since their intersection is zero and $b = (b - ft(b)) + ft(b)$. Now $\operatorname{im} f \cong A$ and $\ker t \cong C$ since we can map $b \in \ker t$ to $g(b) \in C$: this map is injective since its kernel is $\ker t \cap \ker g = \ker t \cap \operatorname{im} f = 0$. It is surjective since $g(b - ft(b)) = g(b)$. Thus $B \cong A \oplus C$ and this commutes with the sequence. ■

Note that this lemma is not true in the category of groups: being (left) split is equivalent to B being the direct product of A and C, but being right split is equivalent to B being the semidirect product of A and C. Thus (left) split implies right split, but not the other way around.

2.2.13 Proposition

An object $P \in \mathcal{A}$ is projective iff every short exact sequence $0 \to A \to B \to P \to 0$ is split.

Proof

If P is projective then we can lift id_P along the surjection $B \to P$ to give us a map which composes with it to give the identity. Thus the sequence is right split, and thus split.

Conversely suppose every such short exact sequence is split. Now $\mathcal{A}$ is finitely complete and thus for a surjection $A \to B$ and a map $P \to B$ we can consider the diagram's pullback C:

$$\begin{tikzcd} C \arrow[r, "f"'] \arrow[d] & P \arrow[r] \arrow[d] & 0 \\ A \arrow[r] & B \arrow[r] & 0 \end{tikzcd}$$

It can be shown that the first row is indeed exact (f is epi), since in Abelian categories epimorphisms are regular (cokernels of their kernels), and we can follow the kernel of the bottom map. Taking K to be the kernel of f, we have an exact sequence $0 \to K \to C \xrightarrow{f} P \to 0$, which must split. Then there is a map $g: P \to C$ such that $fg = \mathrm{id}_P$. Now consider $P \xrightarrow{g} C \to A$: by commutativity we have that it gives the desired lifting of $P \to B$. ■

2.2.14 Definition

Let $M \in \mathcal{A}$, a **left resolution** of M is an exact sequence

$$\cdots \xrightarrow{d} P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} M \to 0$$

The map ϵ is called the **augmentation map**. This is a **projective resolution** if each P_i are projective.

2.2.15 Lemma

If $\mathcal{A}$ has enough projectives, every object M has a projective resolution. In particular every R-module has a projective resolution.

Proof

First take a projective P_0 and a surjection $\epsilon_0: P_0 \to M$, and set $M_0 = \ker \epsilon_0$. Inductively given M_{n-1} choose a

projective P_n and surjection $\epsilon_n: P_n \to M_{n-1}$. Define $M_n = \ker\epsilon_n$ and let d_n be the composite $P_n \to M_{n-1} \to P_{n-1}$. Since $M_{n-1} \to P_{n-1}$ is monic $\ker d_n = \ker\epsilon_n = M_n$. And we have $d_n(P_n) = M_{n-1} = \ker d_{n-1}$, thus the resulting sequence is exact. ■

2.2.16 Proposition

A left resolution $\cdots \to P_1 \to P_0 \to M \to 0$ is the same as a quasi-isomorphism $P \to M$ where P is the chain complex $\cdots \to P_1 \to P_0 \to 0$ and M is considered as a chain complex focused at 0.

Proof

A chain map $P \xrightarrow{\epsilon} M$ is given by a map $P_0 \xrightarrow{\epsilon} M$ such that $\epsilon d = 0$ (for $d: P_1 \to P_0$). This is a quasi-isomorphism if $\epsilon: H_0(P) = P_0/dP_1 \to M$ is an isomorphism, meaning $\ker\epsilon = \operatorname{im} d$, i.e. the sequence is exact. ■

2.2.17 Theorem (Comparison theorem)

Let $P \xrightarrow{\epsilon} M$ be a projective resolution of M and $f': M \to N$ a map in $\mathcal{A}$. Then for every resolution $Q \xrightarrow{\eta} N$, there is a chain map $f: P \to Q$ lifting f' in the sense that $\eta f_0 = f' \circ \epsilon$:

$$\begin{tikzcd}
\cdots \arrow[r] & P_1 \arrow[r] \arrow[d] & P_0 \arrow[r] \arrow[d] & M \arrow[r] \arrow[d, "f'"'] & 0 \\
\cdots \arrow[r] & Q_1 \arrow[r] & Q_0 \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

f is unique up to chain homotopy equivalence.

Alternatively we could say that there exists a unique (up to chain homotopy equivalence) chain map between the chain complexes $P \to M$ to $Q \to N$.

The proof will show that $P \to M$ being a projective resolution is stronger a condition than necessary: it is sufficient to be given a chain complex $P \to M \to 0$ where P_n are all projective.

Proof

We construct f_n inductively; let $f_{-1} = f'$ and $f_{-2} = 0$. Suppose f_i has been constructed for $i \leq n$, so that $f_{i-1}d = df_i$. To construct f_{n+1} we consider the n-cycles of P, Q: if $n = -1$ then set $Z_{-1}(P) = M$ and $Z_{-1}(Q) = N$. Since $f_{n-1}d = df_n$ we know that f_n induces a map $f'_n: Z_n(P) \to Z_n(Q)$, so we have two diagrams with exact rows:

$$\begin{tikzcd}
\cdots \arrow[r] & P_{n+1} \arrow[r] \arrow[d, dashed] & Z_n(P) \arrow[r] \arrow[d, "f'_n"] & 0 \\
\cdots \arrow[r] & Q_{n+1} \arrow[r] & Z_n(Q) \arrow[r] & 0
\end{tikzcd}
\qquad
\begin{tikzcd}
0 \arrow[r] & Z_n(P) \arrow[r] \arrow[d, "f'_n"] & P_n \arrow[r] \arrow[d, "f_n"] & P_{n-1} \arrow[d, "f_{n-1}"] \\
0 \arrow[r] & Z_n(Q) \arrow[r] & Q_n \arrow[r] & Q_{n-1}
\end{tikzcd}$$

Using the first diagram and P_{n+1} being projective allows us to lift $f'_n d$ to $f_{n+1}: P_{n+1} \to Q_{n+1}$ such that $df_{n+1} = f'_n d = f_n d$. And so $\{f_n\}$ is a chain map extending f' as desired.

To show uniquess, suppose $g: P \to Q$ is a chain map extending f' as well. Let $h = f - g$ and we will construct a contraction $\{s_n: P_n \to Q_{n+1}\}$ inductively. For negative n we must have $s_n = 0$. For $n = 0$ we have $\eta h_0 = (f' - f')\epsilon = 0$, and so h_0 maps P_0 to $\ker\eta = Z_0(Q) = dQ_1$. And so we can lift h_0 to $s_0: P_0 \to Q_1$ such that $h_0 = ds_0 = ds_0 + s_{-1}d$.

Inductively suppose for $i < n$ we have maps s_i such that $ds_{n-1} = h_{n-1} - s_{n-2}d$ and then consider $h_n -$

$s_{n-1}d\colon P_n \to Q_n$. Note

$$d(h_n - s_{n-1}d) = dh_n - ds_{n-1}d = dh_n - (h_{n-1} - s_{n-2}d)d = dh_n - h_{n-1}d = 0$$

Thus $h_n - s_{n-1}d$ maps into $Z_n(Q) = dQ_{n+1}$, and so we can lift it to a map $s_{n+1}\colon P_n \to Q_{n+1}$ such that $ds_{n+1} = h_n - s_{n-1}d$. And so we have $h = ds + sd$ as desired. ■

Note that we did not use $P \to M$'s exactness: only $Q \to M$'s.

2.2.18 Lemma (Horeshoe lemma)

Suppose we have the following diagram:

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
\cdots \arrow[r] & P_1' \arrow[r] & P_0' \arrow[r, "\epsilon'"] & A' \arrow[r] \arrow[d, "i_A"] & 0 \\
 & & & A \arrow[d, "\pi_A"] & \\
\cdots \arrow[r] & P_1'' \arrow[r] & P_0'' \arrow[r, "\epsilon''"] & A'' \arrow[r] \arrow[d] & 0 \\
 & & & 0 &
\end{tikzcd}$$

such that the column is exact and the rows are projective resolutions. Set $P_n = P_n' \oplus P_n''$, then P_n form a projective resolution P of A, and the colimn lifts to an exact sequence of complexes $0 \to P' \xrightarrow{i} P \xrightarrow{\pi} P'' \to 0$, where $i_n\colon P_n' \to P_n$ and $\pi_n\colon P_n \to P_n''$ are the inclusion and projection maps.

Proof

Lift ϵ'' to a map $P_0'' \to A$; then the direct sum of this map with $i_A\epsilon' \colon P_0' \to A$ gives a map $\epsilon \colon P_0 \to A$. Then the following diagram commutes:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \ker\epsilon' \arrow[r] \arrow[d] & P_0' \arrow[r, "\epsilon'"] \arrow[d] & A' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \ker\epsilon \arrow[r] \arrow[d] & P_0 \arrow[r, "\epsilon"] \arrow[d] & A \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \ker\epsilon'' \arrow[r] \arrow[d] & P_0'' \arrow[r, "\epsilon''"] \arrow[d] & A'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

The two right columns are exact, and all the top and bottom rows are as well. The snake lemma gives us that the left column is exact, and $\operatorname{coker}\epsilon = 0$ (since $0 = \operatorname{coker}\epsilon' \to \operatorname{coker}\epsilon \to \operatorname{coker}\epsilon'' = 0$ is exact). Thus ϵ is epic, as desired.

Now we have another horseshoe:

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
\cdots \arrow[r] & P_2' \arrow[r] & P_1' \arrow[r, "d'"] & \ker\epsilon' \arrow[r] \arrow[d] & 0 \\
 & & & \ker\epsilon \arrow[d] & \\
\cdots \arrow[r] & P_2'' \arrow[r] & P_1'' \arrow[r, "d''"] & \ker\epsilon'' \arrow[r] \arrow[d] & 0 \\
 & & & 0 &
\end{tikzcd}$$

and we can induct. ■

2.3 Injective resolutions

We can duallize the notion of projective objects:

2.3.1 Definition

An object I in an Abelian category $\mathcal{A}$ is **injective** if it satisfies the following universal lifting property: given a monomorphism $f: A \to B$ and a map $\alpha: A \to I$, we can lift α to a map $\beta: B \to I$ such that $\alpha = \beta \circ f$.

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "f"'] \arrow[d, "\alpha"'] & B \arrow[dl, dashed, "\beta"] \\ & I & \end{tikzcd}$$

$\mathcal{A}$ has enough injectives if for every object $A \in \mathcal{A}$ there is an injective $I \in \mathcal{A}$ and a monomorphism $A \to I$.

Note that if $\{I_\alpha\}$ are injectives, so is their product (if it exists).

2.3.2 Lemma

An injective object in $\mathcal{A}$ is precisely a projective object in $\mathcal{A}^{\mathsf{op}}$. $\mathcal{A}$ has enough injectives iff $\mathcal{A}^{\mathsf{op}}$ has enough projectives.

2.3.3 Corollary

I is injective iff $\hom_{\mathcal{A}}(\bullet, I)$ is exact.

Proof

$I \in \mathcal{A}$ is injective iff $I \in \mathcal{A}^{\mathsf{op}}$ is projective iff $\hom_{\mathcal{A}^{\mathsf{op}}}(I, \bullet) = \hom_{\mathcal{A}}(\bullet, I)$ is exact. ■

2.3.4 Proposition (Baer's criterion)

A right R-module E is injective iff for every right ideal J of R, every map $J \to E$ can extend to a map $R \to E$.

Proof

If E is injective then we lift $J \to E$ along the injection $J \to R$. Conversely suppose we have an R-module B and a submodule A along with a map $\alpha: A \to E$. Let $\mathcal{E}$ be the poset of all extensions of α: maps $\alpha': A' \to E$ for $A \subseteq A' \subseteq B$, partially ordered by $\alpha' \leq \alpha''$ iff α'' extends α'. This clearly satisfies the criteria for Zorn's lemma, as such it has a maximal element, $\alpha': A' \to E$. We claim that $A' = B$.

Suppose not; there is some $b \in B$ not in A'. Define then $J = \{r \in R \mid br \in A'\}$: a right ideal of R. Then the composition $J \xrightarrow{b} A' \xrightarrow{\alpha'} E$ extends to a map $f: R \to E$. Let $A'' = A' + bR$, and define $\alpha'': A'' \to E$ by $\alpha''(a + br) = \alpha'(a) + f(r)$. This is well-defined since for $br \in A'$ we know $\alpha'(br) = f(r)$ by definition. As such α'' is a proper extension of α', a contradiction. ■

Recall that a semisimple R-module is one for which every submodule has a complement: if $A \leq M$ is a submodule then there exists $A' \leq M$ such that $M = A \oplus A'$. A semisimple ring is one which is semisimple as a module over itself (every ideal has a complement). Clearly then modules over semisimple rings are injective: let J' be J's complement and extend $J \to E$ to $R = J \oplus J' \to E$.

2.3.5 Corollary

Let R be semisimple, then every R-module is injective.

2.3.6 Corollary

Let R be a PID. Then an R-module M is injective iff it is **divisible**: for every nonzero $r \in R$ and $a \in M$ there is a $b \in M$ such that $a = br$.

Proof

If M is injective then for $r \in R$ and $a \in M$ we define $f: R \to M$ by $f(1) = a$ and extend by linearity: $f(x) = xa$. And define $g: R \to R$ by $g(1) = r$ (so $g(x) = rx$). Since R is an integral domain g is injective, and thus we can lift f to $f': R \to M$ such that $f'g = f$. Then $f'g(1) = f'(r) = rf'(1) = a$ as desired.

If M is divisible then for every $x \in R$, a map $f: (x) \to M$ is determined by its value on x: $f(rx) = rf(x)$. We want to find a value $f(1)$ such that $xf(1) = f(x)$, which exists precisely because M is divisible. And so we can extend f to $f: R \to M$ by $f(r) = rf(1)$. All ideals of R are of this form (principal) since R is a PID, so by Baer M is injective as desired. ■

In particular $\mathbb{Q}, \mathbb{Z}_{p^\infty} = \mathbb{Z}[1/p]/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}$ are injective Abelian groups ($\mathbb{Z}[1/p]$ are all rationals of the form a/p^n).

2.3.7 Proposition

Ab has enough injectives.

Proof

For an Abelian group A, define $I(A)$ to be the product of $\mathbb{Q}/\mathbb{Z}$ indexed by $\hom_{\mathrm{Ab}}(A, \mathbb{Q}/\mathbb{Z})$. This is a product of injectives and is thus also injective. Define $e_A: A \to I(A)$ by $e_A(a) = (f(a))_{f: A \to \mathbb{Q}/\mathbb{Z}}$.

We claim that e_A is an injection, so we must show that $\ker e_A = 0$. To do this we show that for every $0 \neq a \in A$, there is some $f: A \to \mathbb{Q}/\mathbb{Z}$ such that $f(a) \neq 0$. Let $\langle a \rangle \leq A$ be the subgroup generated by a, then by injectivity of $\mathbb{Q}/\mathbb{Z}$ we can lift any map $f: \langle a \rangle \to \mathbb{Q}/\mathbb{Z}$ to a map $A \to \mathbb{Q}/\mathbb{Z}$ (following the inclusion $\langle a \rangle \subseteq A$). Now $\langle a \rangle \cong \mathbb{Z}/m\mathbb{Z}$ or $\mathbb{Z}$ depending on a's order; in the first case define $f(a) = 1/m$ and in the second define $f(a) = 1/2$ say. In both cases these are well-defined maps where $f(a) \neq 0$, as desired. ■

Note that since $A \to (\mathbb{Q}/\mathbb{Z})^{\hom(A, \mathbb{Q}/\mathbb{Z})}$ is injective, if $\hom(A, \mathbb{Q}/\mathbb{Z}) = 0$ the right-hand side is zero and so too is A. Thus $A = 0$ iff $\hom(A, \mathbb{Q}/\mathbb{Z}) = 0$.

2.3.8 Proposition

$I \in \mathcal{A}$ is injective iff every short exact sequence $0 \to I \to A \to B \to 0$ splits.

Proof

By the duality between injectives and projectives. ■

2.3.9 Definition

Let $M \in \mathcal{A}$, then a **right resolution** of M is a cochain complex I with $I^n = 0$ for $n < 0$ and a map $M \to I^0$ such that the augmented complex

$$0 \to M \to I^0 \xrightarrow{d} I^1 \xrightarrow{d} \cdots$$

is exact. This is the same as a quasi-isomorphism $M \to I$. This is a **injective resolution** if each I_n is injective.

A right resolution in $\mathcal{A}$ is the same as a left resolution in A^{op}; an injective resolution in $\mathcal{A}$ is a projective resolution in $\mathcal{A}^{\mathsf{op}}$. Thus by dualizing our previous results we have the following:

2.3.10 Lemma

If $\mathcal{A}$ has enough injectives, every object in $\mathcal{A}$ has an injective resolution.

2.3.11 Theorem (Comparison theorem)

Let $N \to I$ be an injective resolution of N, and $f' \colon M \to N$ a map in $\mathcal{A}$. Then for every resolution $M \to E$ there is a cochain map $f \colon E \to N$ lifting f', unique up to chain homotopy equivalence.

2.3.12 Proposition

$I \in \mathsf{Ch}(\mathcal{A})$ is injective iff it is a split exact chain complex of injectives. Furthermore if $\mathcal{A}$ has enough injectives, so does $\mathsf{Ch}(\mathcal{A})$.

Proof

This follows immediately from previous results on projectives and the isomorphism $\mathsf{Ch}(\mathcal{A})^{\mathsf{op}} \cong \mathsf{Ch}(\mathcal{A}^{\mathsf{op}})$. This can be seen as follows: $\mathsf{Ch}(\mathcal{A})$ can be viewed as the functor category $[\mathbb{Z}_0, \mathcal{A}]$ (of additive functors) where $\mathbb{Z}_0$ is the category whose objects are integers and $\hom(i,j) = \mathbb{Z}$ when $j = i-1$ and 0 otherwise. Now in general $[\mathcal{C}, \mathcal{D}]^{\mathsf{op}} \cong [\mathcal{C}^{\mathsf{op}}, \mathcal{D}^{\mathsf{op}}]$ by mapping $F \colon \mathcal{C} \to \mathcal{D}$ to its opposite functor $F^{\mathsf{op}} \colon \mathcal{C}^{\mathsf{op}} \to \mathcal{D}^{\mathsf{op}}$ (which acts precisely as F does). Note that natural transformations $F^{\mathsf{op}} \Rightarrow G^{\mathsf{op}}$ are precisely natural transformations $G \Rightarrow F$, so this is a well-defined functor and clearly an isomorphism.

Note that $\mathbb{Z}_0^{\mathsf{op}} \cong \mathbb{Z}_0$ by mapping $n \mapsto -n$, and so

$$\mathsf{Ch}(\mathcal{A})^{\mathsf{op}} \cong [\mathbb{Z}_0, \mathcal{A}]^{\mathsf{op}} \cong [\mathbb{Z}_0, \mathcal{A}^{\mathsf{op}}] \cong \mathsf{Ch}(\mathcal{A}^{\mathsf{op}})$$

as desired.

Explicitly, we map a chain complex $C_\bullet$ to its cochain complex $C^\bullet$ (recall $C^n = C_{-n}$), and a chain map to its dual ($f^n = f_{-n}$). ■

We now will show that every category of modules has enough injectives. Note that since the opposite of a category of modules is itself not a category of modules, this is not a trivial result.

If A is an Abelian group and B a left R-module, then $\hom_{\mathrm{Ab}}(B, A)$ can be given a right R-module structure by $fr \colon b \mapsto f(rb)$.

2.3.13 Lemma

For every right R-module M, we have a natural isomorphism

$$\tau\colon \hom_{\mathrm{Ab}}(M, A) \to \hom_{\mathrm{Mod}_R}(M, \hom_{\mathrm{Ab}}(R, A))$$

given by $(\tau f)(m)\colon r \mapsto f(mr)$.

Proof

We define τ's inverse μ as follows: for $g\colon M \to \hom_{\mathrm{Ab}}(R, A)$ define $(\mu g)(m) = g(m)(1)$. For $f \in \hom_{\mathrm{Ab}}(M, A)$ then $(\mu\tau f)(m) = (\tau f)(m)(1) = f(m)$ so $\mu\tau$ is the identity. And for $f \in \hom_{\mathrm{Mod}_R}(M, \hom_{\mathrm{Ab}}(R, A))$ then $(\tau\mu f)(m)\colon r \mapsto (\mu f)(mr) = f(mr)(1)$. Notice $f(m)(r) = (f(m)r)(1) = (fmr)(1)$ as desired, so $\tau\mu f = f$ as desired. ■

Thus $\hom_{\mathrm{Ab}}(R, \bullet)\colon \mathrm{Ab} \to \mathrm{Mod}_R$ is **right-adjoint** to the forgetful functor $U\colon \mathrm{Mod}_R \to \mathrm{Ab}$:

2.3.14 Definition

Functors $L\colon \mathcal{A} \rightleftarrows \mathcal{B}\colon R$ form an **adjunction** if there is a natural isomorphism

$$\tau_{AB}\colon \hom_{\mathcal{B}}(L(A), B) \cong \hom_{\mathcal{A}}(A, R(B))$$

(i.e. a natural isomorphism of the bifunctors given by the compositions of the hom-functors and L and R.) L is the **left adjoint** and R the **right adjoint**. This is commonly written $L \dashv R$.

2.3.15 Proposition

Suppose we have an adjunction of additive functors $L\colon \mathcal{A} \rightleftarrows \mathcal{B}\colon R$, then

(1) if L is exact and $I \in \mathcal{B}$ injective then $R(I) \in \mathcal{A}$ is injective as well;

(2) dually if R is exact and $P \in \mathcal{A}$ projective then $L(P) \in \mathcal{B}$ is projective.

Proof

We must show that $\hom_{\mathcal{A}}(\bullet, R(I))$ is exact, but by assumption this is isomorphic to $\hom_{\mathcal{B}}(L\bullet, I)$. Since I is injective, $\hom_{\mathcal{B}}(\bullet, I)$ is exact, and the composition of exact functors is obviously exact. Duality follows because $L \dashv R$ iff $R^{\mathsf{op}} \dashv L^{\mathsf{op}}$. ■

We showed that $U \dashv \hom_{\mathrm{Ab}}(R, \bullet)$ and the forgetful functor U is clearly exact. Thus $\hom_{\mathrm{Ab}}(R, I)$ is an injective R-module for any injective Abelian group I.

2.3.16 Theorem

Mod_R has enough injectives.

Proof

For an R-module M, define $I(M)$ to be the product of $I_0 = \hom_{\mathrm{Ab}}(R, \mathbb{Q}/\mathbb{Z})$ with itself indexed by $\hom_R(M, I_0)$. As explained, I_0 is injective and thus so is $I(M)$. Now define $e_M\colon M \to I(M)$ by $e_M(m) = (f(m))_{f:M\to I_0}$. Let $\alpha\colon R \to \mathbb{Q}/\mathbb{Z}$ be non-zero (which exists by extending a non-zero map $\langle 1\rangle \to \mathbb{Q}/\mathbb{Z}$ where $\langle 1\rangle$ is the subgroup generated by 1, which exists as per our previous argument). Now for $m \in M$ define

$f: mR \to I_0$ by $f(m) = \alpha$, and since I_0 is injective this extends to $M \to I_0$ where $f(m) \neq 0$. Thus e_M has trivial kernel and is thus an injection. ■

2.3.17 Example

Consider $\mathsf{sh}(X)$, the category of sheaves (of Abelian groups) over a topological space X. We claim that $\mathsf{sh}(X)$ has enough injectives.

For a sheaf $\mathcal{F}$ and a point $x \in X$, $\mathcal{F}$'s **stalk** at x is the colimit $\mathcal{F}_x = \operatorname{colim}\{\mathcal{F}(U) \mid x \in U\}$ (i.e. the limit of the restriction of $\mathcal{F}$ to the poset of open subsets containing x). This is functorial $\mathsf{sh}(X) \to \mathrm{Ab}$ for every $x \in X$ (since colim is). If A is an Abelian group and $x \in X$, we define the **skyscraper sheaf** x_*A at $x \in X$ to be the presheaf $(x_*A)(U) = A$ iff $x \in U$ and 0 otherwise. x_* acts on inclusions by either maps to zero or the identity, that is if $U \subseteq V$ then $\bullet|_U: (x_*A)(V) \to (x_*A)(U)$ is the identity if $x \in U$ and the zero map otherwise.

The skyscraper sheaf is indeed a sheaf. If $\{U_i\}$ is an open cover of U, suppose $x \in U$; then a_i, a_j ($a_i \in (x_*A)(U_i)$) agreeing on their intersection means that $a_i = a_j \in A$ if $x \in U_i, U_j$. Let a be the common element of all open sets containing x, this is the unique extension. Otherwise if $x \notin U$ all $a_i = 0$, so $a = 0$.

Note that $\{U \mid x \in U\}$ is a directed set and as such $\mathcal{F}_x$ is a direct limit which we can explicitly write out. $\mathcal{F}_x$ has underlying set $\bigsqcup \mathcal{F}(U)$ quotiented by an equivalence, where $s_i \in \mathcal{F}(U_i), s_j \in \mathcal{F}(U_j)$ are equated iff they have a common restriction (on some neighborhood of x).

Now we claim that $\mathcal{F}_x \dashv (x_*A)$ (functorial in x) for every sheaf $\mathcal{F}$ and Abelian group A. That is, we need a natural isomorphism

$$\hom_{\mathrm{Ab}}(\mathcal{F}_x, A) \cong \hom_{\mathsf{sh}(X)}(\mathcal{F}, x_*A)$$

Let $\phi: \mathcal{F}_x \to A$ be an Abelian morphism, then define $\phi'_U: \mathcal{F}(U) \to x_*(A)$ by $\phi'_U(s) = \phi([s])$ (where $[s]$ is the equivalence class of s in $\mathcal{F}_x$), this defines a morphism of sheaves $\phi': \mathcal{F} \to x_*A$. Indeed: if $x \in U \subseteq V$ then $\phi'_U(s|_U) = \phi([s]) = \phi'_V(s)$ (the restriction of $x_*A(V)$ to $x_*A(U)$ is the identity. If $x \notin U$ then all maps compose to zero.

Conversely given $\psi: \mathcal{F} \to x_*A$ defining $\hat{\psi}: \mathcal{F}_x \to A$ by $\hat{\psi}([s]) = \psi_U(s)$ where $s \in \mathcal{F}(U)$. This is well-defined: if $s \in \mathcal{F}(U)$ and $s' \in \mathcal{F}(V)$ have the same restriction to a neighborhood of x, W ($s|_W = s'|_W$) then because ψ is a sheaf map we have that $\psi_U(s)|_W = \psi_V(s')|_W$. But restriction in the skyscraper sheaf is the identity: $\psi_U(s) = \psi_V(s')$ as desired. The composition of $\hat{\bullet}$ and $\bullet'$ gives the identities clearly, so we have a natural isomorphism, as desired.

Note that $\mathcal{F} \mapsto \mathcal{F}_x$ is an exact functor: for a morphism of sheaves $\phi: \mathcal{F} \to \mathcal{G}$ the induced map $\phi_x: \mathcal{F}_x \to \mathcal{G}_x$ is given by $\phi_x([s]) = [\phi_U(s)]$ for $s \in \mathcal{F}(U)$. The kernel of ϕ can be given by the sheaf $\ker\phi(U) = \ker\phi_U$, with restriction inherited from $\mathcal{F}$ (since $\ker\phi(U) \subseteq \mathcal{F}(U)$). And its image is given by $\operatorname{im}\phi(U) = \operatorname{im}\phi_U$, with restriction inherited from $\mathcal{G}$. Then $\ker\phi_x = \{[s] \in \mathcal{F}_x \mid s|_U = 0 \text{ for some } x \in U\} = \bigcup_{x \in U}[\ker\phi_U]$, and $\operatorname{im}\phi_x = \bigcup_{x \in U}[\operatorname{im}\phi_U]$. So if $\operatorname{im}\phi = \ker\psi$ we see that $\operatorname{im}\phi_x = \ker\psi_x$ as desired.

Thus x_*A is right-adjoint to an exact functor; it preserves injectives. So for every $x \in X$ choose an injective Abelian group, then x_*A_x is injective in $\mathsf{sh}(X)$, and thus so is $\prod_{x \in X} x_*A_x$.

Now given a sheaf $\mathcal{F}$, for every $x \in X$ choose an injection $\mathcal{F}_x \to I_x$ with I_x injective. Composing $\mathcal{F} \to x_*\mathcal{F}_x$ ($\phi_U(s) = [s]$ if $x \in U$ and 0 otherwise) with $x_*\mathcal{F}_x \to x_*I_x$ gives a map $\mathcal{F} \to \prod_{x \in X} x_*I_x = \mathcal{I}$. This is an injection: for $0 \neq s \in \mathcal{F}(U)$, note that there must be some $x \in V$ such that $s|_W \neq 0$ for all $W \subseteq V$ (by the sheaf axiom), and so $\phi_V(s) \neq 0$. Thus its kernel is zero.

So we have shown $\mathsf{sh}(X)$ has enough injectives.

2.3.18 Example

Let I be a small category and $\mathcal{A}$ Abelian and complete with enough injectives, we will show the functor category $\mathcal{A}^I = [I, \mathcal{A}]$ has enough injectives.

Consider the k**th coordinate functor** for $k \in I$: map $A: I \to \mathcal{A}$ to $A(k) \in \mathcal{A}$. This is an exact functor.

Now for $k \in I$ and $A \in \mathcal{A}$ define $k_*A: I \to A$ mapping $i \in I$ to

$$k_*A(i) = \prod_{\hom_I(i,k)} A$$

and if $f: i \to j$ is a map in I, the map $k_*A(f): k_*A(i) \to k_*A(j)$ is determined by $f^*: \hom(j,k) \to \hom(i,k)$. That is, the coordinate $k_*A(i) \to A$ of this map corresponding to $\phi \in \hom_I(i,k)$ is the projection of $k_*A(i)$ onto the factor corresponding to $f^*\phi = \phi f$. And for $f: A \to B \in \mathcal{A}$ then define $k_*f: k_*A \to k_*B$ which maps each component $A \in k_*A$ to the corresponding term $B \in k_*B$ via f. This makes k_* a functor $\mathcal{A} \to \mathcal{A}^I$.

Now, k_* is right-adjoint to the kth coordinate functor:

$$\hom_{\mathcal{A}}(A(k), B) \cong \hom_{\mathcal{A}^I}(A, k_*B), \qquad A \in \mathcal{A}^I, B \in \mathcal{A}$$

Map $\phi: A(k) \to B$ to $\phi': A \Rightarrow k_*B$ where $\phi'_i: A(i) \to k_*B(i)$ is defined by mapping $A(i) \xrightarrow{A(f)} A(k) \xrightarrow{\phi} B$ for each component $f: i \to k$. And for $\psi: A \Rightarrow k_*B$ map to $\hat{\psi}: A(k) \to B$ by the composition $A(k) \xrightarrow{\psi_k} k_*B(k) \xrightarrow{\pi_{\mathrm{id}}} B$ where π_{id} is the projection to the component associated with $\mathrm{id} \in \hom(k,k)$.

Thus k_* preserves injectives. For $F \in \mathcal{A}^I$, for each $k \in I$ embed $F(k) \to A_k$ for A_k injective. Then $E = \prod_{k \in I} k_*A_k$ exists (functor categories over complete categories are complete) and is injective. Further, $\eta: F \Rightarrow E$ is given by $\eta_i: F(i) \to \prod_{k \in I} k_*A_k(i)$ whose projection $F(i) \to A_k$ for $\phi: i \to k$ is given by the composition $F(i) \xrightarrow{F\phi} F(k) \to A_k$. This is a monomorphism.

Thus when $\mathcal{A}$ is complete Abelian and has enough injectives, so does $\mathcal{A}^I$.

Conversely if $\mathcal{A}$ is cocomplete and has enough projectives, $\mathcal{A}^{\mathsf{op}}$ has enough injectives and thus so does $[I^{\mathsf{op}}, \mathcal{A}^{\mathsf{op}}] \cong (\mathcal{A}^I)^{\mathsf{op}}$. Thus $\mathcal{A}^I$ has enough projectives.

2.4 Left derived functors

2.4.1 Definition

Let $F: \mathcal{A} \to \mathcal{B}$ be a right exact functor between Abelian categories, where $\mathcal{A}$ has enough projectives. Define the **left derived functors** of F, $\{L_iF\}_{i \geq 0}$ as follows. If A is an object of $\mathcal{A}$, choose a projective resolution $P \to A$ and define $L_iF(A) = H_i(F(P))$. For a map $f: A \to B$ in $\mathcal{A}$, suppose we chose a resolution $Q \to B$ then by comparison we have a unique extension of f to a chain map between the complexes (up to chain homotopy), then Ff is a chain map and as such we have a unique map $L_iF(f)$ from $H_i(F(P)) \to H_i(F(Q))$ (since chain homotopic maps induce the same morphisms on homology modules).

Note that $H_i(F(P))$ corresponds to the homology module of the complex $\cdots \to F(P_1) \to F(P_0) \to 0$, not the augmented complex containing A.

These are indeed functors: clearly $\mathrm{id}: A \to A$ lifts to $\mathrm{id}: P \to P$ and thus $L_iF(\mathrm{id})$ is the identity. And given $P \to A, P' \to A', P'' \to A''$ and maps $A \xrightarrow{f} A' \xrightarrow{f'} A''$, f lifts to $P \to P'$ and f' lifts to $P' \to P''$, then the composition of these lifts is a lift of $f' \circ f$. They are additive functors, too: for $f, g: A \to B$ if $\tilde{f}, \tilde{g}$ are their lifts then $\tilde{f} + \tilde{g}$ lifts $f + g$.

Now notice that since F is right exact, $F(P_1) \xrightarrow{d} F(P_0) \xrightarrow{d'} F(A) \to 0$ is exact, and as such such $H_0(F(P_0)) = F(P_0)/\mathrm{im}d$ and we know $\mathrm{im}d = \ker d'$. Since d' is a projection we have $H_0(F(P_0)) = F(P_0)/\ker d' \cong F(A)$. Thus $L_0F(A) \cong F(A)$, and since $L_0F(f)$ is just the induced map of f, we have that $L_0 \cong F$ naturally.

2.4.2 Lemma

Let F have left derived functors L_iF, then $L_0F \cong F$.

Also we show the naturality of L_iF.

2.4.3 Lemma

Different choices of projective resolutions result in naturally isomorphic left derived functors. That is, left derived functors are well-defined up to a unique natural isomorphism.

Proof

Let $P \to A$ and $Q \to A$ be two projective resolutions. Then there are chain maps $f\colon P \rightleftarrows Q\colon g$ lifting id_A, and these are unique up to chain homotopy. But then $gf\colon P \to P$ and $fg\colon Q \to Q$ are lifts of id_A, and since $\mathrm{id}_P, \mathrm{id}_Q$ are lifts as well, gf and fg are chain homotopic to the identities. Consider then the induced maps $F(f)_*\colon H_i(F(P)) \rightleftarrows H_i(F(Q))\colon F(g)_*$. Additive functors preserve chain homotopy, so $F(fg), F(gf)$ are chain homotopic to the identities, and as such $F(fg)_* = F(f)_*F(g)_*$ is the identity, similar for gf. Thus we have an isomorphism $H_iF(P) \cong H_iF(Q)$ as desired. ■

2.4.4 Corollary

If A is projective, $L_iF(A) = 0$ for $i \neq 0$.

Proof

There is a simple projective resolution of A: $\cdots \to 0 \to 0 \to A \xrightarrow{\mathrm{id}} A \to 0$. Then applying F gives us the chain complex $\cdots \to 0 \to 0 \to FA \to 0$, whose nonzero homology groups are all 0. ■

2.4.5 Definition

An object $Q \in \mathcal{A}$ is **F-acyclic** if $L_iFQ = 0$ for all $i \neq 0$. An **F-acyclic resolution** of $A \in \mathcal{A}$ is a left resolution $Q \to A$ where the Q_is are all F-acyclic.

We will later show that $L_iF(A) \cong H_i(F(Q))$ for any F-acyclic resolution Q of A.

2.4.6 Theorem

The derived functors $L_\bullet F$ form a homological δ-functor.

Proof

Given a short exact sequence $0 \to A' \to A \to A'' \to 0$ choose projective resolutions $P' \to A'$ and $P'' \to A''$. By the horseshoe lemma there is a projective resolution $P \to A$ fitting into a short exact sequence $0 \to P' \to P \to P'' \to 0$ of projective complexes in $\mathcal{A}$. Since P''_n is projective, each $0 \to P'_n \to P_n \to P''_n \to 0$ is split exact, and thus so is $0 \to FP'_n \to FP_n \to FP''_n \to 0$, in particular we have a short exact sequence $0 \to FP' \to FP \to FP'' \to 0$. We thus have a long exact homology sequence

$$\cdots \xrightarrow{\partial} L_iF(A') \to L_iF(A) \to L_iF(A'') \xrightarrow{\partial} L_{i-1}F(A') \to L_{i-1}F(A) \to L_{i-1}F(A'') \xrightarrow{\partial} \cdots$$

These ∂s are natural (and thus form the δ-morphisms of the δ-functor). Indeed, suppose we have a morphism of exact sequences

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r] \arrow[d, "f'"'] & A \arrow[r] \arrow[d, "f"'] & A'' \arrow[r] \arrow[d, "f''"'] & 0 \\
0 \arrow[r] & B' \arrow[r, "i_B"] & B \arrow[r, "\pi_B"] & B'' \arrow[r] & 0
\end{tikzcd}$$

and choose projective resolutions $\epsilon'\colon P' \to A', \epsilon\colon P'' \to A'', \eta'\colon Q' \to B', \eta''\colon Q'' \to B''$. The connecting morphisms are obtained via the horseshoe lemma, which give $\epsilon\colon P \to A, \eta\colon Q \to B$. The horseshoe lemma gives us ϵ is the direct sum of the lift of ϵ'' with $i_A\epsilon'$. The comparison lemma gives us unique lifts of f', f''

to $F'\colon P' \to Q'$ and $F''\colon P'' \to Q''$. We show that there is a lift of f to $F\colon P \to Q$ such that the following diagram commutes

$$\begin{tikzcd}
0 \arrow[r] & P' \arrow[r] \arrow[d, "F'"'] & P \arrow[r] \arrow[d, "F"'] & P'' \arrow[r] \arrow[d, "F''"'] & 0 \\
0 \arrow[r] & Q' \arrow[r] & Q \arrow[r] & Q'' \arrow[r] & 0
\end{tikzcd}$$

To construct F, we define $\gamma_n\colon P''_n \to Q'_n$ (not a chain map) such that

$$F_n = \begin{pmatrix} F'_n & \gamma_n \\ 0 & F''_n \end{pmatrix} : P'_n \oplus P''_n \to Q'_n \oplus Q''_n, \qquad F(p', p'') = (F'p' + \gamma(p''), F''(p''))$$

If F is a chain map, then this diagram will indeed commute. In order for F to lift f we must have that $\eta F_0 - f\epsilon$ is zero on $P_0 = P'_0 \oplus P''_0$. This is true for any γ on P'_0, so we must require just that

$$i_B \eta' \gamma_0 = f\lambda_P - \lambda_Q F''_0$$

where λ_P, λ_Q are the restrictions of ϵ, η to P''_0 and Q''_0, i_B is the inclusion of $B' \to B$. Note that in B'' we have

$$\pi_B(f\lambda - \lambda F''_0) = f''\pi_A\lambda_P - \pi_B\lambda F''_0 = f''\epsilon'' - \eta'' F''_0 = 0$$

thus there is a map $\beta\colon P''_0 \to B'$ such that $i_B\beta = f\lambda - \lambda F''_0$ (since it is zero on B''). Lift β to $\gamma_0\colon P''_0 \to Q'_0$: $\beta = \eta'\gamma_0$. Now we have

$$dF - Fd = \begin{pmatrix} d'F' - F'd' & (d'\gamma - \gamma d'' + \lambda F'' - F'\lambda') \\ 0 & d''F'' - F''d'' \end{pmatrix}$$

and for F to be a chain map, this must vanish. So we must have $d'\gamma = \gamma d'' - \lambda F'' + F'\lambda'$. This can be constructed inductively. ■

2.4.7 Proposition (Dimension shifting)

If $0 \to M \to P \to A \to 0$ is exact with P projective (or F-acyclic), then $L_iF(A) \cong L_{i-1}F(M)$ for $i \geq 2$, and $L_1F(A)$ is the kernel of $F(M) \to F(P)$.

In general if we have an exact sequence

$$0 \to M_m \to P_m \to P_{m-1} \to \cdots \to P_0 \to A \to 0$$

with P_i projective (or in general F-acyclic), then $L_iF(A) \cong L_{i-m-1}F(M_m)$ for $i \geq m+2$, and $L_{m+1}F(A)$ is the kernel of $F(M_m) \to F(P_m)$.

Proof

If $0 \to M \to P \to A \to 0$ is short exact, then since derived functors are δ-functors, we have a long exact sequence

$$\cdots \to L_iF(P) \to L_iF(A) \to L_{i-1}F(M) \to L_{i-1}F(P) \to \cdots$$

and for $i \geq 2$ we have that $L_iF(P) = L_{i-1}F(P) = 0$ and so we have the isomorphism. For $i = 1$ (recalling $L_0F \cong F$) this gives us $0 \to L_1F(A) \to F(M) \to F(P) \to 0$ as desired.

Now we induct on m, we have shown it for $m = 0$. Given an exact sequence $0 \to M_m \to P_m \to \cdots \to P_0 \to A \to 0$, consider the two exact sequences

$$0 \to P_m/M_m \to P_{m-1} \to \cdots P_0 \to A \to 0, \qquad 0 \to M_m \to P_m \to P_m/M_m \to 0$$

then inductively we have $L_iF(A) \cong L_{i-m}F(P_m/M_m) \cong L_{i-m-1}F(M_m)$ (the second isomorphism due to our base case). ■

In particular suppose we have an F-acyclic resolution

$$\cdots P_{m+1} \xrightarrow{f_{m+1}} P_m \xrightarrow{f_m} \cdots \to P_0 \to A \to 0$$

we can turn this into

$$0 \to \operatorname{coker}(f_{m+1}) \to P_{m-1} \cdots \to P_0 \to A \to 0$$

so that $L_m F(A)$ is the kernel of the map $F\operatorname{coker}(f_{m+1}) \to FP_{m-1}$, by our above proposition. Since F is right-exact, it preserves cokernels, and in general the homology group of $A \to B \to C$ is the kernel of $\operatorname{coker}(A \to B) \to C$, as desired.

2.4.8 Proposition

If $P \to A$ is an F-acyclic resolution of A, then $L_m F(A) \cong H_m(FP)$.

2.4.9 Theorem

If $\mathcal{A}$ has enough projectives, the left derived functors of any right exact functor $F\colon \mathcal{A} \to \mathcal{B}$ form a universal homological δ-functor.

Proof

Suppose we have another homological δ-functor T_* and a natural map $\phi_0\colon T_0 \to F \cong L_0 F$. We need to show that ϕ_0 can be uniquely extended to a map $\phi_\bullet\colon T_\bullet \to L_\bullet F$. We construct ϕ_n inductively: suppose ϕ_i is constructed for $0 \le i < n$, such that they commute with the appropriate δs.

Given $A \in \mathcal{A}$, choose an exact sequence $0 \to K \to P \to A \to 0$ with P projective. Since $L_n F(P) = 0$ we have a commutative diagram with exact rows:

$$\begin{tikzcd}
 & T_n A \arrow[r, "\delta_n"'] & T_{n-1} K \arrow[d, "\phi_{n-1}"] & T_{n-1} P \arrow[d, "\phi_{n-1}"] \\
0 & L_n F A \arrow[r, "\delta_n"] & L_{n-1} F K & L_{n-1} F P
\end{tikzcd}$$

Since the bottom δ_n is monic, $\delta_n \circ \phi_n = \phi_{n-1} \circ \delta_n$ has a unique solution (explicitly, if a solution exists it is unique, we can check that such a solution exists in Mod_R via diagram chase). So we have defined ϕ_n on A: we must now show that the resulting map is natural $T_n \to L_n F$. Note that our ϕ_n ostensibly depends on choice of K, P; we will show that this is not the case.

So suppose we have a map $f\colon A' \to A$ and an exact sequence $0 \to K' \to P' \to A' \to 0$ with P' projective. Since P' is projective, we can lift $P' \to A$ (composing $P' \to A'$ with f) to $g\colon P' \to P$ (via the epic $P \to A$). This in turn induces a map $h\colon K' \to K$ such that the diagram commutes

$$\begin{tikzcd}
0 \arrow[r] & K' \arrow[r] \arrow[d, "h"] & P' \arrow[r] \arrow[d, "g"] & A' \arrow[r] \arrow[d, "f"] & 0 \\
0 \arrow[r] & K \arrow[r] & P \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

We now show that ϕ_n commutes with f. Note that in the following diagram, each small quadrilateral commutes:

$$\begin{tikzcd}
T_n(A') \arrow[rrr, "T_n f"'] \arrow[dr, "\delta"] \arrow[ddd, "\phi_n(A')"'] & & & T_n(A) \arrow[dl, "\delta"'] \arrow[ddd, "\phi_n(A)"] \\
& T_{n-1}(K') \arrow[r, "T_n h"'] \arrow[d, "\phi_{n-1}"] & T_{n-1}(K) \arrow[d, "\phi_{n-1}"] & \\
& L_{n-1}F(K') \arrow[r, "L_n F h"] & L_{n-1}F(K) & \\
L_nF(A') \arrow[ur, "\delta"] \arrow[rrr, "L_n F f"'] & & & L_nF(A) \arrow[ul, "\delta"']
\end{tikzcd}$$

And so we have

$$\delta \circ L_n(f) \circ \phi_n(A') = \delta \circ \phi_n(A) \circ T_n(f)$$

both of these δs are $L_nF(A) \to L_{n-1}F(K)$ which is monic, and thus we can cancel it from both sides. Thus we have that ϕ_n is indeed natural. Note that choosing $A = A'$ and $f = \mathrm{id}_A$ we see that $\phi_n(A)$ is independent on choice of short exact sequence.

We also need to show that ϕ_n commutes with δ_n. Given a short exact sequence $0 \to A' \to A \to A'' \to 0$, choose a short exact sequence $0 \to K'' \to P'' \to A'' \to 0$ with P'' projective. Then we can construct f, g making the following diagram commute

$$\begin{tikzcd}
0 \arrow[r] & K'' \arrow[r] \arrow[d, "g"] & P'' \arrow[r] \arrow[d, "f"] & A'' \arrow[r] \arrow[d, "="] & 0 \\
0 \arrow[r] & A' \arrow[r] & A \arrow[r] & A'' \arrow[r] & 0
\end{tikzcd}$$

This gives us a commutative diagram

$$\begin{tikzcd}
T_n(A'') \arrow[r, "\delta"'] \arrow[d, "\phi_n"] & T_{n-1}(K'') \arrow[r, "Tg"'] \arrow[d, "\phi_{n-1}"] & T_{n-1}(A') \arrow[d, "\phi_{n-1}"] \\
L_nF(A'') \arrow[r, "\delta"] & L_{n-1}F(K'') \arrow[r, "LF(g)"] & L_{n-1}F(A')
\end{tikzcd}$$

the horizontal composites are the δs of $0 \to A' \to A \to A'' \to 0$ (by the δ-functorality of $T_\bullet$, $L_\bullet F$), giving us the desired commutativity of ϕ_n with δ. ■

2.4.10 Definition

An additive functor $F\colon \mathcal{A} \to \mathcal{B}$ is **effaceable** if for every $A \in \mathcal{A}$ there is a monomorphism $u\colon A \to I$ such that $F(u) = 0$. F is **coeffaceable** if for every A there is an epimorphism $u\colon P \to A$ such that $F(u) = 0$.

A δ-functor is (co)effaceable if every component (other than 0) is (co)effaceable.

2.4.11 Theorem

A homological coeffaceable δ-functor is universal. Dually a cohomological effaceable δ-functor is universal.

Proof

In the above proof, instead of choosing P projective choose P with $P \to A$ epic whose image is 0. The proof then proceeds the same. ■

Note that left derived functor are coeffaceable (choosing $P \to A$ with P projective), so this is a generalization of the above theorem.

Furthermore, homology $H_\bullet \colon \mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathcal{A}$ is coeffaceable. Indeed, recall that a map $f \colon B \to C$ induces a short exact sequence $0 \to C \to \operatorname{cone}(f) \to B[-1] \to 0$. In particular for a chain complex A we have an epic $\sigma_{\geq 0}\operatorname{cone}(A)[1] \to A$ which maps $(a, a') \mapsto -a$. Since id_A is a quasi-isomorphism we have that $\operatorname{cone}(A)$ is acyclic and in particular the induced homology map of this epimorphism is zero (except at the zero index, which is unimportant). So we have shown

2.4.12 Theorem

Homology and cohomology are universal δ-functors $\mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathcal{A}$ and $\mathsf{Ch}^{\geq 0}(\mathcal{A}) \to \mathcal{A}$ respectively.

2.5 Right derived functors

2.5.1 Definition

Let $F \colon \mathcal{A} \to \mathcal{B}$ be a left exact functor between additive categories. If $\mathcal{A}$ has enough injectives, we define F's **right derived functors** $R^i F \colon \mathcal{A} \to \mathcal{B}$ as follows. For $A \in \mathcal{A}$, choose an injective resolution $A \to I$ and define

$$R^i F(A) = H^i(F(I))$$

Note that since $0 \to F(A) \to F(I^0) \to F(I^1)$ is exact, we have that $R^0 F \cong F$.

Furthermore, $F \colon \mathcal{A} \to \mathcal{B}$ defines a functor $F^{\mathsf{op}} \colon \mathcal{A}^{\mathsf{op}} \to \mathcal{B}^{\mathsf{op}}$ and $\mathcal{A}^{\mathsf{op}}$ has enough projectives so we have left derived functors $L_i F^{\mathsf{op}}$. Since I is a projective resolution of A in $\mathcal{A}^{\mathsf{op}}$ we clearly have that

$$R^i F = (L_i F^{\mathsf{op}})^{\mathsf{op}}$$

thus all results about left derived functors hold dually for right derived functors. In particular $R^i F$'s construction is independent on choice of injective resolution, $R^\bullet F$ is a universal cohomological δ-functor, and $R^i F(I) = 0$ for injective I when $i \neq 0$.

2.5.2 Definition

For an R-module A, the functor $\hom_R(A, \bullet)$ is left-exact. Thus it has right derived functors, called **Ext groups**:

$$\operatorname{Ext}^i_R(A, \bullet) = R^i \hom_R(A, \bullet)$$

In particular we have $\operatorname{Ext}^0_R(A, B) = \hom(A, B)$.

2.5.3 Proposition

The following are equivalent:

(1) B is injective,

(2) $\hom_R(\bullet, B)$ is exact,

(3) $\operatorname{Ext}^i_R(A, B)$ vanishes for all $i \neq 0$ and all A,

(4) $\mathrm{Ext}^1_R(A, B)$ vanishes for all A.

Proof

As already explained (1) and (2) are equivalent, and these imply the rest. Now if $\mathrm{Ext}^1_R(\bullet, B) = 0$ then consider a short exact sequence $0 \to C \to D \to E \to 0$ then we have a long exact sequence

$$0 \to \hom(C, B) \to \hom(D, B) \to \hom(E, B) \to \mathrm{Ext}^1(C, B) = 0$$

thus $\hom(\bullet, B)$ is exact as desired. ■

Dually we have

2.5.4 Proposition

The following are equivalent:

(1) A is projective,

(2) $\hom_R(A, \bullet)$ is exact,

(3) $\mathrm{Ext}^i_R(A, B)$ vanishes for all $i \neq 0$ and all B,

(4) $\mathrm{Ext}^1(A, B)$ vanishes for all B.

We can consider derived functors of contravariant functors. For example, left $F\colon \mathcal{A}^{\mathsf{op}} \to \mathcal{B}$ be left exact, then if $\mathcal{A}$ has enough projectives ($\mathcal{A}^{\mathsf{op}}$ has enough injectives), F has right derived functors via cohomology of $F(P)$ for $P \to A$ a projective resolution.

In particular we have left exact $\hom_R(\bullet, B)$ and thus right derived functors $R^i \hom_R(\bullet, B)$. We will show later that these are not new:

$$R^\bullet \hom_R(\bullet, B)(A) \cong R^\bullet \hom_R(A, \bullet)(B) = \mathrm{Ext}^\bullet_R(A, B)$$

2.5.5 Example

Consider the **global sections** functor $\Gamma\colon \mathsf{sh}(X) \to \mathrm{Ab}$ via $\Gamma(\mathcal{F}) = \mathcal{F}(X)$. It can be shown that this is right adjoint to the constant sheaves functor, and in particular is left exact. Thus Γ has right derived functors, called the **cohomology functors** on X:

$$H^i(X, \mathcal{F}) = R^i\Gamma(\mathcal{F})$$

2.6 Adjointness and exactness

Note that we can characterize (left/right) exactness purely categorically:

2.6.1 Lemma

A functor $F\colon \mathcal{A} \to \mathcal{B}$ is left exact iff it preserves finite limits, right exact iff it preserves finite colimits, and exact iff it preserves finite limits and colimits.

Proof

It is a classic result that a functor is additive iff it preserves finite biproducts, and note that an exact diagram $0 \to A \to B \to C$ is simply a kernel diagram. So preserving diagrams of this form is the same as preserving kernels, and thus a functor is left exact iff it preserves finite limits (finite biproducts and kernels). Similarly exact $A \to B \to C \to 0$ are cokernel diagrams. ■

This result allows us to generalize the concept of exactness to arbitrary categories, but this is not necessary here. More importantly (for us) it gives us the following.

2.6.2 Corollary

Left adjoints are right exact, right adjoints are left exact.

Proof

Left adjoints preserve (arbitrary) colimits, and right adjoints preserve (arbitrary) limits. ■

In particular (assuming the existence of enough projectives/injectives) left adjoints have left derived functors and right adjoints have right derived functors.

Another interesting result is the following:

2.6.3 Lemma

The following are equivalent for an exact functor $F\colon \mathcal{A} \to \mathcal{B}$:

(1) F reflects exact sequences: if $FA \to FB \to FC$ is exact, so is $A \to B \to C$.

(2) F is faithful.

(3) F reflects zero morphisms: if $Ff = 0$ then $f = 0$.

(4) F reflects zero objects: if $FA = 0$ then $A = 0$.

Proof

Clearly F reflects zero morphisms iff it is faithful as $Ff = Fg$ iff $F(f - g) = 0$ iff $f = g$. If F reflects zero morphisms then suppose $FA = 0$; then $F\mathrm{id}_A = 0$ and so $\mathrm{id}_A = 0$, meaning $A = 0$.

If F reflects zero objects, suppose $f\colon A \to B$ is non-zero: we want to show Ff is non-zero as well. Consider the exact sequence

$$0 \to \ker f \to A \xrightarrow{f} B \to \operatorname{coker} f \to 0$$

which maps to the exact sequence

$$0 \to F\ker f \to FA \xrightarrow{Ff} FB \to F\operatorname{coker} f \to 0$$

0 is the unique map whose kernel and cokernel are isomorphisms, so we want to show that the induced maps $F\ker f \to FA, FB \to F\operatorname{coker} f$ are not isomorphisms. But indeed, $\operatorname{coker}(F\ker f \to FA) \cong F\operatorname{coker}(\ker f \to A)$ is non-zero since $\ker f \to A$ is non-zero; so $F\ker f \to FA$ is not an isomorphism. Similarly $\ker(FB \to F\operatorname{coker} f) \cong F\ker(B \to \operatorname{coker} f)$ is non-zero and thus $FB \to F\operatorname{coker} f$ is not an isomorphism. So Ff is non-zero as well, meaning F reflects zero morphisms if it reflects zero objects.

Thus we have shown the equivalence of (2), (3), (4). If F reflects exact sequences then it reflects zero objects, as if $FA = 0$ then $0 \to FA \to 0$ is exact and so $0 \to A \to 0$ is as well.

And if F is faithful then it must reflect exact sequences: it is sufficient to show that F reflects kernels and cokernels. Let $A \xrightarrow{f} B \xrightarrow{g} C$ such that $Ff = \ker Fg$, and let $h\colon K \to B$ be a kernel of g. Since $F(gf) = 0$ and F is faithful, $gf = 0$, so there is a unique $i\colon K \to A$ such that $f = hi$; we want to show i is an isomorphism. Similarly since $F(gh) = 0$ there is a unique $j\colon FA \to FK$ such that $Fh = (Ff)j$. Then we have that

$$F(f)jF(i) = F(h)F(i) = F(f)$$

and since $F(f)$ is monic we have $jF(i) = \mathrm{id}$, in particular $F(i)$ is monic. Faithfulness reflects monomorphisms, so i is monic as well.

Let $p\colon K \to K/A$ be a cokernel of i, and $q\colon B \to B/A$ a cokernel of f. Now $qhi = qf = 0$, which means there exists a unique $h'\colon K/A \to B/A$ such that $qh = h'p$. Since h is monic, so is h'. And note

$$F(h'p) = F(qh) = F(q)F(f)j = 0$$

So $h'p = 0$ and thus $p = 0$ since h' is monic. But $p = \mathrm{coker} i$, thus i is an epic monic, thus an isomorphism as desired. ■

Recall that in particular we have the tensor-hom adjunction $\bullet \otimes_R B\colon \mathrm{Mod}_R \rightleftarrows \mathrm{Ab}\colon \hom_{\mathrm{Ab}}(B, \bullet)$ for all $B \in {}_R\mathrm{Mod}$. If B is an (R, S)-bimodule then this adjunction lands in Mod_S and not just Ab. So $\bullet \otimes_R B$ is left adjoint and thus right exact.

2.6.4 Definition

For an (R, S)-bimodule B, define its **Tor modules** to be the right S-modules

$$\mathrm{Tor}_n^R(A, B) = (L_n(\bullet \otimes_R B))(A)$$

for $A \in \mathrm{Mod}_R$.

If $S' \subseteq S$ is a subring then for an (R, S)-bimodule B we can also view it as an (R, S')-bimodule. Thus $\mathrm{Tor}_n^R(\bullet, B)$ can be viewed to land in either Mod_S or $\mathrm{Mod}_{S'}$. But the forgetful functor $U\colon \mathrm{Mod}_S \to \mathrm{Mod}_{S'}$ is exact, and thus

$$U(L_\bullet(\otimes_R B)) \cong L_\bullet(U \otimes_R B)$$

thus the underlying S'-module is the same in either case.

Note that for a right R-module A, $A \otimes_R \bullet$ is also right-exact (due to a similar adjunction), and thus we can construct its left derived functors as well. We will show soon that $L_\bullet(A \otimes_R \bullet)(B) \cong L_\bullet(\bullet \otimes_R B)(A)$, so this yields nothing new.

2.6.5 Example

Consider the inclusion functor $I\colon \mathsf{sh}(X) \to \mathsf{Psh}(X)$. It has a left adjoint called **sheafification**, and is thus left exact and has right derived functors. Further consider the exact "section" functors $\mathsf{Psh}(X) \to \mathrm{Ab}$ mapping a presheaf $\mathcal{F}$ to $\mathcal{F}(U)$, call this Γ_U. Then we have

$$\Gamma_U R^n(I) \cong R^n(\Gamma_U \circ I)$$

note that Γ_U is just the section functor restricted to sheaves. $\Gamma_U \circ I$ just maps a sheaf $\mathcal{F}$ to $\Gamma\mathcal{F}|_U$ (the global section of the restriction of $\mathcal{F}$ to U), and thus by definition

$$\Gamma_U R^n(I) \cong H^n(U, \mathcal{F}|_U)$$

meaning that $R^n(I)$ maps a sheaf $\mathcal{F}$ to the presheaf $U \mapsto H^n(U, \mathcal{F}|_U)$.

2.6.6 Example

Let $f\colon X \to Y$ be continuous, then define the **direct image** functor $f_*\colon \mathsf{Psh}(X) \to \mathsf{Psh}(Y)$ by $(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}V)$ for V open in Y. For $V' \subseteq V$, we have $f^{-1}V' \subseteq f^{-1}V$ so this defines a presheaf: map $s \in f_*\mathcal{F}(V)$ to $s|_{f^{-1}V'} \in f_*\mathcal{F}(V')$. Note that if $\mathcal{F} \in \mathsf{sh}(X)$ is a sheaf, then consider the exact diagram for an open cover $\{V_i\}$ of $V \subseteq Y$. $\{f^{-1}V_i\}$ is an open cover of $f^{-1}V$ and so we have an exact diagram

$$0 \to \mathcal{F}(f^{-1}V) \to \prod_i \mathcal{F}(f^{-1}V_i) \to \prod_{i,j} \mathcal{F}(f^{-1}V_i \cap f^{-1}V_j)$$

The maps are precisely the maps in the diagram for $f_*\mathcal{F}$, thus $f_*\mathcal{F}$ is a sheaf. So f_* restricts to a map $\mathsf{sh}(X) \to \mathsf{sh}(Y)$.

We also define the **inverse image** functor $f^{-1}\colon \mathsf{sh}(Y) \to \mathsf{sh}(X)$ to map $\mathcal{G} \in \mathsf{sh}(Y)$ to the sheafification of the presheaf mapping U to the direct colimit $\operatorname{colim} \mathcal{G}(V)$ over the poset of all $V \subseteq Y$ containing $f(U)$. Then we have that $f^{-1} \dashv f_*$ form an adjunction.

Now, let $*$ be the one-point topological space. Then we have $\mathsf{sh}(*) \cong \mathrm{Ab}$. Consider the unique map $f\colon X \to *$. The direct image functor $f_*\colon \mathsf{sh}(X) \to \mathrm{Ab}$ maps $\mathcal{F} \in \mathsf{sh}(X)$ to $\mathcal{F}(f^{-1}*) = \mathcal{F}(X)$, i.e. f_* is the global section functor. The inverse image functor maps $A \in \mathrm{Ab}$ to the sheafification of $\mathcal{F}(U) = A$, i.e. the constant sheaf. Thus we have that $\mathrm{const} \dashv \Gamma$: the constant sheaf functor is a left adjoint to the global section functor. This shows us that Γ is indeed left exact and so sheaf cohomology is well-defined.

For $x \in X$ consider the map $x\colon * \to X$ mapping the point to $x \in X$. Then $x_*\colon \mathrm{Ab} \to \mathsf{sh}(X)$ maps an Abelian group A to the sheaf defined by $\mathcal{F}(U) = A$ if $x \in U$ (so $x^{-1}U = *$) and 0 otherwise (so $x^{-1}U = \emptyset$), i.e. the skyscraper sheaf. And $x^{-1}\colon \mathsf{sh}(X) \to \mathrm{Ab}$ maps $\mathcal{F} \in \mathsf{sh}(X)$ to the direct colimit $\operatorname{colim}_{x \in U} \mathcal{F}(U)$, i.e. the stalk of $\mathcal{F}$ at x. So we have shown that $\bullet_x \dashv x_*\bullet$.

2.6.7 Example

For a small category I, we have the **diagonal functor** $\Delta_I\colon \mathcal{A} \to \mathcal{A}^I$ which maps $A \in \mathcal{A}$ to the **constant functor** ΔA mapping $i \in I$ to A, and $i \to j$ to id_A. A classical result is $\operatorname{colim}_I \dashv \Delta_I \dashv \lim_I$ when all co/limits of shape I exist. This is pretty immediate: $\lim_I F$ is an object $A \in \mathcal{A}$ along with a natural map (the limit cone) $\lambda\colon \Delta A \to F$ which is universal in a particular way (terminal in the category of cones). So then for $A \in \mathcal{A}$ and functor F, a map $\Delta A \to F$ defines a unique map $A \to \lim F$ commuting with the limit cone (by terminal-ness), and vice versa.

In particular colim is right exact, lim left exact, and Δ exact. lim, colim are not necessarily exact though.

Recall that all limits can be computed via products and equalizers. In Abelian categories this means that all limits can be computed via products and kernels. Similarly colimits can be computed via coproducts (direct sums) and cokernels. Thus we have the following.

2.6.8 Theorem

An Abelian category is cocomplete (all colimits exist) iff all direct sums exist, and is complete (all limits exist) iff all products exist.

Note that if $\mathcal{D}$ is a full subcategory of a cocomplete category $\mathcal{C}$ such that the inclusion $i\colon \mathcal{D} \to \mathcal{C}$ has a left adjoint ($\mathcal{D}$ is a **reflective subcategory**), $\mathcal{D}$ is also cocomplete. Indeed, let $s\colon \mathcal{C} \to \mathcal{D}$ be the left adjoint then the composition $\mathcal{D} \xrightarrow{i} \mathcal{C} \xrightarrow{\Delta} \mathcal{C}^I$ is just the diagonal functor on $\mathcal{D}$, and since we can compose adjunctions, it has a left adjoint $s \circ \operatorname{colim}$. So $s \circ \operatorname{colim}$ is the colimit functor of $\mathcal{D}$, meaning $\mathcal{D}$ is cocomplete. We can also show that if $\mathcal{C}$ is complete, so is $\mathcal{D}$.

In particular it is simple to show that $\mathsf{Psh}(X)$ is co/complete, and since $\mathsf{sh}(X)$ is a reflective subcategory, it too is co/complete. Thus we have the following co/complete categories:

$$\mathrm{Mod}_R, \qquad \mathsf{Psh}(X), \qquad \mathsf{sh}(X)$$

2.6.9 Definition

An Abelian category $\mathcal{A}$ satisfies axiom **(AB4)** if it is cocomplete and direct sums of monics are monic.

The direct sum of maps $f_i\colon A_i \to B_i$ is the induced map $\oplus_i f_i\colon \oplus_i A_i \to \oplus_i B_i$ defined by composing f_i with the inclusion $B_i \to \oplus_i B_i$, and then utilizing the universal property of direct sums. That is, $\oplus_i f_i$ is the unique map where composing with inclusion $A_i \to \oplus A_i$ is equal to f_i composed with inclusion $B_i \to \oplus B_i$.

Axiom (AB4) is equivalent to saying that direct sums are exact. Since direct sums are colimits, they are automatically right exact. The direct sums of monics being monic is equivalent to saying that direct sums preserve monomorphisms: direct sums are thus exact.

Since homology commutes with exact functors we have the following.

2.6.10 Theorem

Suppose $\mathcal{A}$ satisfies (AB4) and has enough projectives. Let $F\colon \mathcal{A} \to \mathcal{B}$ be left adjoint, then for any collection $\{A_i\}$ in $\mathcal{A}$, we have

$$L_\bullet F\left(\bigoplus_i A_i\right) \cong \bigoplus_i L_\bullet F(A_i)$$

Proof

The direct sum of projectives is projective ($\oplus P_i \to C$ factors as $P_i \to C$ which then lifts to $P_i \to B$, giving us $\oplus P_i \to B$). Since direct sums are exact, if $P_i \to A_i$ are projective resolutions then so is $\oplus P_i \to A_i$. Then since F is left adjoint and thus commutes with colimits, in particular direct sums, we have

$$L_\bullet F\left(\bigoplus_i A_i\right) = H_\bullet(F(\oplus_i P_i) \cong H_\bullet(\oplus_i F(P_i)) \cong \bigoplus_i H_\bullet(F(P_i)) = \bigoplus_i L_\bullet F(A_i)$$

as desired. ■

We can define the dual axiom of AB4:

2.6.11 Definition

An Abelian category $\mathcal{A}$ satisfies axiom **(AB4*)** if the product of epimorphisms is epic.

$\mathcal{A}$ satisfies (AB4*) iff $\mathcal{A}^{\mathsf{op}}$ satisfies (AB4) clearly. Thus by duality we have:

2.6.12 Corollary

Suppose $\mathcal{A}$ satisfies (AB4*) and has enough injectives. Let $F\colon \mathcal{A} \to \mathcal{B}$ be right adjoint, then for any $\{A_i\}$ in $\mathcal{A}$, we have

$$R_\bullet F\left(\prod_i A_i\right) \cong \prod_i R_\bullet F(A_i)$$

It is not hard to see that Mod_R satisfies both (AB4) and (AB4*), thus we have

2.6.13 Corollary

In Mod_R:

$$\mathrm{Tor}_\bullet\left(A, \bigoplus_i B_i\right) \cong \bigoplus_i \mathrm{Tor}_\bullet(A, B_i), \qquad \mathrm{Ext}_\bullet\left(A, \prod_i B_i\right) \cong \prod_i \mathrm{Ext}_\bullet(A, B_i)$$

2.6.14 Definition

A nonempty category I is **filtered** if

(1) for every $i, j \in I$, there is some $k \in I$ with arrows $i \to k \leftarrow j$.

(2) for every two parallel arrows $u, v\colon i \rightrightarrows j$ there is an arrow $w\colon j \to k$ such that $wu = wv$.

A **filtered colimit** is the colimit of a functor $I \to \mathcal{A}$ where I is filtered. A filtered colimit is denoted by $\varinjlim$.

We can dualize this to **cofiltered** categories and **cofiltered limits**. Note that posets always satisfy (2) (since in such a case $u = v$), so a filtered poset is just one where every two elements can be bound. Filtered posets are called **directed**. Filtered colimits from directed posets are often called directed limits, but for clarity I will call them directed *colimits*.

2.6.15 Lemma

Let I be filtered, and $A\colon I \to \mathrm{Mod}_R$ a functor.

(1) Every element $a \in \varinjlim(A_i)$ is the image of some element $a_i \in A_i$ under the canonical map $A_i \to \varinjlim A_i$.

(2) For every $i \in I$, the kernel of the canonical map $A_i \to \varinjlim A_i$ is the union of kernels of the maps $\phi\colon A_i \to A_j$ (where $\phi\colon i \to j$).

I explain quickly what we mean by "canonical map". Recall that a colimit is given by a cone $\mu\colon A \to \Delta(\operatorname{colim} A)$, and thus we have maps $\mu_i\colon A_i \to \operatorname{colim} A$; these are the canonical maps.

Proof

Recall that in general the colimit of A is given by the coequalizer diagram

$$\bigoplus_{\phi: i \to j} A_i \overset{f}{\underset{g}{\rightrightarrows}} \bigoplus_{i \in I} A_i \to \varinjlim A$$

where $f = \bigoplus_\phi f_\phi$ and $g = \bigoplus_\phi g_\phi$ for $f_\phi, g_\phi\colon A_i \to \bigoplus_i A_i$ by setting f_ϕ to be the inclusion of A_i, and g_ϕ to be the composition of the inclusion of A_j with ϕ. I.e.

$$\varinjlim A = \operatorname{coker}\left(\bigoplus_{\phi: i \to j} \lambda_i - \lambda_j \phi\right)$$

where $\lambda_i\colon A_i \to \bigoplus_i A_i$ is inclusion. The canonical maps $A_i \to \varinjlim A_i$ are the compositions of λ_i with the coequalizer arrow.

Thus $\varinjlim A$ is the quotient of $\bigoplus_i A_i$ by the image of $\bigoplus_{\phi: i \to j} \lambda_i - \lambda_j \phi$. Elements of $\bigoplus_i A_i$ are of the form $\sum_{j \in J} \lambda_j(a_j)$ for a finite set $J \subseteq I$. Since I is filtered, J has a bound, let it be $J \leq i$. Let $\phi_i\colon j \to i$ for $j \in J$ then we have that $\lambda_j = \lambda_i \phi$ in the filtered colimit (we quotient by these direct sums), so we have

$$\sum_{j \in J} \lambda_j(a_j) = \sum_{j \in J} \lambda_i(\phi a_j) = \lambda_i\left(\sum_j \phi a_j\right)$$

proving our first point.

If $a_i \in A_i$ vanishes in $\varinjlim A_i$ then this means that $\lambda_i(a_i) = \sum_{j,k} \lambda_j(a_j) - \lambda_k(\phi_{jk}a_j)$ for finitely many j, k, $\phi_{jk}\colon j \to k$, and $a_j \in A_j$. Let t be an upper bound for all these i, j, k, then we have

$$\lambda_t(\phi_{it}a_i) = \sum_{j,k} \lambda_t(\phi_{jt}a_j) - \lambda_t(\phi_{kt}\phi_{jk}a_j)$$

for $\phi_{it}\colon i \to t$. We can assume that $\phi_{kt}\phi_{jk} = \phi_{jt}$, potentially by increasing t (due to the second condition of filtered categories). Thus we have

$$\lambda_t(\phi_{it}a_i) = \sum_{j,k} \lambda_t(\phi_{jt}a_j) - \lambda_t(\phi_{jt}a_j) = 0$$

since λ_t is injective, this means that $a_i \in \ker\phi_{it}$ as desired. The converse (if an element is in the union of kernels, it vanishes) is because $\lambda_i(a_i) = \lambda_j(\phi_{ij}a_i) = 0$. ■

This lemma shows that we can define a surjection $\bigsqcup A_i \to \varinjlim A_i$, by mapping $a_i \in A_i$ to $\lambda_i(a_i)$. Then a_i, a_j define the same element in the filtered colimit iff $a_i - a_j$ is in the kernel of one of the maps. Bounding i, j by k, this means that $a_i - a_j$ is in the kernel of some $\phi\colon A_k \to A_{k'}$, i.e. $\phi(a_i) = \phi(a_j)$. That is,

$$\varinjlim A_i \cong \bigsqcup A_i \Big/ \sim, \qquad a_i \sim a_j \text{ iff there exists some } i, j \to k \text{ such that } \phi_{ik}(a_i) = \phi_{jk}(a_j)$$

2.6.16 Theorem

Filtered colimits (in particular directed colimits) of R-modules are exact, as functors $\mathrm{Mod}_R^I \to \mathrm{Mod}_R$.

Proof

Let $\mathcal{A} = \mathrm{Mod}_R$, and I be a filtered category. Colimits are in general right exact, and so $\varinjlim_I\colon \mathcal{A}^I \to \mathcal{A}$ is right exact. So we need only show that $\varinjlim_I$ preserves monics. So suppose $f\colon A \to B$ is monic in $\mathcal{A}^I$ (when $\mathcal{A} = \mathrm{Mod}_R$ this just means that every component of f is monic). We want to show that the induced map $f_*\colon \varinjlim A_i \to \varinjlim B_i$ is monic as well. So suppose $a \in \varinjlim A_i$ vanishes in $\varinjlim B_i$, by the above lemma a is the image of some $a_i \in A_i$. Thus $f_i(a_i) \in B_i$ vanishes in $\varinjlim B_i$, meaning there is some $\phi\colon i \to j$ such that $\phi(f_i(a_i)) = f_j(\phi(a_i)) = 0$ in B_j. Since f_j is monic, $\phi(a_i) = 0$, so a_i vanishes in $\varinjlim A_i$, i.e. $a = 0$. ■

This leads us to the following definition:

2.6.17 Definition

An Abelian category $\mathcal{A}$ satisfies the axiom **(AB5)** if it is cocomplete and filtered colimits are exact. Dually it satisfies **(AB5*)** if it is complete and cofiltered limits are exact.

Mod_R is not (AB5*): $\mathrm{Mod}_R^{\mathsf{op}}$ is not (AB5).

2.6.18 Example

Recall that for continuous $f\colon X \to Y$, the inverse image functor $f^{-1}\colon \mathsf{sh}(Y) \to \mathsf{sh}(X)$ is given by the sheafification of $f^{-1}\mathcal{G}(U) = \mathrm{colim}_{f(U)\subseteq V}\, \mathcal{G}(V)$. Sheafification is exact, so we need only show that $f^{-1}\colon \mathsf{Psh}(Y) \to \mathsf{Psh}(X)$ is exact to show that the inverse image functor on sheaves is.

A sequence of presheaves (not sheaves) is exact iff it is exact at every open set. Note that the set of open sets V containing $f(U)$ is directed, and thus $f^{-1}\mathcal{G}(U)$ is a filtered colimit, and thus exact. That

is to say if $\mathcal{G} \to \mathcal{G}' \to \mathcal{G}''$ is exact, then these define an exact sequence of functors on the directed set of open subsets containing $f(U)$. Then taking their filtered colimit over this directed set gives us that $f^{-1}\mathcal{G}(U) \to f^{-1}\mathcal{G}'(U) \to f^{-1}\mathcal{G}''(U)$ is exact. This is true for all U, and thus f^{-1} is exact, as desired.

Since filtered colimits are exact, they commute with homology. And left adjoints commute with colimits in general, so we have the following.

2.6.19 Theorem

Suppose $\mathcal{A}, \mathcal{B}$ satisfy (AB5), $\mathcal{A}$ has enough projectives. If $F\colon \mathcal{A} \to \mathcal{B}$ is left adjoint such that filtered limits of projectives in $\mathcal{A}$ are F-acyclic, then for every $A\colon I \to \mathcal{A}$ with I filtered,

$$L_\bullet F(\varinjlim A_i) \cong \varinjlim L_\bullet F(A_i)$$

Note that the filtered limit of projective modules are $\bullet \otimes A$-acylic for every R-module A. Indeed, $\bullet \otimes (\varinjlim P_i)$ is exact, as it is isomorphic to $\varinjlim(\bullet \otimes P_i)$ which is the filtered colimit of exact functors, and is thus exact by (AB5). This gives us the result (after balancing Tor, i.e. showing that it can be viewed as a bifunctor). Thus we have the following corollary:

2.6.20 Corollary

For every filtered $B\colon I \to \mathrm{Mod}_R$ and $A \in {}_R\mathrm{Mod}$,

$$\mathrm{Tor}_\bullet(A, \varinjlim B_i) \cong \varinjlim \mathrm{Tor}_\bullet(A, B_i)$$

2.7 Balancing Tor and Ext

Recall that since tensor products and hom-sets are bifunctors, which are (left/right) exact in both their coordinates, they both provide two derived functors. I.e. $L_\bullet(\bullet \otimes_R B), L_\bullet(A \otimes_R \bullet)$ for tensor products. We claimed that $L_\bullet(\bullet \otimes_R B)(A) \cong L_\bullet(A \otimes_R \bullet)(B)$ (similar for Ext); we show this now.

2.7.1 Definition

Let P, Q be chain complexes of right and left R-modules, respectively. Form the bicomplex $P \otimes_R Q$ whose components are $P_p \otimes_R Q_q$, with horizontal differential $d_P \otimes 1$, and vertical differential $(-1)^p \otimes d_Q$. This is called the **tensor product bicomplex**, and $\mathrm{Tot}^\oplus(P \otimes_R Q)$ is the **tensor product chain complex** of P and Q.

This is indeed a bicomplex: $(d \otimes 1) \circ (1 \otimes d)$ maps $p \otimes q$ to $dp \otimes dq$, and thus these differentials form a positive bicomplex, we use the sign trick to give us a signed bicomplex.

If P is a chain complex of right R-modules and A is a left R-module, then we can view A as a chain complex concentrated at 0, thus defining $P \otimes A$. Note then that $\mathrm{Tot}^\oplus(P \otimes_R A)$ has coefficient at index n, $P_n \otimes_R A$, since all other coefficients in A are 0. The differential is $d_P \otimes 1 + (-1)^p \otimes d_A = d_P \otimes 1$.

If we have chain maps $f\colon P \to A$ and $g\colon Q \to B$, these define chain (bicomplex) maps $f \otimes g\colon P \otimes Q \to A \otimes B$ whose components are $f_p \otimes g_q$.

2.7.2 Theorem

$L_n(A \otimes_R \bullet)(B) \cong L_n(\bullet \otimes_R B)(A) = \mathrm{Tor}_n^R(A, B)$ for all A, B and n.

Proof

Choose projective resolutions $P \xrightarrow{\epsilon} A$ in Mod_R and $Q \xrightarrow{\eta} B$ in ${}_R\mathrm{Mod}$. Consider A, B as complexes concentrated at degree 0, then we have three bicomplexes: $P \otimes Q$, $P \otimes B$, and $A \otimes Q$. The augmentations ϵ and η define maps from $P \otimes Q$ to $A \otimes Q$ and $P \otimes B$ respectively. The **acyclic assembly lemma** (below) will help us show that the maps

$$A \otimes Q = \mathrm{Tot}(A \otimes Q) \xleftarrow{\epsilon \otimes Q} \mathrm{Tot}(P \otimes Q) \xrightarrow{P \otimes \eta} \mathrm{Tot}(P \otimes B) = P \otimes B$$

are quasi-isomorphisms, inducing the desired isomorphisms on homology:

$$L_\bullet(A \otimes_R)(B) \xleftarrow{\cong} H_\bullet(\mathrm{Tot}(P \otimes Q)) \xrightarrow{\cong} L_\bullet(\otimes_R B)(A)$$

(Recall that $L_\bullet(A \otimes_R)(B)$ is computed by taking a projective resolution $Q \to B$ and then computing the homology of $A \otimes Q$.)

Consider the bicomplex C obtained by adding $A \otimes Q$ in the column $p = -1$ of $P \otimes Q$, so that $C_{-1,q} = A \otimes Q_q$, and the horizontal differential $C_{0,q} = P_0 \otimes Q_q \to C_{-1,q} = A \otimes Q_q$ is given by $\epsilon \otimes 1$. Now consider $\mathrm{Tot}(C)$: its nth coefficient is $(A \otimes Q_{n+1}) \oplus \mathrm{Tot}(P \otimes Q)_n$. Its differentials are the direct sum of $\epsilon \otimes 1 - 1 \otimes d_Q$ with the differential of $\mathrm{Tot}(P \otimes Q)$, d^{Tot}. So $\mathrm{Tot}(C)[-1]$ has its nth coefficient $(A \otimes Q_n) \oplus \mathrm{Tot}(P \otimes Q)_{n-1}$ with differential $-d^{\mathrm{Tot}} \oplus (1 \otimes d_Q - \epsilon \otimes 1)$.

Consider the mapping cone of $\epsilon \otimes 1 \colon \mathrm{Tot}(P \otimes Q) \to A \otimes Q$. Its nth component is $\mathrm{Tot}(P \otimes Q)_{n-1} \oplus (A \otimes Q_n)$, and its differential is $-d^{\mathrm{Tot}} \oplus (1 \otimes d_Q - \epsilon \otimes 1)$. Thus $\mathrm{Tot}(C)$ is isomorphic to the mapping cone of $\epsilon \otimes 1$. So to show that $\epsilon \otimes 1$ is a quasi-isomorphism (equivalently its mapping cone is acyclic), it is sufficient to show C is acyclic.

Since $\otimes Q_q$ is an exact functor, the rows of C are exact, and C is an upper half plane bicomplex. Thus the acyclic assembly lemma tells us $\mathrm{Tot}C$ is acyclic, as desired.

Similarly we can add $P \otimes B$ to the row $q = -1$ of $P \otimes Q$ and the resulting bicomplex's total complex is acyclic (by the assembly lemma) and isomorphic to the mapping cone of $P \otimes \eta$. ■

2.7.3 Lemma (Acyclic assembly lemma)

Let C be a bicomplex of R-modules. Then

(1) $\mathrm{Tot}^{\Pi}(C)$ is acylic if either of the following hold:

- **(i)** C is an upper half-plane complex with exact columns, or
- **(ii)** C is a right half-plane complex with exact rows.

(2) $\mathrm{Tot}^{\oplus}(C)$ is acylic if either of the following hold:

- **(i)** C is an upper half-plane complex with exact rows, or
- **(ii)** C is a right half-plane complex with exact columns.

Proof

Consider the transpose of C, $C^\top$. Then $\mathrm{Tot}(C^\top) = \mathrm{Tot}(C)$, and $C^\top$ is right half-plane iff C is upper half-plane so (1) and (2) are equivalent, and (3) and (4) are equivalent.

We claim that (1) implies (4): suppose C is right half-plane with exact columns. Define $\tau_n C$ by truncating the columns of C at level n:

$$(\tau_n C)_{pq} = \begin{cases} C_{pq} & \text{if } q > n \\ \ker(d^v \colon C_{pn} \to C_{p,n-1}) & \text{if } q = n \\ 0 & \text{else} \end{cases}$$

Each τ_n is (up to translation) a first-quadrant bicomplex, and so its total complexes coincide: $\mathrm{Tot}^{\oplus}(\tau_n C) = \mathrm{Tot}^{\Pi}(\tau_n C)$. Furthermore $\tau_n C$ has exact columns (its columns are the truncations of the chain complex $C_{p,\bullet}$ which has the same homology of $C_{p,\bullet}$; zero). Assuming (1), this means that $\mathrm{Tot}(\tau_n C)$ is acylic.

Now $\mathrm{Tot}(\tau_n C)$ are subcomplexes of $\mathrm{Tot}^{\oplus}(C)$, and their union (union of each component) is all of $\mathrm{Tot}^{\oplus}(C)$. Thus every cycle of $\mathrm{Tot}^{\oplus}(C)$ is a cycle in some $\mathrm{Tot}(\tau_n C)$, and thus also a boundary since $\mathrm{Tot}(\tau_n C)$ is acylic. This means that $\mathrm{Tot}^{\oplus}(C)$ is acyclic, as desired.

To show (1), by translating C left and right, it is sufficient to show that $H_0(\mathrm{Tot}(C)) = 0$. Let $c = (\dots, c_{-p,p}, \dots, c_{-1,1}, c_{0,0})_p \in \prod_p C_{-p,p} = \mathrm{Tot}(C)_0$ be a 0-cycle. This means that $d^v(c_{-pp}) + d^h(c_{-p+1,p-1}) = 0$. We find by induction elements $b_{-p,p+1}$ such that $d^v(b_{-p,p+1}) + d^h(b_{-p+1,p}) = c_{-p,p}$. The differential of b is then c, meaning that c is a boundary; and then $H_0(\mathrm{Tot}(C)) = 0$ as desired.

For $p = -1$ choose $b_{1,0} = 0$. And for $p = 0$, since $C_{0,-1} = 0$ we must have that $d^v(c_{00}) = 0$, and since the zeroth column is exact there is a $b_{0,1} \in C_{0,1}$ such that $d^v(b_{0,1}) = c_{00}$. This gives us $d^v(b_{0,1}) + d^h(b_{1,0}) = c_{00}$ as desired.

Inductively, we have

$$d^v(c_{-p,p} - d^h(b_{-p+1,p})) = d^v(c_{-pp}) + d^h d^v(b_{-p+1,p}) = d^v(c_{-pp}) + d^h(c_{-p+1,p-1}) - d^h d^h(b_{-p+2,p-1}) = d^v(c_{-pp}) + d^h(c_{-p+1,p-1}) = 0$$

Since the $-p$th column is exact, $c_{-p,p} - d^h(b_{-p+1,p}$ must be in the image of d^v: there is a $b_{-p,p+1} \in C_{-p,p+1}$ such that

$$d^v(b_{-p,p+1}) = c_{-p,p} - d^h(b_{-p+1,p})$$

as desired. ■

2.7.4 Definition

Let P be a chain complex, and I a cochain complex. We define their **hom bicomplex** to be the cochain bicomplex $\hom(P, I)$ with components $\hom(P, I)_{p,q} = \hom(P_p, I^q)$ and differentials

$$d^h\colon \hom(P_p, I^q) \to \hom(P_{p+1}, I^q),\ f \mapsto f \circ d_P, \qquad d^v\colon \hom(P_p, I^q) \to \hom(P_p, I^{q+1}),\ f \mapsto (-1)^{p+q+1} d_Q \circ f$$

Their **(total) hom cochain complex** is defined to be $\mathrm{Tot}^{\Pi}(\hom(P, I))$.

Suppose we have two chain complexes, C and D. Reindexing D as a cochain complex, then an n-cycle f of the hom cochain complex $\hom(C, D)$ is a collection of maps $f_p\colon C_p \to D^{n-p} = D_{p-n}$ whose differential is zero. The differential is

$$d(f_p) = (f_p d_C\colon C_{p+1} \to D_{p-n}) + ((-1)^{n+1} d_D f_p\colon C_p \to D_{p-n-1})$$

and so is zero when $f_p d = (-1)^n d f_{p+1}$. Equivalently, n-cycles are chain maps $C \to D[-n]$. n-boundaries are null homotopic chain maps. And so the cohomology grousp of the hom cochain complex, $H^n \hom(C, D)$, are the chain homotopy classes of chain maps $C \to D[-n]$, i.e. the hom-set of maps $C \to D[-n]$ in K.

2.7.5 Proposition

There is an internal tensor-hom adjunction, that is a natural isomorphism

$$\mathrm{Tot}^{\Pi} \hom_{\mathrm{Ab}}(\mathrm{Tot}^{\oplus}(P \otimes Q), I) \cong \mathrm{Tot}^{\Pi} \hom_R(P, \mathrm{Tot}^{\Pi} \hom_{\mathrm{Ab}}(Q, I))$$

for P, Q right and left R-module chain complexes, and I a cochain complex of Abelian groups.

2.7.6 Theorem

For all R-modules and n, we have isomorphisms

$$\mathrm{Ext}^n_R(A, B) = R^n \hom_R(A, \bullet)(B) \cong R^n \hom_R(\bullet, B)(A)$$

Proof

Choose a projective resolution $P \to A$ and an injective resolution $B \to I$. Then we have a first-quadrant double cochain complex $\hom(P, I)$. The augmentations induce maps $\hom(A, I) \to \hom(P, I) \leftarrow \hom(P, B)$. As for tensor products, the mapping cones of $\hom(A, I) \to \operatorname{Tot}\hom(P, I)$ and $\hom(P, B) \to \operatorname{Tot}\hom(P, I)$ are translates of the total complex obtained by adding $\hom(A, I)$ and $\hom(P, B)$ to $\hom(P, I)$. By the acylic assembly lemma (or rather its dual for cochain bicomplexes), these are exact and thus quasi-isomorphisms. Taking cohomology gives us

$$R^\bullet \hom(A, \bullet)(B) = H^\bullet(\hom(A, I)) \cong H^\bullet \operatorname{Tot}\hom(P, I) \cong H^\bullet \hom(P, B) = R^\bullet \hom(\bullet, B)(A)$$

as desired. ■

We can naturally extend these results to a broader class of derived functors.

2.7.7 Definition

Let T be a multivariate functor $\mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \mathcal{B}$ between Abelian categories such that T is left exact (equivalently left exact in each of its components). T is then **right balanced** if whenever any of its components are held constant at an injective object, T becomes exact (in each of its other components).

Dually a right exact functor is **left balanced** if it becomes exact when its components are held constant at a projective object.

For example $\hom(\bullet, \bullet)$ is right balanced (since injectives in $\mathcal{A}^{\mathrm{op}}$ are projectives in $\mathcal{A}$), and $\bullet \otimes \bullet$ are left balanced.

If T is right balanced, then all of its right derived functors are isomorphic in the sense that $R^\bullet T(A_1, \ldots, \hat{A}_i, \ldots, A_n)(A_i)$ are all isomorphic ($\hat{A}_i$ means the omission of A_i, replaced by an indeterminant $\bullet$). Thus for T right balanced, we can construct its **right derived functors** $R^\bullet T \colon \mathcal{A}_1 \times \cdots \times A_n \to \mathcal{B}$. Dually for T left balanced.

The proof proceeds as above. It is sufficient to consider only bifunctors (since the rest of the variables are held constant) $T \colon \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$. For cochain complexes $P^\bullet, Q^\bullet$ we can construct a (positive) cochain bicomplex $T(P^p, Q^q)$ with horizontal and vertical differentials as expected:

$$d^h = T(d_P, Q^\bullet) \colon T(P^p, Q^q) \to T(P^{p+1}, Q^q), \qquad d^v = T(P^\bullet, d_Q) \colon T(P^p, Q^q) \to T(P^p, Q^{q+1})$$

Let $A_i \in \mathcal{A}_i$ and injective resolutions $A_i \to I^i_\bullet$, then we can consider the bicomplex $T(I^1_p, I^2_q)$. Appending $T(A_1, I^2_\bullet)$ and $T(I^1_\bullet, A_2)$ give us bicomplexes whose total complexes are isomorphic to the mapping cones $T(A_1, I^2_\bullet), T(I^1_\bullet, A_2) \to \operatorname{Tot}(T(I^1_\bullet, T^2_\bullet))$. By the acylic assembly lemma, these bicomplexes are acylic and thus these maps are quasi-isomorphisms. In particular the cohomology of $T(A_1, I^2_\bullet)$ and $T(I^1_\bullet, A_2)$, which are $R^\bullet T(A_1, \bullet)(A_2)$ and $R^\bullet T(\bullet, A_2)(A_1)$, are isomorphic as desired.

3 Tor and Ext

3.1 Tor for Abelian groups

Consider the following projective resolution of $\mathbb{Z}/p\mathbb{Z}$:

$$0 \to \mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \mathbb{Z}/p \to 0$$

Then for any Abelian group B, computing Tor amounts to computing the homology of $0 \to B \xrightarrow{p} B \to 0$ (recall $R \otimes_R M \cong M$, and the tensor of multiplication by p with id_B is multiplication by p in B). This shows us that $\operatorname{Tor}_0^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, B) = B/pB$ and $\operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, B) = {}_pB = \{b \in B \mid pb = 0\}$; all higher Tor-groups are trivial.

3.1.1 Proposition

For all Abelian A, B:

(1) $\mathrm{Tor}_1^{\mathbb{Z}}(A, B)$ is torsion,

(2) All higher ($n \geq 2$) Tor-groups are trivial.

Proof

A is the directed colimit of its finitely-generated subgroups A_i, and since Tor commutes with filtered colimits, $\mathrm{Tor}_\bullet(A, B) \cong \varinjlim \mathrm{Tor}_\bullet(A_i, B)$. The direct colimit of torsion groups is torsion (consider the construction of the colimit), so it is sufficient to consider only finitely-generated A. Thus we have $A \cong \mathbb{Z}^m \oplus \mathbb{Z}/p_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/p_n\mathbb{Z}$ for some integers m, p_i. Tor again commutes with direct sums, and since $\mathbb{Z}^m$ is projective, we have for $n > 0$,

$$\mathrm{Tor}_n(A, B) \cong \mathrm{Tor}_n(\mathbb{Z}/p_1\mathbb{Z}, B) \oplus \cdots \oplus \mathrm{Tor}_n(\mathbb{Z}/p_n\mathbb{Z}, B)$$

These are all torsion for $n = 1$ and 0 for $n \geq 2$, as computed right before this proposition. ■

3.1.2 Proposition

$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B)$ is the torsion subgroup of B for all Abelian B.

Proof

$\mathbb{Q}/\mathbb{Z}$'s finitely-generated subgroups are all of the form $\mathbb{Z}/p\mathbb{Z}$ for some integer p. $\mathbb{Q}/\mathbb{Z}$ is the direct colimit of these subgroups, and thus

$$\mathrm{Tor}_1(\mathbb{Q}/\mathbb{Z}, B) \cong \varinjlim \mathrm{Tor}_1(\mathbb{Z}/p\mathbb{Z}, B) \cong \varinjlim_p B = \bigcup_p \{b \in B \mid pb = 0\}$$

as desired. ■

3.1.3 Proposition

If A is torsion-free, then $\mathrm{Tor}_n(A, B) = 0$ for all $n \neq 0$.

Proof

Finitely-generated subgroups of A are isomorphic to $\mathbb{Z}^m$, all projective. ■

Note that since $\otimes_R$ is commutative (up to isomorphism) for any commutative ring R, $\mathrm{Tor}_\bullet^R(A, B) \cong \mathrm{Tor}_\bullet^R(B, A)$ naturally for all R-modules A and B. This is because for fixed B, both form universal δ-functors over $\bullet \otimes B \cong B \otimes \bullet$. Thus we have,

3.1.4 Corollary

A is torsion-free iff $\mathrm{Tor}_1(A, \bullet) = 0$ iff $\mathrm{Tor}_1(\bullet, A) = 0$.

Note that these computations are not as simple over arbitrary commutative rings. For example take $R = \mathbb{Z}/m\mathbb{Z}$ and $A = \mathbb{Z}/d\mathbb{Z}$ for a divisor d of m. Consider then the free resolution

$$\cdots \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{m/d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{\epsilon} \mathbb{Z}/d\mathbb{Z} \to 0$$

Then for all $\mathbb{Z}/m\mathbb{Z}$-module B we have

$$\mathrm{Tor}_n^{\mathbb{Z}/m\mathbb{Z}}(\mathbb{Z}/d\mathbb{Z}, B) = \begin{cases} B/dB & \text{if } n = 0 \\ {}_dB/(m/d)B & \text{if } n \text{ is odd} \\ {}_{m/d}B/dB & \text{if } n \text{ is even} \end{cases}$$

3.1.5 Example

Let $r \in R$ be a left-nonzero divisor, i.e. ${}_rR = \{s \in R \mid rs = 0\}$ is zero. Then we have a free resolution $0 \to R \xrightarrow{r} R \to R/rR \to 0$, giving us

$$\mathrm{Tor}_0(R/rR, B) \cong B/rB, \qquad \mathrm{Tor}_1^R(R/rR, B) \cong {}_rB$$

and all higher Tor-modules are zero.

3.1.6 Example

Now if ${}_rR$ is nonzero, then we have an exact sequence,

$$0 \to {}_rR \to R \xrightarrow{r} R \to R/rR \to 0$$

since R are projective, then by dimension shifting we have $\mathrm{Tor}_n^R(R/rR, B) \cong \mathrm{Tor}_{n-2}^R({}_rR, B)$ for $n \geq 3$. $\mathrm{Tor}_2^R(R/rR, B)$ is the kernel of ${}_rR \otimes B \to B$ given by multiplication: $s \otimes b \mapsto sb$, so it can be viewed as ${}_rR \otimes B \to {}_rB$.

3.1.7 Example

Let R be an integral domain whose field of fractions is F. The finitely-generated submodules of F/R are isomorphic to R/rR (generalizing $\mathbb{Q}/\mathbb{Z}$), and so

$$\mathrm{Tor}_1^R(F/R, B) \cong \varinjlim \mathrm{Tor}_1^R(R/rR, B) \cong \varinjlim {}_rB = \bigcup_r {}_rB$$

i.e. the torsion submodule of B.

3.2 Tor and flatness

3.2.1 Definition

A left R-module is **flat** if $\bullet \otimes_R B$ is exact. Similarly a right R-module A is **flat** if $A \otimes_R \bullet$ is exact.

Projective modules are flat, but flat modules need not be projective: any torsion-free Abelian group is flat.

3.2.2 Proposition

An R-module B is flat iff $\mathrm{Tor}_n^R(\bullet, B) = 0$ for all $n > 0$ iff $\mathrm{Tor}_1^R(\bullet, B) = 0$.

Proof

If B is flat then $\bullet \otimes B$ is exact and thus its higher derived functors are trivial. Conversely if $\mathrm{Tor}_1^R(\bullet, B) = 0$ then if $0 \to A_1 \to A_2 \to A_3 \to 0$ is exact, then the long exact sequence given by $\mathrm{Tor}(\bullet, B)$ shows us that $0 \to A_1 \otimes B \to A_2 \otimes B \to A_3 \otimes B \to 0$ is exact. ■

So flat R-modules are precisely those which are $A \otimes \bullet$-acylic for all R-modules A. Recall that when showing Tor commutes with filtered colimits we showed that filtered colimits of projective R-modules make the tensor product exact. That is to say, filtered colimits of projectives are flat.

3.2.3 Theorem

If S is a central multiplicatively closed set in R (meaning it is closed under multiplication, contains 1, and commutes with all elements of R), then $S^{-1}R$ is flat.

Proof

Let I be the filtered category whose objects are elements of S and $\hom_I(s_1, s_2) = \{s \in S \mid s_1 s = s_2\}$. This is filtered, since s_1, s_2 can be bound by $s_1 s_2$ and if $s_1 s = s_1 s'$ then we can just take the arrow s_1 to equalize.

Define the functor $F\colon I \to \mathrm{Mod}_R$ by $F(s) = R$ and $F(s\colon s_1 \to s_2)$ is multiplication by s. Now if we define the map $F(s) = R \to S^{-1}R$ by mapping 1 to $1/s$, this defines a surjection $\bigsqcup F(s) \cong R \times S \to S^{-1}R$ mapping (r, s) to r/s. Recall that $r/s = r'/s'$ iff there is a $t \in S$ such that $(rs' - r's)t = 0$. Now in the colimit (r, s) and (r', s') are equivalent iff there are t_1, t_2 such that $st_1 = s't_2$ and $rt_1 = r't_2$. Then $(rs' - r's)t_1 t_2 = 0$ meaning $r/s = r'/s'$, so this map can be quotient to a surjection $\varinjlim F \to S^{-1}R$. And conversely if $(rs' - r's)t = 0$ then $t_1 = s't$ and $t_2 = st$ show that (r, s) and (r', s') are equivalent, so this surjection is injective and thus an isomorphism.

Thus $S^{-1}R \cong \varinjlim F$, a filtered colimit of free R-modules and is therefore flat. ■

Recall that localization of an R-module by a multiplicative set $S \subseteq R$ is equivalent to tensoring with $S^{-1}R$; $S^{-1}M \cong M \otimes_R S^{-1}R$. Thus this theorem shows us that

$$S^{-1}\bullet\colon \mathrm{Mod}_R \to \mathrm{Mod}_{S^{-1}R}$$

is an exact functor.

3.2.4 Example

Let $0 \to A \to B \to C \to 0$ be short exact such that B, C are flat, then A is flat as well. This is true in general for F-acylic objects by taking the long exact sequence; we have $0 = L_{n+1}F(C) \to L_n F(A) \to L_n F(B) = 0$ exact.

3.2.5 Definition

The **Pontrjagin dual** of a left R-module B is the right R-module $B^* = \hom_{\mathrm{Ab}}(B, \mathbb{Q}/\mathbb{Z})$ where $r \in R$ acts by $(fr)(b) = f(rb)$.

Recall that $\mathbb{Q}/\mathbb{Z}$ is an injective Abelian group, so $\hom_{\mathrm{Ab}}(\bullet, \mathbb{Q}/\mathbb{Z})$ is exact. But duality is stronger in that it reflects exactness as well: it is exact, and we showed (when discussing having enough injectives) that $A^* = 0$ iff $A = 0$. Thus duality reflects zero objects and is exact; we showed this means it reflects exactness. In particular it reflects and preserves monics and epics: since it is contravariant this means that f is monic (resp. epic) iff f^* is epic (resp. monic).

3.2.6 Proposition

The following are equivalent for every left R-module B:

(1) B is flat,

(2) B^* is injective,

(3) $I \otimes_R B \cong IB = \{x_1 b_1 + \cdots + x_n b_n \mid x_i \in I, b_i \in B\}$ for every right ideal I of R,

(4) $\mathrm{Tor}_1^R(R/I, B) = 0$ for every right ideal I of R.

Proof

We have a short exact sequence $0 \to I \to R \to R/I \to 0$ which gives us a long exact sequence

$$0 \to \mathrm{Tor}_1^R(R/I, B) \to I \otimes B \to B \to R/I \otimes B \cong B/IB \to 0$$

Where the isomorphism is given by $(r + I) \otimes b \mapsto rb + IB$. Then $\mathrm{Tor}_1^R(R/I, B) = 0$ iff $0 \to I \otimes B \to B \to B/IB \to 0$ is exact, which is iff $I \otimes B \cong IB$. Thus (3) and (4) are equivalent.

B^* is injective iff $\hom(\bullet, B^*) \cong \hom(\bullet \otimes B, \mathbb{Q}/\mathbb{Z}) = (\bullet \otimes B)^*$ is exact. Since duality reflects and preserves exactness, this is iff $\bullet \otimes B$ is exact, i.e. B is flat. (In general if F preserves and relects exactness, then $F \circ G$ is exact iff G is.)

By Baer, B^* is injective iff every $I \to B^*$ extends to $R \to B^*$ for every I. That is, $\hom(R, B^*) \to \hom(I, B^*)$ is surjective. Meaning $(R \otimes B)^* \to (I \otimes B)^*$ is surjective for all I, and since the dual map maps epics to monics and vice versa, this is iff $I \otimes B \to R \otimes B \cong B$ is injective. This is equivalent to $I \otimes B \cong IB$ (its image) for all I, as desired. ■

3.2.7 Definition

An R-module M is **finitely presented** if it has a presentation using finitely many generators and relations. That is, M is isomorphic to the quotient of $R(e_1, \ldots, e_n)$ (the free R-module on $e_1, \ldots, e_n$) by the submodule generated by $\sum_j \alpha_{ij} e_j$ for $i = 1, \ldots, m$.

Equivalently M is finitely presented iff there exists an $m \times n$ matrix α and an exact sequence $R^n \xrightarrow{\alpha} R^m \to M \to A$.

There is a natural map $\sigma\colon A^* \otimes_R M \to \hom_R(M, A)^*$ given by $\sigma(f \otimes m)$ being the map $h \mapsto f(h(m))$. This is well-defined: $(fr) \otimes m$ maps to $h \mapsto (fr)(h(m)) = f(rh(m)) = f(h(rm))$ which is the same as the image of $f \otimes (rm)$.

3.2.8 Lemma

σ is an isomorphism for every finitely presented M and all A.

Proof

For $M = R$, σ reduces to a map $\sigma\colon A^* \to A^*$ mapping f to f; the identity. For $M = R^n$, $\sigma\colon (A^*)^n \to (A^*)^n$ is the identity as well. Now if $R^n \to R^m \to M \to 0$ is exact, then by right exactness of $A^* \otimes \bullet$, left exactness of $\hom(\bullet, A)^*$, we get that the following diagram commutes and has exact rows:

$$\begin{tikzcd}
A^* \otimes R^n \arrow[r] \arrow[d, "\sigma"] & A^* \otimes R^m \arrow[r] \arrow[d, "\sigma"] & A^* \otimes M \arrow[r] \arrow[d, "\sigma"] & 0 \\
\hom(R^n, A)^* \arrow[r] & \hom(R^m, A)^* \arrow[r] & \hom(M, A)^* \arrow[r] & 0
\end{tikzcd}$$

By the five-lemma, the rightmost σ is an isomorphism as well. ■

3.2.9 Theorem

Every finitely presented flat R-module M is projective.

Proof

We show that $\hom(M, \bullet)$ is exact: it is sufficient to show that it preserves surjections (as it is already left exact). Suppose $B \to C$ is surjective, then $C^* \to B^*$ is an injection, and since M is flat

$$\hom(M, C)^* \cong C^* \otimes M \to B^* \otimes M \cong \hom(M, B)^*$$

is an injection. And then $\hom(M, C) \to \hom(M, B)$ is a surjection, as desired. ■

Recall that we can compute the derived functor of F using the homology of F-acylic resolutions. In particular:

3.2.10 Theorem

If $F \to A$ is a flat resolution of A, then $\mathrm{Tor}_\bullet(A, B) \cong H_\bullet(F \otimes B)$.

3.2.11 Proposition

Suppose $R \to T$ is a ring map which gives T the structure of a flat R-module. Then for all R-modules A and all T-modules C,

$$\mathrm{Tor}^R_\bullet(A, C) \cong \mathrm{Tor}^T_\bullet(A \otimes_R T, C)$$

Proof

Let $P \to A$ be a projective R-module resolution. Since T is R-flat and $P_n \otimes_R T$ is T-projective (since $\hom_T(P_n \otimes_R T, \bullet) \cong \hom_R(P_n, \hom_T(T, \bullet))$ is the compostion of exact functors), we have that $P \otimes_R T \to A \otimes T$ is a T-projective resolution. Thus $\mathrm{Tor}^T_\bullet(A \otimes T, C) = H_\bullet((P \otimes_R T) \otimes_T C) \cong H_\bullet(P \otimes_R C) = \mathrm{Tor}^R_\bullet(A, C)$. ■

3.2.12 Corollary

If R is commutative and T is a flat R-algebra, then for all $A, B \in \mathrm{Mod}_R$,

$$T \otimes_R \mathrm{Tor}_\bullet^R(A, B) \cong \mathrm{Tor}_\bullet^T(A \otimes_R T, T \otimes_R B)$$

Proof

Let $C = T \otimes_R B$, by the above proposition it is sufficient to show $\mathrm{Tor}_\bullet^R(A, T \otimes B) \cong T \otimes \mathrm{Tor}_\bullet^R(A, B)$. Since T is flat, $T \otimes \bullet$ is exact and derived functors commute with exact functors, as desired. ■

We assume R is commutative so that $\mathrm{Tor}_\bullet^R$ map into Mod_R.

3.2.13 Lemma

If $\mu: A \to A$ is multiplication by an element $r \in R$, then so are the induced maps $\mu_\bullet: \mathrm{Tor}_\bullet^R(A, \bullet) \to \mathrm{Tor}_\bullet^R(A, \bullet)$.

Proof

Let $P \to A$ be a projective resolution, then multiplication by $r \in R$ is a chain map $\tilde{\mu}: P \to P$. And $\tilde{\mu} \otimes B$ is multiplication by r on $P \otimes B$, and since $\mathrm{Tor}_\bullet^R(A, B)$ are subquotients of these modules, the induced map is still multiplication by r. ■

3.2.14 Corollary

Let A be an R/I module, then for every R-module B, $\mathrm{Tor}_\bullet^R(A, B)$ is an R/I-module as well.

This just says that if $IA = 0$ then $I\mathrm{Tor}_\bullet^R(A, \bullet) = 0$ as well. This is clearly true since $\mu = 0$ is multiplication by $r \in I$, and so $\mu_\bullet = 0$ is also multiplication by $r \in I$.

3.2.15 Corollary (Localization for Tor)

If R is commutative and $A, B \in \mathrm{Mod}_R$, then the following are equivalent for each n:

(1) $\mathrm{Tor}_n^R(A, B) = 0$,

(2) For every prime ideal $\mathfrak{p}$ of R, $\mathrm{Tor}_n^{R_\mathfrak{p}}(A_\mathfrak{p}, B_\mathfrak{p}) = 0$.

(3) For every maximal ideal $\mathfrak{m}$ of R, $\mathrm{Tor}_n^{R_\mathfrak{m}}(A_\mathfrak{m}, B_\mathfrak{m}) = 0$.

Proof

An R-module M is zero iff all of its localizations by prime ideals are zero iff all of its localizations by maximal ideals are zero. For $M = \mathrm{Tor}_n^R(A, B)$ we have

$$M_\mathfrak{p} = R_\mathfrak{p} \otimes_R M = \mathrm{Tor}^{R_\mathfrak{p}}(A_\mathfrak{p}, B_\mathfrak{p})$$

since $R_\mathfrak{p}$ is flat: we showed that $S^{-1}R$ is always flat. ■

3.3 Ext for nice rings

Recall that R-modules are injective iff they are divisible when R is a PID.

3.3.1 Lemma

$\mathrm{Ext}^n_{\mathbb{Z}}(A, B) = 0$ for all Abelian groups A, B when $n \geq 2$.

Proof

Embed B in an injective I^0, and since divisibility is preserved by quotients, the quotient I^1 is divisible and thus injective. So we have an injective resolution $0 \to B \to I^0 \to I^1 \to 0$, giving the desired result (take the homology of the induced complex). ■

For $A = \mathbb{Z}/p\mathbb{Z}$ we have the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0$$

This is a projective resolution of $\mathbb{Z}/p\mathbb{Z}$ and thus we can compute its Ext-groups via the cohomology of $0 \leftarrow B \xleftarrow{p} B \leftarrow 0$ (recall Ext in the first variable is computed via projective resolutions, and injective resolutions in the second variable). Thus

$$\mathrm{Ext}^0_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, B) = {}_pB, \qquad \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, B) = B/pB$$

and all higher Ext-groups vanish.

Since Ext commutes with finite direct sums, for a finitely generated $A \cong \mathbb{Z}^m \oplus \mathbb{Z}/p_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/p_n\mathbb{Z}$, we have

$$\mathrm{Ext}^\bullet_{\mathbb{Z}}(A, B) \cong \bigoplus_i \mathrm{Ext}^\bullet_{\mathbb{Z}}(\mathbb{Z}/p_i\mathbb{Z}, B)$$

3.3.2 Example

We have an injective resolution $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. Thus $\mathrm{Ext}^\bullet(A, \mathbb{Z})$ is the cohomology of $0 \to \hom(A, \mathbb{Q}) \to A^* \to 0$. If A is torsion, then $\hom(A, \mathbb{Q}) = 0$ and so we have $\mathrm{Ext}^0(A, \mathbb{Z}) = 0$ and $\mathrm{Ext}^1(A, \mathbb{Z}) = A^*$.

3.3.3 Proposition

Ext commutes with arbitrary direct products in its second component, and maps direct sums to direct products in its first. That is,

$$\mathrm{Ext}^\bullet_R\left(\bigoplus_i A_i, \bullet\right) \cong \prod_i \mathrm{Ext}^\bullet_R(A_i, \bullet), \qquad \mathrm{Ext}^\bullet_R\left(\bullet, \prod_i B_i\right) \cong \prod_i \mathrm{Ext}^\bullet_R(\bullet, B_i)$$

Proof

If $P_i \to A_i$ are projective resolutions, so is $\oplus P_i \to \oplus A_i$. And if $B_i \to I_i$ are injective resolutions, so is $\prod_i B_i \to \prod_i I_i$. Since $\hom(\oplus P_i, \bullet) \cong \prod \hom(P_i, \bullet)$ and $\hom(\bullet, \prod I_i) \cong \prod \hom(\bullet, I_i)$, this results from cohomology commuting with direct products. ■

3.3.4 Example

Let $B = \prod_p \mathbb{Z}/p\mathbb{Z}$, then

$$\mathrm{Ext}^1(A, B) \cong \prod_p A/pA$$

which vanishes iff A is torsion-free.

3.3.5 Lemma

Let R be a commutative ring, so we can view $\hom_R(A, B), \mathrm{Ext}^\bullet_R(A, B)$ as R-modules. If $\mu\colon A \to A, \nu\colon B \to B$ are multiplication by $r \in R$, so are $\mu^\bullet, \nu_\bullet$ of $\mathrm{Ext}^\bullet_R(A, B)$.

Proof

Pick a projective resolution $P \to A$, then μ extends to a chain map $\tilde{\mu}\colon P \to P$ given by multiplication by r (which is the up-to-homotopy unique extension). Then $\hom(\tilde{\mu}, B)$ is multiplication by r on $\hom(P, B)$: f maps to $f\tilde{\mu}$ which maps $p \in P_n$ to $f(rp) = rf(p)$. Thus $\tilde{\mu}$ is multiplication by r on subquotients, in particular $\mathrm{Ext}^\bullet_R(A, B)$. ■

3.3.6 Corollary

If R is commutative and $A \in \mathrm{Mod}_R$ is an R/I-module (i.e. $IA = 0$), then $\mathrm{Ext}^\bullet_R(A, \bullet)$ are R/I-modules.

We want to show, similar to Tor, that Ext commutes with localization. There is a natural map $\Phi\colon S^{-1}\hom_R(A, B) \to \hom_{S^{-1}R}(S^{-1}A, S^{-1}B)$, as we can view $f/s \in S^{-1}\hom(A, B)$ as mapping $a/s' \mapsto f(a)/ss'$.

3.3.7 Lemma

If A is finitely presented, then for every multiplicative set $S \subseteq R$, Φ is an isomorphism.

Proof

When $A = R$ then Φ is a map $S^{-1}B \to S^{-1}B$ which maps b/s (viewed as $f(r) = rb$) to b/s (since 1 maps to b/s), so Φ is an isomorphism. By additivity of hom, Φ is an isomorphism when $A = R^n$. If $R^n \to R^m \to A \to 0$ is exact, we can apply the 5-lemma to the diagram (since $\hom(\bullet, B)$ is left exact)

$$\begin{tikzcd}
0 \arrow[r] & S^{-1}\hom(A, B) \arrow[r] \arrow[d, "\Phi"] & S^{-1}\hom(R^m, B) \arrow[r] \arrow[d, "\cong"] & S^{-1}\hom(R^n, B) \arrow[d, "\cong"] \\
0 \arrow[r] & \hom(S^{-1}A, S^{-1}B) \arrow[r] & \hom(S^{-1}R^m, S^{-1}B) \arrow[r] & \hom(S^{-1}R^n, S^{-1}B)
\end{tikzcd}$$

and we get that Φ is an isomorphism, as desired. ■

3.3.8 Definition

A ring R is **(right) Noetherian** if it satisfies the **ascending chain condition**: there exists no infinite proper ascending chain of (right) ideals.

There is an obvious definition of left Noetherian rings. The dual condition, that R satisfies the **descending chain condition**, makes R an **Artinian ring**.

A classic result is the following:

3.3.9 Lemma

A ring R is (right) Noetherian iff every (right) ideal is finitely generated.

Furthermore, it is a well-known result that every finitely-generated module over a Noetherian ring is finitely presented. In particular every finitely-generated module has a left presentation by finitely-generated free modules (since when constructing the left resolution we can always choose finitely-generated free modules).

3.3.10 Proposition

Let A be a finitely generated module over a commutative Noetherian ring R, then for every multiplicative set $S \subseteq R$ and $B \in \mathrm{Mod}_R$, Φ extends to an isomorphism $S^{-1}\mathrm{Ext}^\bullet_R(A, B) \cong \mathrm{Ext}^\bullet_{S^{-1}R}(S^{-1}A, S^{-1}B)$.

Proof

Let $F \to A$ be a resolution by finitely-generated free R-modules. Then since $S^{-1}\bullet\colon \mathrm{Mod}_R \to \mathrm{Mod}_{S^{-1}R}$ is exact and maps finitely-generated free R-modules to finitely-generated free $S^{-1}R$-modules, we get the desired result. ■

3.3.11 Corollary (Localization for Ext)

If R is commutative Noetherian and $A \in \mathrm{Mod}_R$ is finitely-generated, then the following are equivalent for all $B \in \mathrm{Mod}_R$ and n:

(1) $\mathrm{Ext}^n_R(A, B) = 0$.

(2) For every prime ideal $\mathfrak{p} \leq R$, $\mathrm{Ext}^n_{R_\mathfrak{p}}(A_\mathfrak{p}, B_\mathfrak{p}) = 0$.

(3) For every maximal ideal $\mathfrak{m} \leq R$, $\mathrm{Ext}^n_{R_\mathfrak{m}}(A_\mathfrak{m}, B_\mathfrak{m}) = 0$.

3.4 Ext and extensions

3.4.1 Definition

An **extension** ξ of A by B is an exact sequence $0 \to B \to X \to A \to 0$. Two extensions ξ, ξ' are **equivalent** if there is a map $X \to X'$ commuting with the short exact sequences:

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, "="] & X \arrow[r] \arrow[d] & B \arrow[r] \arrow[d, "="] & 0 \\
0 \arrow[r] & A \arrow[r] & X' \arrow[r] & B \arrow[r] & 0
\end{tikzcd}$$

Note that by the 5-lemma such a map must be an isomorphism, so this is indeed an equivalence.

An extension is **split** if it is equivalent to $0 \to B \to A \oplus B \to A \to 0$.

3.4.2 Lemma

If $\mathrm{Ext}^1(A,B)$ then every extension of A by B is split.

Proof

Let $\xi\colon 0 \to B \to X \to A \to 0$ be an extension, then applying $\mathrm{Ext}^\bullet(A,\bullet)$ gives the exact sequence

$$\hom(A,X) \to \hom(A,A) \xrightarrow{\partial} \mathrm{Ext}^1(A,B) = 0$$

meaning the map $\hom(A,X) \to \hom(A,A)$ is surjective. In particular we can lift id_A to a map $\sigma\colon A \to X$; i.e. σ is a section of the above short exact sequence, so it splits. ■

By naturality of the connecting morphism δ, $\Theta(\xi) = \partial(\mathrm{id}_A)$ is well-defined on the equivalence classes of extensions. Furthermore ξ is split iff id_A lifts to $\hom(A,X)$ iff $\Theta(\xi) = \partial(\mathrm{id}_A) = 0$ (since this means $\mathrm{id}_A \in \ker\partial$, the image of $\hom(A,X) \to \hom(A,A)$). We can thus interpret $\Theta(\xi)$ as an *obstruction* to ξ being split.

3.4.3 Theorem

Given two R-modules A, B, the mapping $\Theta\colon \xi \mapsto \partial(\mathrm{id}_A)$ is a bijection between (equivalence classes of) extensions of A by B and $\mathrm{Ext}^1(A,B)$, which maps split extensions to 0.

Proof

Fix an exact sequence $0 \to M \xrightarrow{j} P \to A \to 0$ with P projective. Applying $\hom(\bullet,B)$ gives an exact sequence (since $\mathrm{Ext}^1(P,B) = 0$)

$$0 \to \hom(A,B) \to \hom(P,B) \to \hom(M,B) \xrightarrow{\partial} \mathrm{Ext}^1(A,B) \to 0$$

Given $x \in \mathrm{Ext}^1(A,B)$, choose $\beta \in \hom(M,B)$ such that $\partial(\beta) = x$. Let X be the pushout of j, β, i.e. the cokernel of $M \to P \oplus B, m \mapsto (j(m), -\beta(m))$. Then we have a diagram

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r, "j"'] \arrow[d, "\beta"] & P \arrow[r] \arrow[d, "\sigma"] & A \arrow[r] \arrow[d, "="] & 0 \\
0 \arrow[r] & B \arrow[r, "i"] & X \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

The map $X \to A$ is induced by $B \xrightarrow{0} A$ and $P \to A$. The bottom sequence is exact, and by naturality of ∂ we see that $\partial(\mathrm{id}_A) = \partial(\beta) = x$, so Θ is surjective.

Now notice that if $\beta' \in \hom(M,B)$ is another lift of x, then there is an $f \in \hom(P,B)$ so that $\beta' = \beta + fj$. If X' is the pushout of β', j then $i\colon B \to X, \sigma + if\colon P \to X$ induce an isomorphism $X' \cong X$ giving an equivalence between ξ' and ξ. Thus we have a map Ψ from $\mathrm{Ext}^1(A,B)$ to the set of equivalences of extensions, which satisfies $\Psi\Theta(\xi) = \xi$, meaning Θ is injective and thus bijective. ■

3.4.4 Definition

Let $\xi\colon 0 \to B \to X \to A \to 0$ and $\xi'\colon 0 \to B \to X' \to A \to 0$ be two extensions of A by B. Let X'' be the pullback of $X \to A, X' \to A$: $X'' = \{(x,x') \in X \times X' \mid \bar{x} = \bar{x}' \in A\}$. Then X'' contains three copies of B: $B \times 0, 0 \times B$ and the skew diagonal $\{(-b,b) \mid b \in B\}$ (since $\bar{b} = 0$ by exactness). Let Y be the quotient of X'' by the skew diagonal; this identifies $B \times 0$ and $0 \times B$.

Thus we have a natural embedding $B \to Y, b \mapsto [b,0]$, and the induced epimorphism $Y \to A$ given by the pullback. The kernel of $Y \to A$ are $[x,x']$ such that $\bar{x} = \bar{x}' = 0$, which means that $x, x' \in B$ and so the

kernel is the image of B. Thus the sequence $\phi: 0 \to B \to Y \to A \to 0$ is exact, and is called the **Baer sum** of ξ and ξ'.

3.4.5 Corollary

The set of extensions of A by B is an Abelian group with the Baer sum, isomorphic to $\mathrm{Ext}^1(A, B)$ via Θ.

Proof

We will show $\Theta(\phi) = \Theta(\xi) + \Theta(\xi')$ which is sufficient: the isomorphic structure is given by $\xi + \xi' = \Theta^{-1}(\Theta(\xi) + \Theta(\xi'))$, and so if we show this we show that Baer summation is well-defined, as well as the isomorphism.

We can lift $P \to A$ via the surjection $X \to A$ to give us a map $\tau: P \to X$, similarly $\tau': P \to X'$. These induce maps $\gamma, \gamma': M \to B$ which make the short exact sequences commute. Further, we have an induced map $\bar{\tau}: P \to Y$ whose restriction to M is induced by $\gamma + \gamma': M \to B$ so we have another map of short exact sequences. Thus $\Theta(\phi) = \partial(\gamma + \gamma') = \partial(\gamma) + \partial(\gamma') = \Theta(\xi) + \Theta(\xi')$ as desired. ■

This construction allows us to define $\mathrm{Ext}^1(A, B)$ in *any* Abelian category, no matter the presence of projectives or injectives. The Freyd-Mitchell embedding theorem shows us that $\mathrm{Ext}^1(A, B)$ is always an Abelian group; but it can be shown directly as well. We can also generalize $\mathrm{Ext}^n(A, B)$ due to Yoneda.

$\mathrm{Ext}^n(A, B)$ is the equivalence class of exact sequences of the form

$$\xi: B \to X_n \to \cdots \to X_1 \to A \to 0$$

by the equivalence $\xi \sim \xi'$ iff there is a morphism of complexes $\xi \to x'$, whose maps $B \to B, A \to A$ are the respective identities. This is not obviously an equivalence (since the morphisms need not be isomorphisms, so transitivity is not clear), but it can be shown.

To add ξ, ξ', let X_1'' be the pullback of X_1, X_1' over A and X_n'' the pusout of X_n, X_n'' under B. Then define Y_n to be the quotient of X_n'' by the skew diagonal copy of B. Then $\xi + \xi'$ is the class of the extension

$$0 \to B \to Y_n \to X_{n-1} \oplus X_{n-1}' \to \cdots \to X_2 \oplus X_2' \to X_1'' \to A \to 0$$

If $\mathcal{A}$ has enough projectives and we take a projective resolution $P \to A$, the comparison theorem gives us a chain map $P \to \xi$, thus a diagram

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta"] & P_{n-1} \arrow[r] \arrow[d, "\gamma_n"] & \cdots \arrow[r] & P_0 \arrow[r] \arrow[d, "\gamma_1"] & A \arrow[r] \arrow[d, "="] & 0 \\
0 \arrow[r] & B \arrow[r] & X_n \arrow[r] & \cdots \arrow[r] & X_1 \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

Dimension shifting gives us $\mathrm{Ext}^n(A, B) \cong \hom(M, B)$: $\Theta(\xi) = \partial(\beta)$ is the bijection.

3.5 Derived functors of the inverse limit

Recall that we showed $\mathcal{A}^I$ has enough injectives when $\mathcal{A}$ is complete and has enough injectives (see the last example in the chapter on injective resolutions). In particular Mod_R^I has enough injectives.

We are interested in the case $\mathcal{A} = \mathrm{Ab}$ and $I = \omega^{\mathsf{op}}$ (the opposite poset of natural numbers). Elements of $\mathrm{Ab}^{\omega^{\mathsf{op}}}$ are called **towers** of Abelian groups: they are sequences

$$\{A_i\}_{i<\omega}: \cdots \to A_2 \to A_1 \to A_0$$

Note that since ω^{op} has a maximum (terminal) element, colimits over it are uninteresting (the colimit from a category with a terminal element is just the image of that element, if it exists). ω is a directed set, and thus ω^{op} is cofiltered, and so limits from it are **inverse limits** $\varprojlim: \mathrm{Ab}^{\omega^{\mathsf{op}}} \to \mathrm{Ab}$.

Recall the axiom (AB4*): $\mathcal{A}$ is complete and the product of any set of surjections is surjective. That is, $\prod$ is an exact functor. It is not hard to see that Mod_R satisfies (AB4*).

Recall that the limit of a diagram $A\colon \mathcal{A}^I \to \mathcal{A}$ is the kernel of

$$\prod_{i\in I} A_i \xrightarrow{\Delta} \prod_{\phi: i\to j} A_j$$

where $\Delta_\phi\colon \prod_{i\in I} A_i \to A_j$ is $p_j - A(\phi)p_i$, where $p_i\colon \prod_{i\in I} A_i \to A_i$ is the projection map.

3.5.1 Proposition

Let $\mathcal{A}$ be (AB4*), and I small. Then $\lim_I\colon \mathcal{A}^I \to \mathcal{A}$ is left exact, and we can define a cohomological δ-functor

$$\lim_I^1 A = \mathrm{coker}\Delta_A, \qquad \lim_I^n A = 0$$

Proof

If $0 \to A \to B \to C \to 0$ is an exact sequence of functors, then by (AB4*) we have a commutative diagram of exact rows

$$\begin{tikzcd}
0 \arrow[r] & \prod_i A_i \arrow[r] \arrow[d, "\Delta_A"] & \prod_i B_i \arrow[r] \arrow[d, "\Delta_B"] & \prod_i C_i \arrow[r] \arrow[d, "\Delta_C"] & 0 \\
0 \arrow[r] & \prod_\phi A_j \arrow[r] & \prod_\phi B_j \arrow[r] & \prod_\phi C_j \arrow[r] & 0
\end{tikzcd}$$

The snake lemma gives us a connecting morphism ∂ so that

$$0 \to \ker\Delta_A \to \ker\Delta_B \to \ker\Delta_C \xrightarrow{\partial} \mathrm{coker}\Delta_A \to \mathrm{coker}\Delta_B \to \mathrm{coker}\Delta_C \to 0$$

is exact. Thus $\lim_I^\bullet$ is a cohomological δ-functor (by naturality of the connecting morphism). ■

Recall that a functor is effaceable if for every object A, there is a monic $A \to I$ mapped to zero. A δ-functor is effaceable if every component (other than zero) is effaceable, and we showed that in such a case the functor is universal. To show that $\lim_I^\bullet$ is the right derived functor of $\lim_I$, it is sufficient then to show it is effaceable. This is not true in general, but it is when $I = \omega^{\mathsf{op}}$.

In this case, let us consider more closely Δ. Given a tower of Abelian groups

$$\cdots \to A_2 \xrightarrow{f_1} A_1 \xrightarrow{f_0} A_0$$

Δ is the map

$$\Delta\colon \prod_{n<\omega} A_n \to \prod_{n\geq m} A_m$$

where $\Delta_{n,m}(a_i)_i = a_m - f_{n,m}a_n$ where $f_{n,m} = f_m \circ \cdots \circ f_{n-1}\colon A_n \to A_m$. The kernel of Δ are those $(a_i)_i$ such that $a_m - f_{n,m}a_n = 0$ for all $n \geq m$, equivalently $a_n - f_n a_{n+1} = 0$ for all n. So we can simplify Δ:

$$\Delta\colon \prod_{n<\omega} A_n \to \prod_{n<\omega} A_n, \qquad \Delta(a_n)_n = (a_n - f_n a_{n+1})_n$$

3.5.2 Lemma

If all maps f_n are surjective, then $\varprojlim^1 A_i = 0$.

Proof

We want to show that Δ is surjective, and thus $\operatorname{coker}\Delta = 0$. So let $(b_i) \in \prod_i A_i$ and choose $a_0 \in A_0$. Then we inductively can choose $a_{i+1} \in A_{i+1}$ such that $f_i a_{i+1} = a_i - b_i$, then $\Delta(a_i)_i = (b_i)_i$ as desired. ■

3.5.3 Corollary

$\varprojlim^\bullet$ is the right derived functor of $\varprojlim$.

Proof

As explained, it is sufficient to show that $\varprojlim^1$ is effaceable. When showing that $\mathcal{A}^I$ has enough injectives, we showed that we can embed every functor F into an injective of the form $\prod_{k\in I} k_* I_k$ for I_k injective. k_*E is defined by

$$k_*E(i) = \prod_{\hom_I(i,k)} E, \qquad k_*E(f\colon i \to j)\colon \prod_{\hom(i,k)} E \to \prod_{hom(j,k)} E, (a_\phi)_{\phi:i\to k} \mapsto (a_{\psi f})_{\phi:j\to k}$$

So in our case k_*E is the tower $\cdots \to E \to E \to 0 \to \cdots \to 0$, where $k_*E(i) = 0$ when $i < k$, and the maps $E \to E$ are identities. In particular k_*E's maps are all surjective, and thus by (AB4*) so are all the maps of $\prod_{k\in I} k_* I_k$. So $\varprojlim^1$ maps $\prod_{k\in I} k_* I_k$ to zero and is thus effaceable, as desired. ■

3.5.4 Example

Let $\{A_i\}$ be a tower such that $A_{i+1} \to A_i$ are all inclusions, and set $A = A_0$. Then A is a topological group where $\{a + A_i \mid a \in A, i \geq 0\}$ form a basis of open sets. Then $\varprojlim A_i = \bigcap A_i$ is zero iff A is Hausdorff: $\{A_i\}$ form a local basis at 0, so if A is Hausdorff then for every $a \neq 0$ there must be some neighborhood of 0 (A_i) not including a, as desired. And conversely, if $a \neq 0$ then let $a \notin A_i$; $a \in a + A_i$ and $0 \in A_i$ are distinct, and thus disjoint, cosets, as desired.

Now $\varprojlim^1 A_i = 0$ iff A is complete (Cauchy sequences being sequences $(a_i)_i$ such that for every open $0 \in U$ there is an N such that for every $n, m \geq N$, $a_n - a_m \in U$). We consider the inverse limit of quotients, $\varprojlim(A/A_i)$. Elements of $\varprojlim(A/A_i)$ are sequences $(a_i + A_i)_i$ such that $a_i - a_j \in A_i$ for all $i \leq j$. Note that if $(a_i + A_i)_i \in \varprojlim(A/A_i)$, then $(a_i)_i$ is Cauchy. We have an exact sequence of towers,

$$0 \to \{A_i\} \to \{A\} \to \{A/A_i\} \to 0$$

which yields

$$0 \to \varprojlim(A_i) \to A \to \varprojlim(A/A_i) \to \varprojlim{}^1 A_i \to 0$$

since $\{A\}$'s maps are all surjective, its derived inverse limit is zero. The map $A \to \varprojlim(A/A_i)$ is the map $a \mapsto (a + A_i)_i$. By exactness, this map is surjective iff $\varprojlim^1 A_i = 0$.

If A is complete, then for $(a_i + A_i)_i \in \varprojlim(A/A_i)$, $(a_i)_i$ must have some limit for it is Cauchy. Call it a, then for every i we must have $a - a_j \in A_i$ for every $j \geq N$ for some N. If $i \geq N$ this means $a - a_i \in A_i$, and if $i < N$ then $a_i - a_N \in A_i$ and $a - a_N \in A_i$ implies $a - a_i \in A_i$ as well. Thus $(a + A_i)_i = (a_i + A_i)_i$, so this map is indeed surjective and $\varprojlim^1 A_i = 0$ as desired.

Conversely if this map is surjective, then consider a Cauchy sequence $(a_n)_n$. For every i, there is an N_i such that $a_n - a_m \in A_i$ for all $n, m \geq N_i$. Since A_i decrease, we can assume that N_i is monotonically increasing, and so $(a_n)_n$ has a limit iff $(a_{N_i})_i$ does (Cauchy sequences have a limit iff some subsequence does). Now $a_{N_i} - a_{N_j} \in A_i$ since $N_j \geq N_i$ and so $(a_{N_i} + A_i)_i \in \varprojlim(A/A_i)$ and thus is equal to $(a + A_i)_i$. But then a is the limit of $(a_{N_i})_i$: $a_{N_j} - a \in A_i$ for $j \geq i$ (since it is true for $j = i$ by definition and $a_{N_j} - a_{N_i} \in A_i$). Thus A is complete, as desired.

3.5.5 Definition

A tower of Abelian groups $\{A_i\}$ satisfies the **Mittag-Leffler condition** if for every k there is some $j \geq k$ such that the image of every $A_i \to A_k$ is constant for all $i \geq j$. Equivalently, we could say that the images of $A_i \to A_k$ satisfies the descending chain condition.

If for each k there is a $j > k$ such that the map $A_j \to A_k$ is zero, then this obviously satisfies the Mittag-Leffler condition. In this case, we say the tower satisfies the **trivial Mittag-Leffler condition**.

3.5.6 Proposition

If $\{A_i\}$ satisfies the Mittag-Leffler condition, then $\varprojlim^1 A_i = 0$.

Proof

We need to show that $\Delta\colon \prod_i A_i \to \prod_i A_i$ is surjective. If the tower satisfies the trivial Mittag-Leffler condition, then let $(b_i)_i \in \prod_i A_i$. Now let $a_k = b_k + \bar{b}_{k+1} + \cdots + \bar{b}_{j-1}$ where $\bar{b}_i$ is the image of $b_i \in A_i$ in A_k. Note that $\bar{b}_i = 0$ for $i \geq j$. Δ maps $(a_i)_i$ to $(a_i - \bar{a}_{i+1})_i = (b_i)_i$, so Δ is surjective as desired.

Now if $\{A_i\}$ satisifes the general condition, let $B_k \subseteq A_k$ be the image of $A_i \to A_k$ for large i. The maps $B_{k+1} \to B_k$ are surjective, and so $\varprojlim^1 B_i = 0$. Furthermore, the map $A_i/B_i \to A_k/B_k$ is zero and so $\{A_k/B_k\}$ satisfies the trivial condition and thus $\varprojlim^1 A_i/B_i = 0$. The short exact sequence

$$0 \to \{B_k\} \to \{A_k\} \to \{A_k/B_k\} \to 0$$

gives us the desired result: $\varprojlim^1 A_i = 0$. ■

Note that Ab can be replaced with Mod_R or $\mathsf{Ch}(\mathrm{Mod}_R)$ in all of these results; the proofs are the same.

3.5.7 Example

If $\{A_i\}$ is a tower of Noetherian modules (e.g. finite Abelian groups, finite-dimensional vector spaces), then it must satisfy the Mittag-Leffler condition since the images of $A_i \to A_k$ are descending. Thus $\varprojlim^1 A_i = 0$ in this case.

3.5.8 Theorem

Let $\cdots \to C_1 \to C_0$ be a tower of chain complexes of Abelian groups satisfying the Mittag-Leffler condition. Set $C = \varprojlim C_i$, then we have an exact sequence for every q:

$$0 \to \varprojlim^1 H_{q+1}(C_i) \to H_q(C) \to \varprojlim H_q(C_i) \to 0$$

Proof

Let $B_i \subseteq Z_i \subseteq C_i$ be the subcomplexes of boundaries and cycles, so that $H_\bullet(C_i) = Z_i/B_i$ with zero differentials. We have an exact sequence $0 \to \{Z_i\} \to \{C_i\} \xrightarrow{d} \{C_i[-1]\}$, and so we can apply the left-exact functor $\varprojlim$ to get $0 \to \varprojlim Z_i \to C \xrightarrow{d} C[-1]$. This means that $Z = \varprojlim Z_i$ is the subcomplex of cycles of C (since these are precisely those chains in the kernel of d).

Let B be the subcomplex $d(C)[1] = (C/Z)[1]$ of boundaries in C, so Z/B is the chain complex $H_\bullet(C)$. We have an exact sequence

$$0 \to \{Z_i\} \to \{C_i\} \xrightarrow{d} \{B_i[-1]\} \to 0$$

which gives us, since $\varprojlim^1 C_i = 0$,

$$0 \to Z = \varprojlim Z_i \to C = \varprojlim C_i \to \varprojlim B_i[-1] \to \varprojlim^1 Z_i \to 0 \to \varprojlim^1 B_i[-1] \to 0$$

So $\varprojlim^1 B_i = (\varprojlim^1 B_i[-1])[1] = 0$. Since $C/Z = B[-1]$ we also have the exactness of

$$0 \to B[-1] \to \varprojlim B_i[-1] \to \varprojlim^1 Z_i \to 0$$

Now we also have an exact sequence of towers $0 \to \{B_i\} \to \{Z_i\} \to \{H_\bullet(C_i)\} \to 0$, which gives us the exact

$$0 \to \varprojlim B_i \to Z \to \varprojlim H_\bullet(C_i) \to 0 \to \varprojlim^1 Z_i \to \varprojlim^1 H_\bullet(C_i) \to 0$$

so $\varprojlim^1 Z_i \cong \varprojlim^1 H_\bullet(C_i)$.
Now if we have a filtration $A \subseteq B \subseteq C$, then we have a short exact sequence $0 \to B/A \to C/A \to C/B \to 0$. We have a filtration here $B \subseteq \varprojlim B_i \subseteq Z$, and our short exact sequences show $\varprojlim B_i/B = \varprojlim^1 Z_i[1] \cong \varprojlim^1 H_\bullet(C_i)[1]$ and $Z/\varprojlim B_i \cong \varprojlim H_\bullet(C_i)$. Since $Z/B \cong H_\bullet(C)$ we get the desired short exact sequence. ■

3.5.9 Corollary

If $\cdots \to C_1 \to C_0$ is a tower of cochain complexes satisfying the Mittag-Leffler condition, the sequences become

$$0 \to \varprojlim^1 H^{q-1}(C_i) \to H^q(C) \to \varprojlim H^q(C_i) \to 0$$

3.5.10 Example

Let X be a topological CW complex, and denote by $H^\bullet(X)$ the integral cohomology of X. Recall that integral cohomology of X is the cohomology of $\hom(S(X), \mathbb{Z})$ where $S(X)$ is the singular chain complex of $\mathbb{Z}$. Now let $\{X_i\}$ be an increasing sequence of subcomplexes with $X = \bigcup X_i$. Let C_i be the cochain complex $\hom(S(X_i), \mathbb{Z})$. Now, $S_n(X_{i+1})/S_n(X_i)$ is free Abelian, and so the inclusion $S(X_i) \subseteq S(X_{i+1})$ splits. Thus $C_{i+1} \to C_i$ is surjective, and so $\cdots \to C_1 \to C_0 \to 0$ satisfies the Mittag-Leffler condition.
Furthermore, $S(X)$ is the union of $S(X_i)$s. Thus $H^\bullet(X)$ is the cohomology of

$$\hom\left(\bigcup S(X_i), \mathbb{Z}\right) = \varprojlim \hom(S(X_i), \mathbb{Z}) = \varprojlim C_i$$

By our above theorem, this tells us we have a short exact sequence

$$0 \to \varprojlim^1 H^{\bullet-1}(X_i) \to H^\bullet(X) \to \varprojlim H^\bullet(X_i) \to 0$$

This is the historic motivation for the usage of $\varprojlim^1$, discovered by Milnor.

3.6 Universal coefficient theorems

Recall that for a short exact sequence of chain complexes $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$, we have connecting morphisms $H_{n+1}(C) \xrightarrow{\partial} H_n(A)$ giving a long exact sequence. ∂ is given by lifting $[z] \in H_{n+1}(C)$ to $b \in B_{n+1}$ (so that $g(b) = z$) and then defining $\partial(z) = f^{-1}(db)$.
If we have an exact sequence

$$A \xrightarrow{f} B \to X \to C \xrightarrow{g} D$$

then we get a short exact sequence

$$0 \to \operatorname{coker} f \to X \to \ker g \to 0$$

3.6.1 Theorem (Künneth formula)

Let P be a chain complex of flat right R-modules such that the boundary submodules $B_n(P) = d(P_{n+1})$ are also flat. Then for every left R-module M and every n, we have an exact sequence

$$0 \to H_n(P) \otimes M \to H_n(P \otimes M) \to \mathrm{Tor}_1^R(H_{n-1}(P), M) \to 0$$

Proof

Since $P_n, d(P_n)$ are flat ($\otimes$-acylic) and we have a short exact sequence $0 \to Z_n \to P_n \to d(P_n) \to 0$, this shows the Z_n are flat as well and that $0 \to Z_n \otimes M \to P_n \otimes M \to d(P_n) \otimes M \to 0$ is exact. Thus we have a short exact sequence of chain complexes $0 \to Z \otimes M \to P \otimes M \to d(P) \otimes M \to 0$.

Thus we have the induced long exact sequence of homology modules,

$$\cdots \to H_{n+1}(dP \otimes M) \xrightarrow{\partial} H_n(Z \otimes M) \to H_n(P \otimes M) \to H_n(dP \otimes M) \xrightarrow{\partial} H_{n-1}(Z \otimes M) \to \cdots$$

since the differentials of dP and Z are zero, this gives us

$$\cdots \to dP_{n+1} \otimes M \xrightarrow{\partial} Z_n \otimes M \to H_n(P \otimes M) \to dP_n \otimes M \xrightarrow{\partial} Z_{n-1} \otimes M \to \cdots$$

In particular we get a short exact sequence

$$0 \to \mathrm{coker}\partial \to H_n(P \otimes M) \to \ker\partial \to 0$$

The connecting morphism is given as follows: given $dp \otimes m \in dP \otimes M$, we lift it to $p \otimes m$ then apply d to get $dp \otimes m \in P \otimes M$, and then lift to $Z \otimes M$. Thus ∂ is just $i \otimes M$, where i is the inclusion $dP_{n+1} \to Z_n$. But

$$0 \to d(P_{n+1}) \xrightarrow{i} Z_n \to H_n(P) \to 0$$

is a flat resolution of $H_n(P)$, so $\mathrm{Tor}_\bullet(H_n(P), M)$ is the homology of

$$0 \to d(P_{n+1}) \otimes M \xrightarrow{\partial} Z_n \otimes M \to 0$$

Thus $H_n(P) \otimes M$ is $\mathrm{coker}\partial$ and $\mathrm{Tor}_1(H_n(P), M)$ is $\ker\partial$. This gives us the desired result. ■

3.6.2 Theorem (Universal coefficient theorem for homology)

Let P be a chain complex of free Abelian groups. Then for every n and every Abelian group M, the above Künneth formula splits noncanonically. Thus we have a non-natural isomorphism

$$H_n(P \otimes M) \cong (H_n(P) \otimes M) \oplus \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(P), M)$$

Proof

Subgroups of free Abelian groups are free, thus P satisfies the condition for the above theorem. Since $d(P_n)$ is free and thus projective, the short exact sequence $0 \to Z_n \to P_n \to d(P_n) \to 0$ splits, giving a noncanonical isomorphism

$$P_n \cong Z_n \oplus d(P_n)$$

Thus $P_n \otimes M \cong (Z_n \otimes M) \oplus (d(P_n) \otimes M)$. In fact, $Z_n \otimes M$ is a direct summand of the kernel of $d_n \otimes M \colon P_n \otimes M \to$

$P_{n-1} \otimes M$. Now $H_n(P) \otimes M$ is the quotient of $Z_n \otimes M$ by the image of $d_{n+1} \otimes M$, and quotienting $\ker d_n \otimes M$ by this image gives $H_n(P \otimes M)$. Thus $H_n(P) \otimes M$ is a direct summand of $H_n(P \otimes M)$. The Künneth formula gives us the rest (since if A is a direct summand of B, then $B \cong A \oplus B/A$; thus $B \cong A \oplus C$ for any short exact sequence $0 \to A \to B \to C \to 0$). ■

Recall if P, Q are chain complexes of right and left R-modules respectively, we defined their tensor product $P \otimes Q$ to be the chain complex

$$(P \otimes Q)_n = \bigoplus_{p+q=n} P_p \otimes Q_q, \qquad d(a \otimes b) = (da) \otimes b + (-1)^p a \otimes d(b)$$

For a left R-module M, $P \otimes M$ is the special case where M is viewed as a chain complex focused at 0.

3.6.3 Theorem (Künneth theorem for complexes)

Let P, Q be chain complexes such that $P_n, d(P_n)$ are flat for all n. Then there is an exact sequence for all n,

$$0 \to \bigoplus_{p+q=n} H_p(P) \otimes H_q(Q) \to H_n(P \otimes Q) \to \bigoplus_{p+q=n-1} \mathrm{Tor}_1^R(H_p(P), H_q(Q)) \to 0$$

If $R = \mathbb{Z}$ and P_n are all free Abelian, then this splits noncanonically.

Proof

As before, letting $Z_p = Z_p(P)$, we have an exact sequence for every p, q:

$$0 \to Z_p \otimes Q_q \to P_p \otimes Q_q \to d(P_p) \otimes Q_q \to 0$$

By the exactness of direct sums (i.e. (AB4)), we get an exact sequence

$$0 \to \bigoplus_{p+q=n} Z_p \otimes Q_q \to \bigoplus_{p+q=n} P_p \otimes Q_q \to \bigoplus_{p+q=n} dP_p \otimes Q_q \to 0$$

This forms into a short exact sequence of chain complexes, $0 \to Z \otimes Q \to P \otimes Q \to dP \otimes Q \to 0$ (it is simple and routine to check that these maps commute with the differentials). Now, notice that since Z's differentials are all zero, we can write

$$Z \otimes Q = \bigoplus_{p \in \mathbb{Z}} (Z_p \otimes Q)[-p]$$

and since homology commutes with direct sums and exact functors (recall Z_n are all flat),

$$H_\bullet(Z \otimes Q) \cong \bigoplus_{p \in \mathbb{Z}} H_{\bullet - p}(Z_p \otimes Q) \cong \bigoplus_{p \in \mathbb{Z}} Z_p \otimes H_{\bullet - p}(Q)$$

Similarly for dP:

$$H_\bullet(dP \otimes Q) \cong \bigoplus_{p \in \mathbb{Z}} dP_p \otimes H_{\bullet - p}(Q)$$

Thus our short exact sequence gives us a long exact sequence

$$\bigoplus_p dP_p \otimes H_{n+1-p}(Q) \xrightarrow{\partial} \bigoplus_p Z_p \otimes H_{n-p}(Q) \to H_n(P \otimes Q) \to \bigoplus_p dP_p \otimes H_{n-p}(Q) \xrightarrow{\partial} \bigoplus_p Z_p \otimes H_{n-1-p}(Q)$$

∂ being the direct sum of the maps $dP_{p+1} \otimes H_{n-p}(Q) \to Z_p \otimes H_{n-p}(Q)$, i.e. $i \otimes 1$ as before. Then again,

$$0 \to dP_{p+1} \xrightarrow{i} Z_p \to H_p(P) \to 0$$

is a flat resolution of $H_p(P)$ and so $\mathrm{Tor}_\bullet^R(H_p(P), H_{n-p}(Q))$ is the homology of

$$0 \to dP_{p+1} \otimes H_{n-p}(Q) \xrightarrow{i\otimes 1} Z_p \otimes H_{n-p}(Q) \to 0$$

That is to say, $\mathrm{coker} i \otimes 1 = H_p(P) \otimes H_{n-p}(Q)$ and $\ker i \otimes 1 = \mathrm{Tor}_1^R(H_p(P), H_{n-p}(P))$
∂ is the direct sum of these $i \otimes 1$, and since the direct sum is exact it preserves kernels and cokernels. That is to say,

$$\mathrm{coker}\partial = \bigoplus_{p\in\mathbb{Z}} H_p(P) \otimes H_{n-p}(Q), \qquad \ker\partial = \bigoplus_{p\in\mathbb{Z}} \mathrm{Tor}_1^R(H_p(P), H_{n-p}(P))$$

and refining our long exact sequence to $0 \to \mathrm{coker}\partial \to H_n(P \otimes Q) \to \ker\partial \to 0$ gives us the desired result. ■

3.6.4 Example

Let $S(X)$ be the singular chain complex of a topological space; $S_n(X)$ are all free Abelian. If M is an Abelian group, the **homology of X with coefficients in** M is the homology of $S(X) \otimes M$:

$$H_\bullet(X; M) = H_\bullet(S(X) \otimes M)$$

We write $H_\bullet(X)$ when $M = \mathbb{Z}$. The above theorems then give us

$$H_n(X, M) \cong (H_n(X) \otimes M) \oplus \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(X), M)$$

This result is called the **universal coefficient theorem** in topology. Note that this isomorphism is noncanonical, and thus does not allow us to compute the long exact sequence associated with the homology groups.
If Y is another topological space, the Eilenberg-Zilber theorem says that $S(X \otimes Y)$ and $S(X) \otimes S(Y)$ are chain-homotopic. Thus $H_\bullet(X \otimes Y)$ can be computed as $H_\bullet(S(X) \otimes S(Y))$ and in particular we have a non-canonical isomorphism

$$H_n(X \otimes Y) \cong \left(\bigoplus_{p=0}^{n} H_p(X) \otimes H_{n-p}(Y)\right) \oplus \left(\bigoplus_{p=1}^{n} \mathrm{Tor}_1^{\mathbb{Z}}(H_{p-1}(X), H_{n-p}(Y))\right)$$

3.6.5 Theorem (Universal coefficient theorem for cohomology)

Let P be a chain complex of projective R-modules such that each dP_n is projective as well. Then for every n and R-module M, there is a noncanonically split exact sequence

$$0 \to \mathrm{Ext}_R^1(H_{n-1}(P), M) \to H^n(\hom_R(P, M)) \to \hom_R(H_n(P), M) \to 0$$

Proof

The short exact sequence $0 \to Z_n \to P_n \to dP_n \to 0$ gives us the long exact sequence

$$0 \to \hom(dP_n, M) \to \hom(P_n, M) \to \hom(Z_n, M) \to 0$$

since $\mathrm{Ext}_R^1(dP_n, M) = 0$ as dP_n is projective. This gives us a short exact sequence of cochain complexes $0 \to \hom(dP, M) \to \hom(P, M) \to \hom(Z, M) \to 0$. As before since the differentials of the complexes

$\hom(dP, M), \hom(Z, M)$ are zero, this gives us a long exact sequence

$$\hom(Z_{n-1}, M) \xrightarrow{\partial} \hom(dP_n, M) \to H^n \hom(P, M) \to \hom(Z_n, M) \xrightarrow{\partial} \hom(dP_{n+1}, M)$$

which then gives us the short exact sequence

$$0 \to \operatorname{coker}\partial \to H^n \hom(P, M) \to \ker\partial \to 0$$

∂ is defined by taking $f \in \hom(Z_n, M)$ and then mapping it to $\hat{f} \in \hom(P_n, M)$ restricting to f on Z_n; then we apply $\hom(d, M)$ to get $\hat{f}d$ and we then need to lift it to $\hom(dP, M)$, which we do by restricting f to dP. Thus $\partial(f) = f|_{dP}$, i.e. ∂ is $\hom(i, M)$ where i is the inclusion of $dP_{n+1} \subseteq Z_n$.
As before

$$0 \to dP_{n+1} \xrightarrow{i} Z_n \to H_n(P) \to 0$$

is a projective resolution of $H_n(P)$ and so $\operatorname{Ext}_R^\bullet(H_n(P), M)$ is the cohomology of $0 \to \hom(Z_n, M) \xrightarrow{\partial} \hom(dP_{n+1}, M) \to 0$. So $\ker\partial = \hom(H_n(P), M)$ and $\operatorname{coker}\partial = \operatorname{Ext}_R^1(H_n(P), M)$, as desired. ■

3.6.6 Example

If X is a topological space and M an Abelian group, the **cohomology of X with coefficients in M** is

$$H^\bullet(X; M) = H^\bullet \hom(S(X), M)$$

Thus we get

$$H^n(X; M) \cong \hom(H_n(X), M) \oplus \operatorname{Ext}_{\mathbb{Z}}^1(H_{n-1}(X), M)$$

If X has n path-connected components then $H_0(X) = \mathbb{Z}^n$, so

$$H^0(X; M) \cong M^n, \qquad H^1(X; M) \cong \operatorname{Ext}^1(H_1(X), \mathbb{Z})$$

Dual to the tensor of chain complexes, recall if P is a chain complex and Q a cochain complex, we define their **internal hom cochain complex** to be

$$\hom(P, Q)^n = \prod_{p+q=n} \hom(P_p, Q^q), \qquad d(f\colon P_p \to Q^q)_{p+q=n} = fd - (-1)^n df$$

Then we have an exact sequence

$$0 \to \prod_{p+q=n-1} \operatorname{Ext}_R^1(H_p(P), H^q(Q)) \to H^n \hom(P, Q) \to \prod_{p+q=n} \hom_R(H_p(P), H^q(Q)) \to 0$$

Note that in our discussion of the splitting of the above short exact sequences, we needed only that dP_n all be projective (so that $0 \to Z \to P \to dP \to 0$ split). In particular, we can replace $R = \mathbb{Z}$ with a **hereditary** ring, and require only that P be a chain complex of projective modules.

3.6.7 Definition

A ring is **hereditary** if every submodules of any free module is projective. Since a module is projective iff it is a direct summand of a free module, a ring is hereditary iff submodules of projective modules are projective.